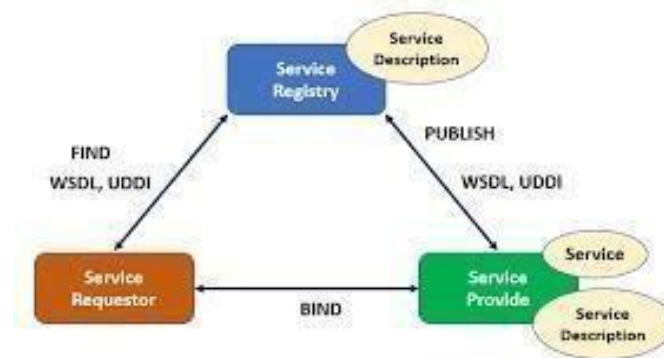


EXP NO: 1 A DEVELOP A NEW WEBSERVICE FOR DESIGNING A CALCULATOR

AIM: To develop a new webservice calculator for designing a calculator using NetBeans IDE.

DESCRIPTION:

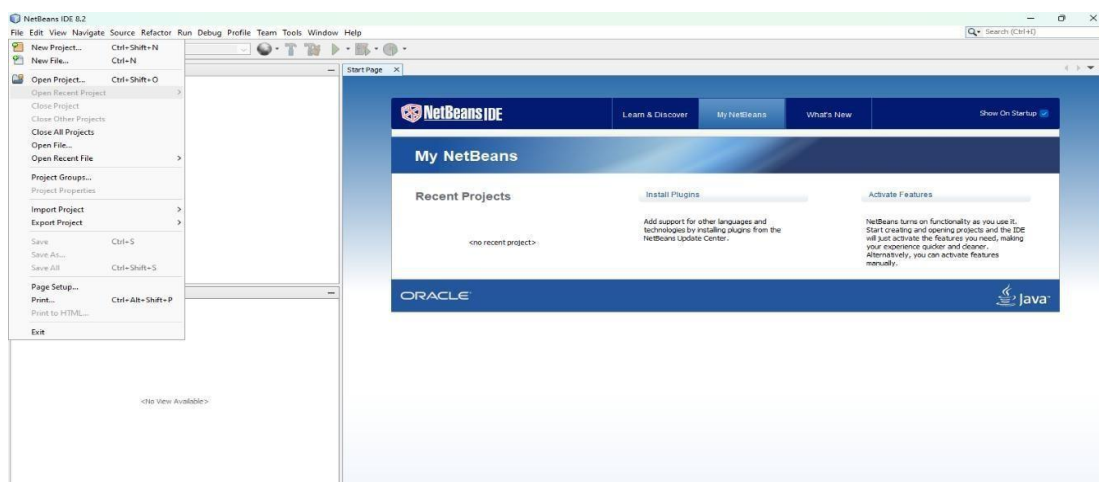


- The Service Provider publishes the WSDL (Web Services Description Language) to a Service Registry (e.g., UDDI), which includes both abstract (interface) and concrete (implementation) definitions. The Service Requestor finds the required service by querying the registry and retrieves the WSDL. Based on the WSDL, the requestor binds to the service using protocols like SOAP/HTTP.
- The four main specifications in WebServices are: XML (Extensible Markup Language), WSDL (Web Service Description Language), SOAP (Simple Object Access Protocol), UDDI (Universal Description Discovery Language).

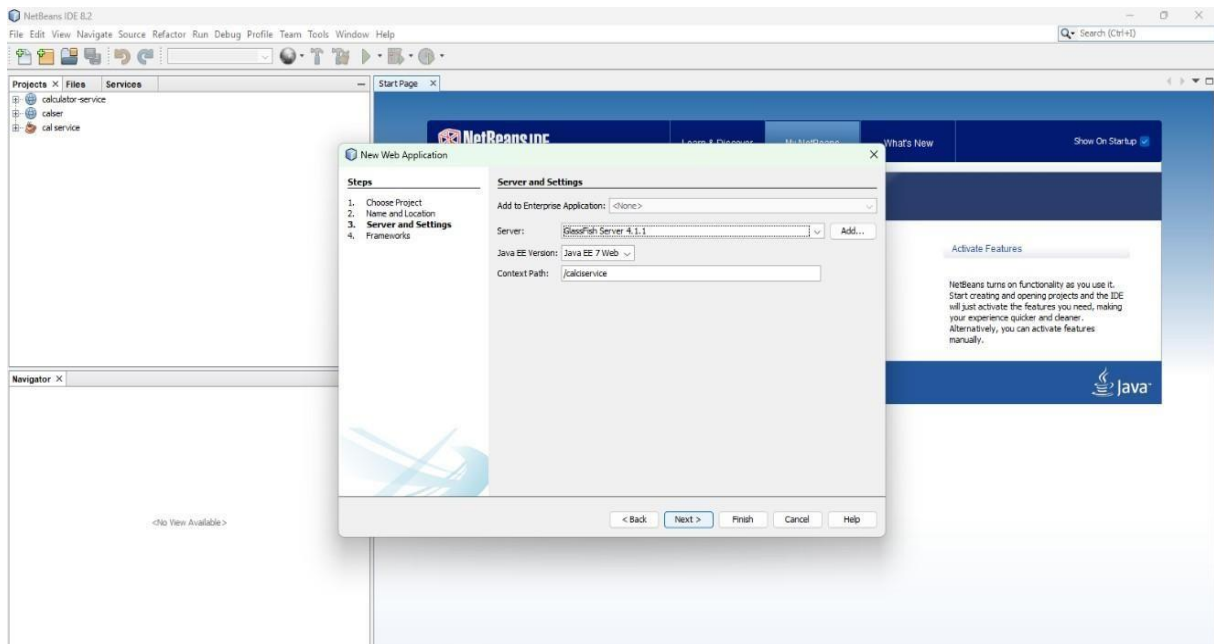
PROCEDURE:

Creating a webservice Calculator

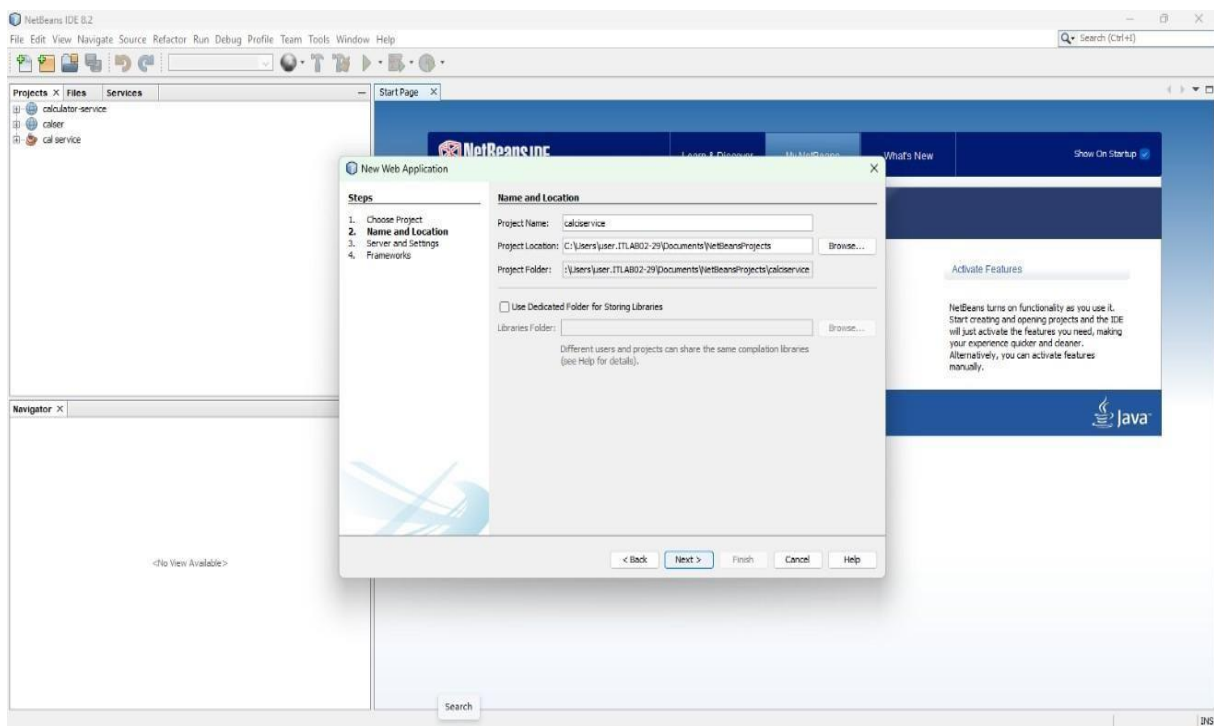
Step 1: Install NetBeans IDE , now go to file and click new project.



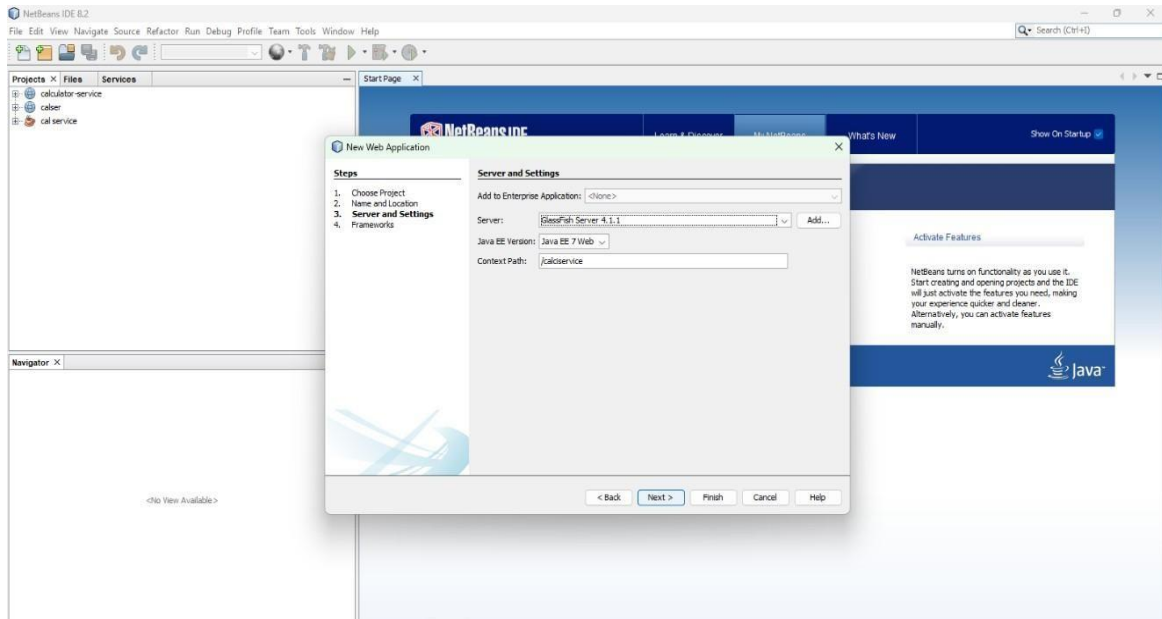
Step 2: Select Java Web Application , click finish and next.



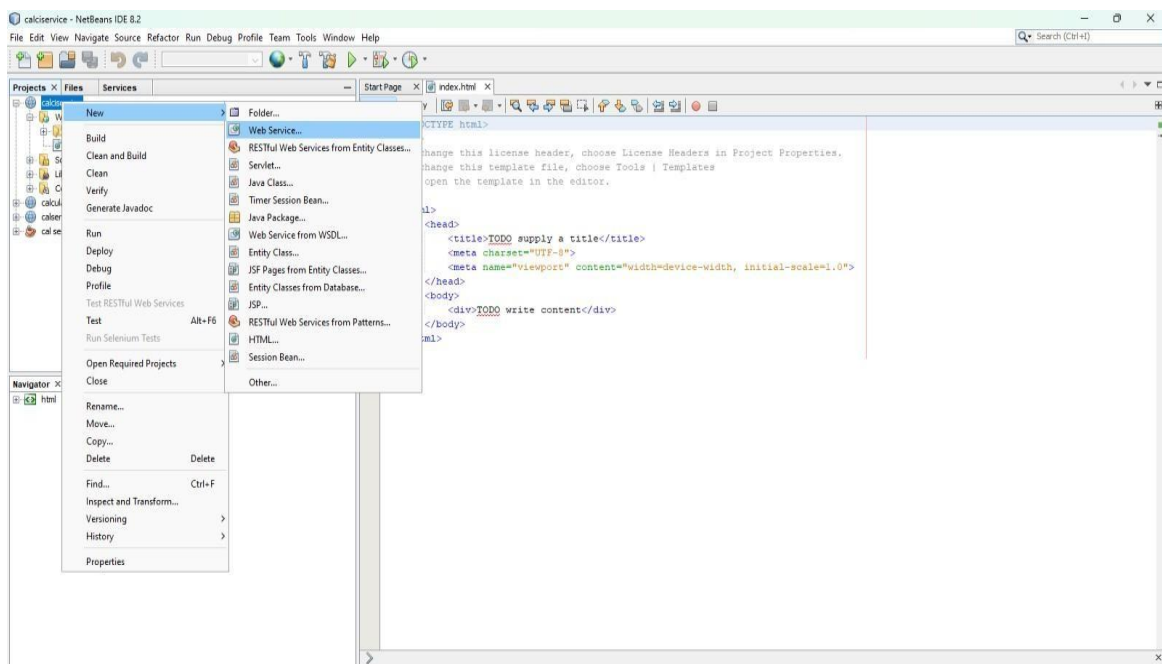
Step 3: Give the project name as CalculatorService and click next .Now, right click the CalculatorService , click new and create a webservice.

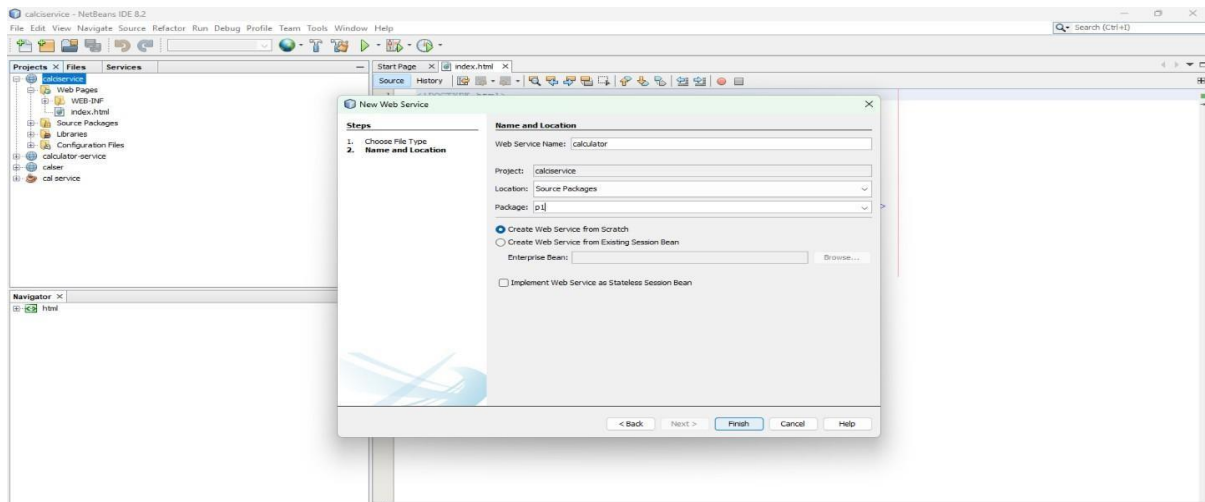


Step 4: In the server and settings, ensure that the server name is set to GlassFish Server and the Java EE version is Java EE 7 Web

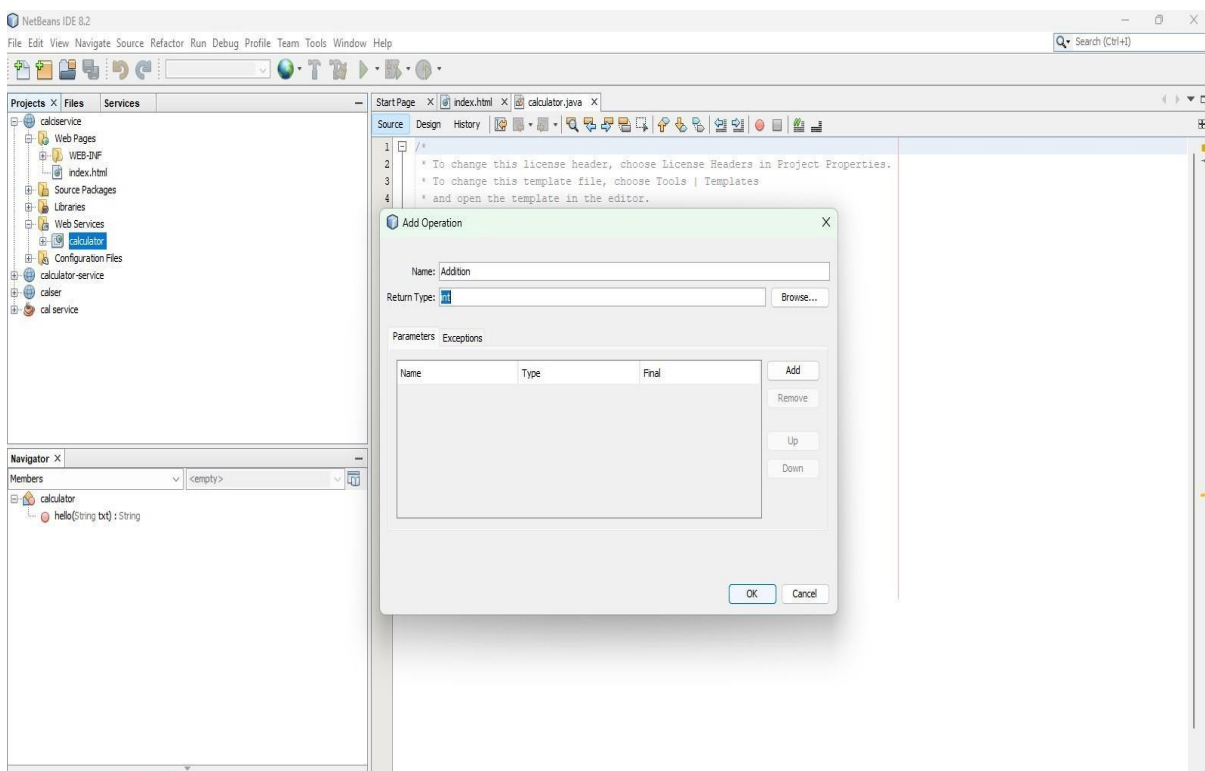


Step 5: Right click the service project, Select->new->web service, Enter the web service name, Assign the package name, Click finish.

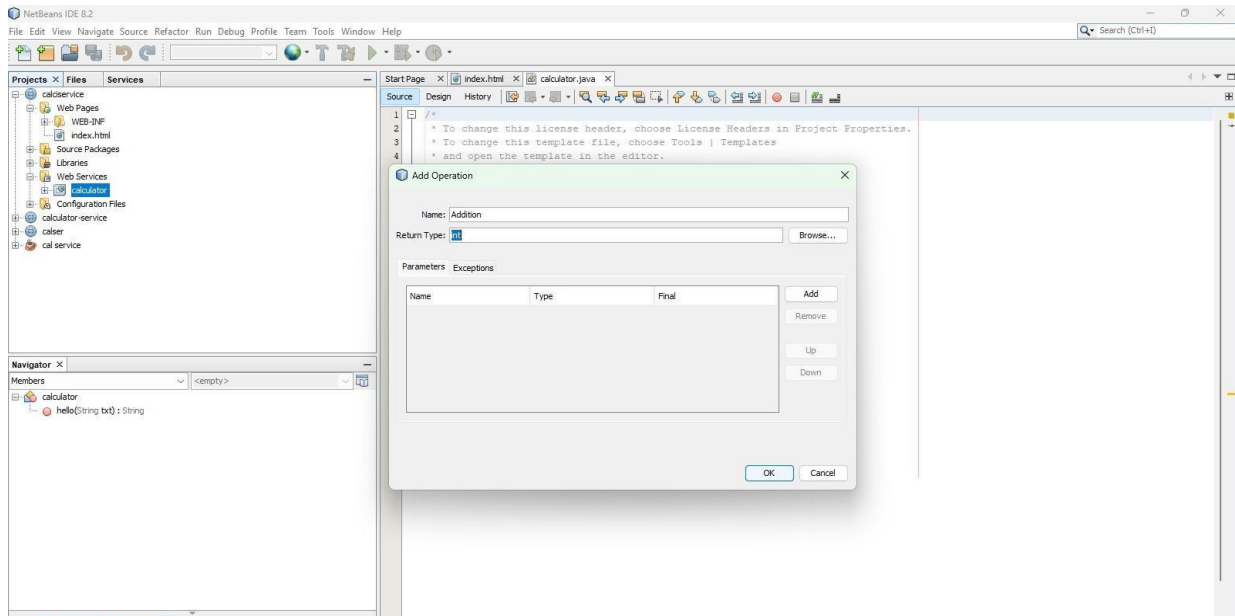




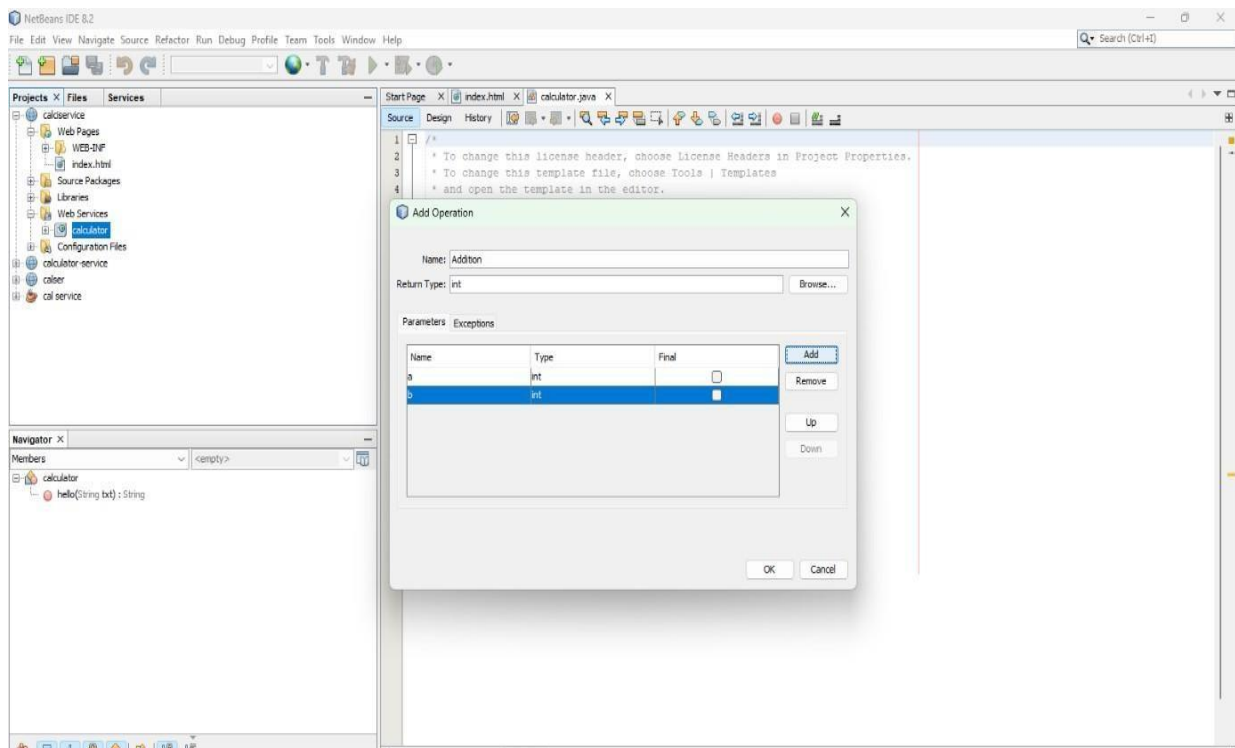
Step 6: Expand the webservice folder, right click on the service you have created and select add operation.



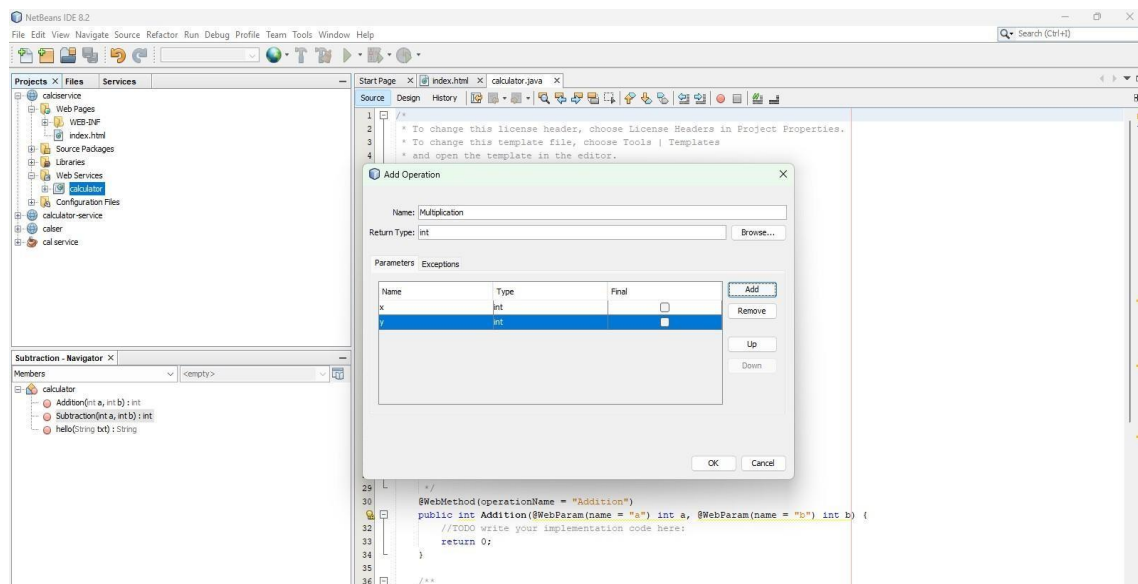
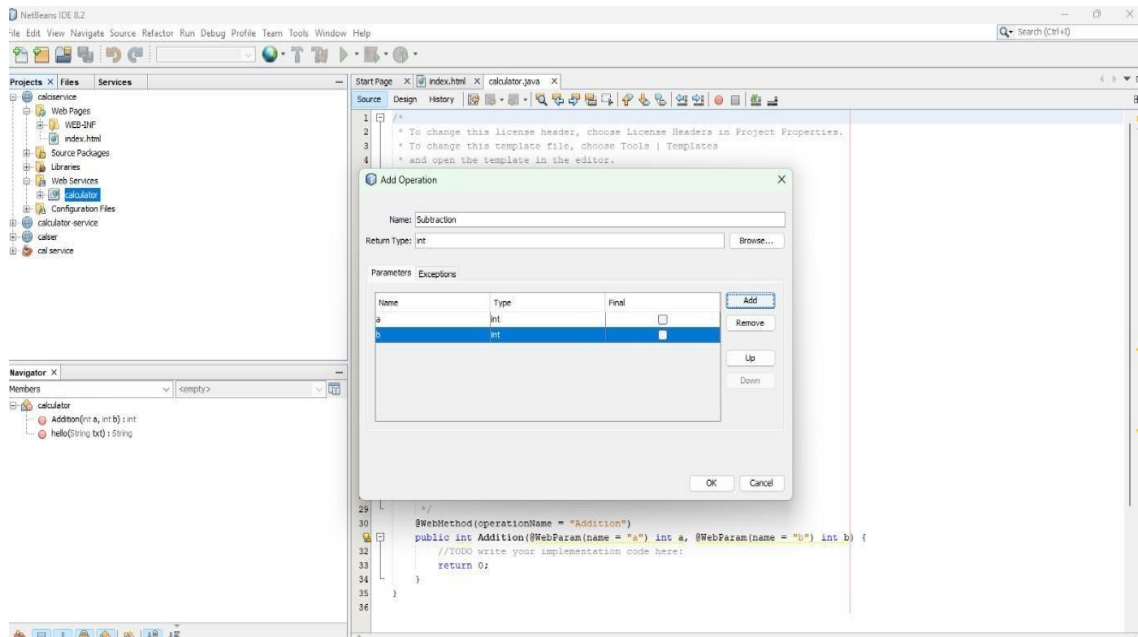
Step 7: Give name of the operations as add, Choose the return type as Integer(int).



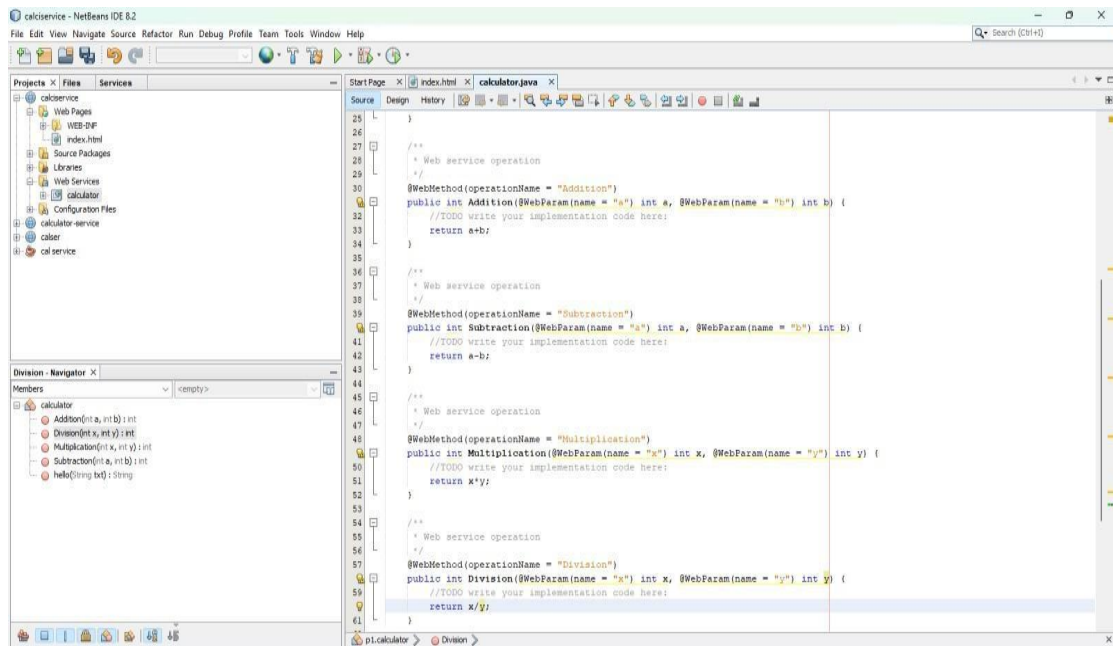
Step 8: Click on the add button, enter the parameters, choose the datatype as Integer, click ok.



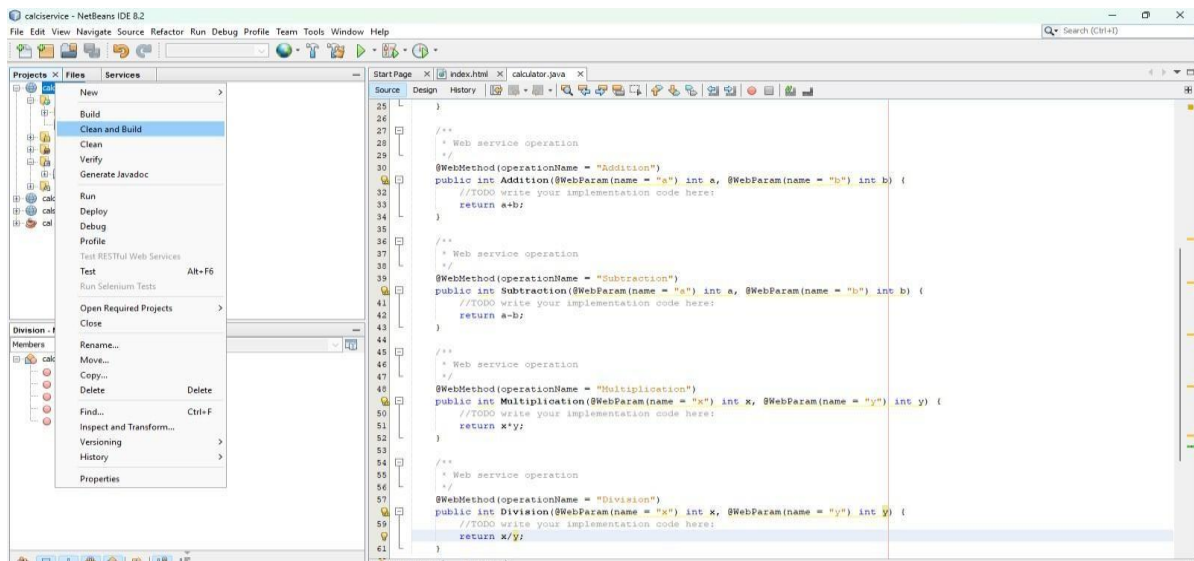
Step 9: Similarly create for Subtraction, Multiplication, Division operations, Add the parameters and change the datatype.



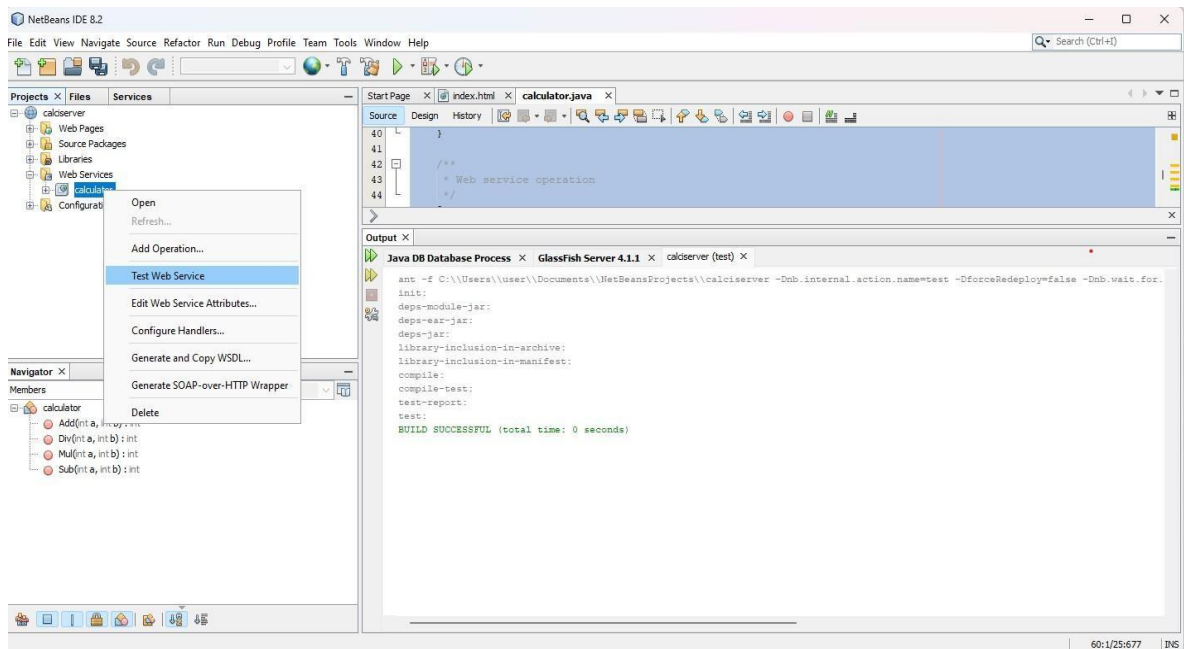
Step 10: In web method code, change return '0' as return a+b and follow the same procedure for other operations.



Step 11: Right click on the service project, click clean and build and also deploy it.



Step 12: Right click the web service page, give Test web service.



<http://localhost:8080/Jc>
Calc Web Service Tester

Calc Web Service Tester

This form will allow you to test your web service implementation ([WSDL File](#))

To invoke an operation, fill the method parameter(s) input boxes and click on the button labeled with the method name.

Methods :

public abstract int p.Calc.add(int,int)

add

public abstract int p.Calc.sub(int,int)

sub

public abstract int p.Calc.mul(int,int)

mul

public abstract int p.Calc.div(int,int)

div

public abstract java.lang.String p.Calc.hello(java.lang.String)

hello

division Method invocation

Method parameter(s)

Type	Value
int	50
int	2

Method returned

int : "25"

SOAP Request

```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body>
    <ns2:division xmlns:ns2="http://p3/">
      <x058/x>
        <y2/y>
      </ns2:division>
    </S:Body>
  </S:Envelope>
```

SOAP Response

```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body>
    <ns2:divisionResponse xmlns:ns2="http://p3/">
      <return>25</return>
    </ns2:divisionResponse>
  </S:Body>
</S:Envelope>
```

addition Method invocation**Method parameter(s)**

Type	Value
int	34
int	23

Method returned

int : "57"

SOAP Request

```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body>
    <ns2:addition xmlns:ns2="http://p3/">
      <a>34</a>
      <b>23</b>
    </ns2:addition>
  </S:Body>
</S:Envelope>
```

SOAP Response

```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body>
    <ns2:additionResponse xmlns:ns2="http://p3/">
      <return>57</return>
    </ns2:additionResponse>
  </S:Body>
</S:Envelope>
```

subtraction Method invocation**Method parameter(s)**

Type	Value
int	55
int	23

Method returned

int : "32"

SOAP Request

```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body>
    <ns2:subtraction xmlns:ns2="http://p3/">
      <x>55</x>
      <y>23</y>
    </ns2:subtraction>
  </S:Body>
</S:Envelope>
```

SOAP Response

```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body>
    <ns2:subtractionResponse xmlns:ns2="http://p3/">
      <return>32</return>
    </ns2:subtractionResponse>
  </S:Body>
</S:Envelope>
```

multiplication Method invocation**Method parameter(s)**

Type	Value
int	34
int	3

Method returned

int : "102"

SOAP Request

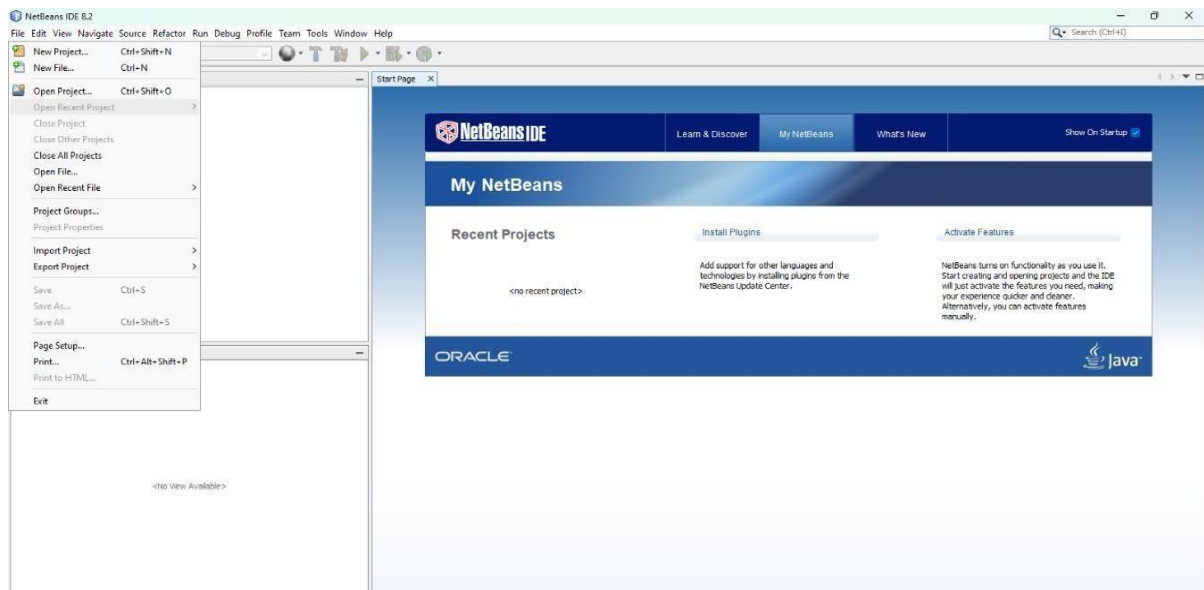
```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body>
    <ns2:multiplication xmlns:ns2="http://p3/">
      <a>34</a>
      <b>3</b>
    </ns2:multiplication>
  </S:Body>
</S:Envelope>
```

SOAP Response

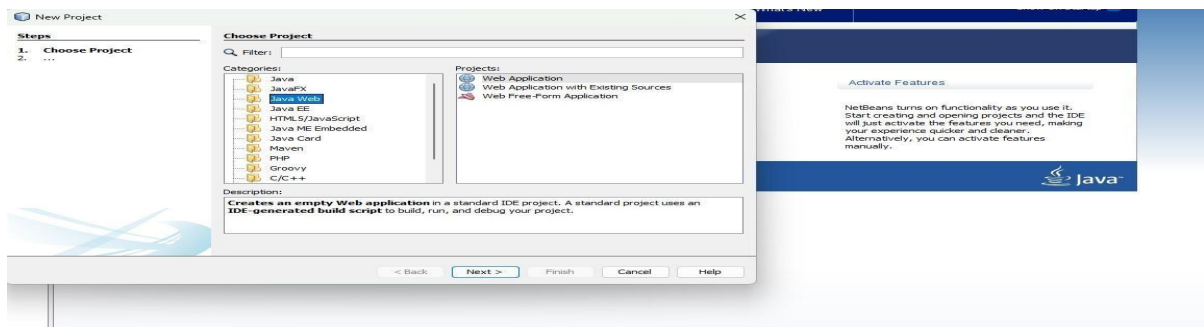
```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body>
    <ns2:multiplicationResponse xmlns:ns2="http://p3/">
      <return>102</return>
    </ns2:multiplicationResponse>
  </S:Body>
</S:Envelope>
```

Creating a Client Application.

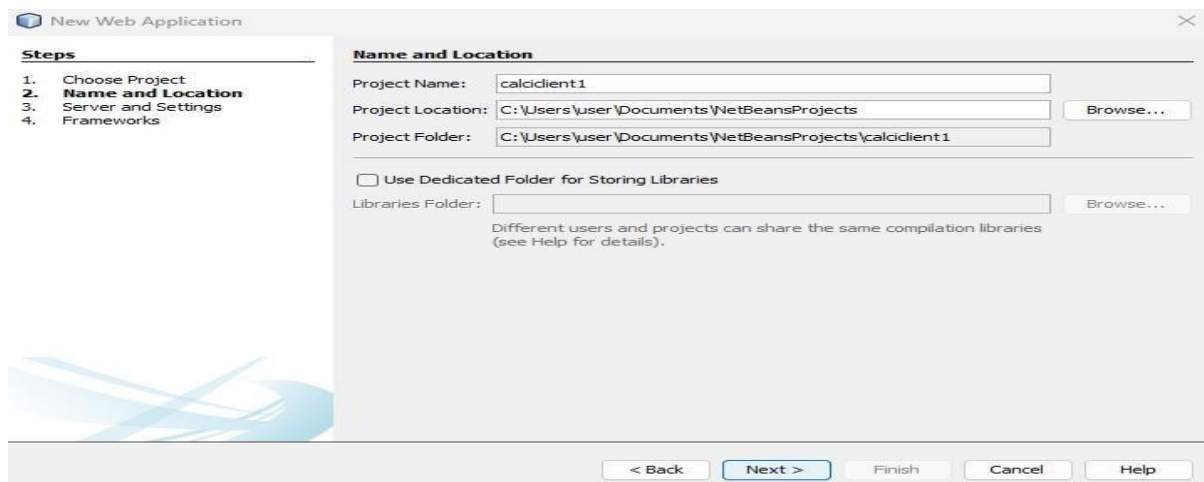
Step 1: Install NetBeans IDE , now go to file and click new project.



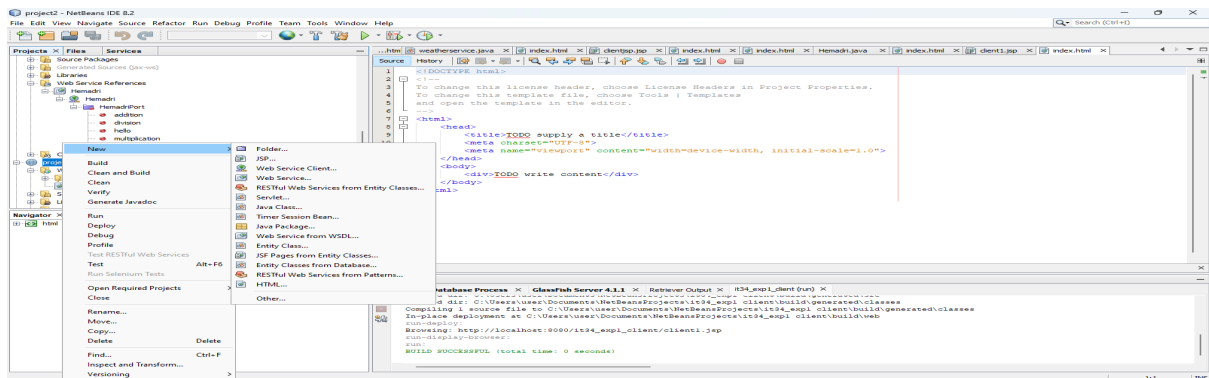
Step 2: Select Java web, Choose web application, Click next.



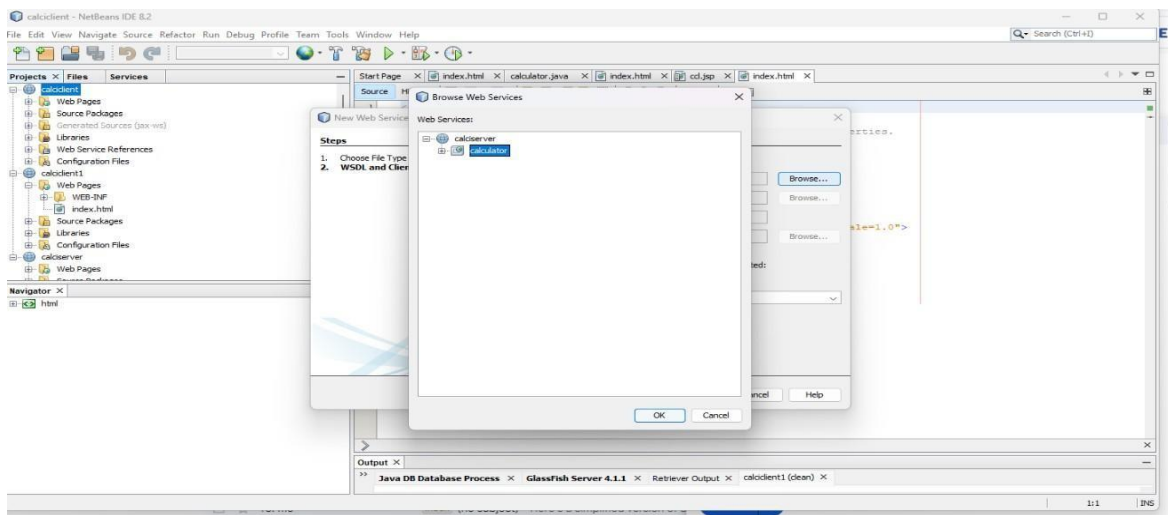
Step 3 : Assign a name for your project, Click next.



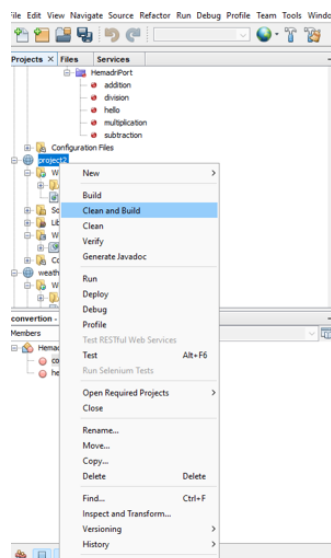
Step 4: Right click the client project, select new web service from the client(If webservice is not listed, click other, choose web services from the category, then choose web service client from file type, click next).



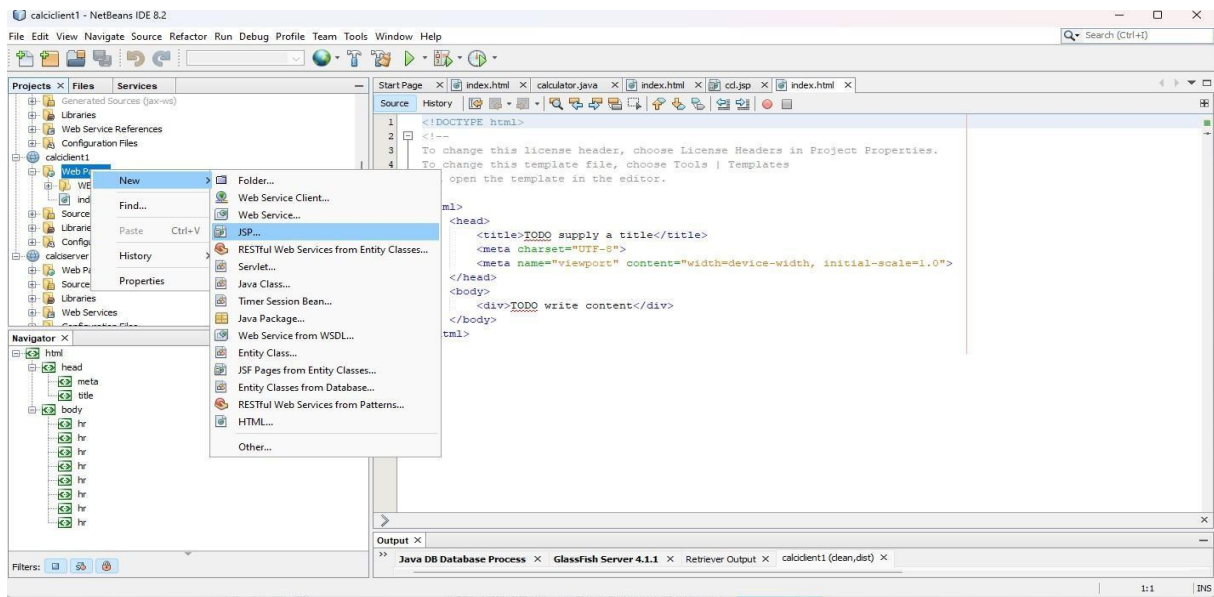
Step 5: Next browse for the project you have created and choose the web service that you have created in the service side, click ok and then click finish.



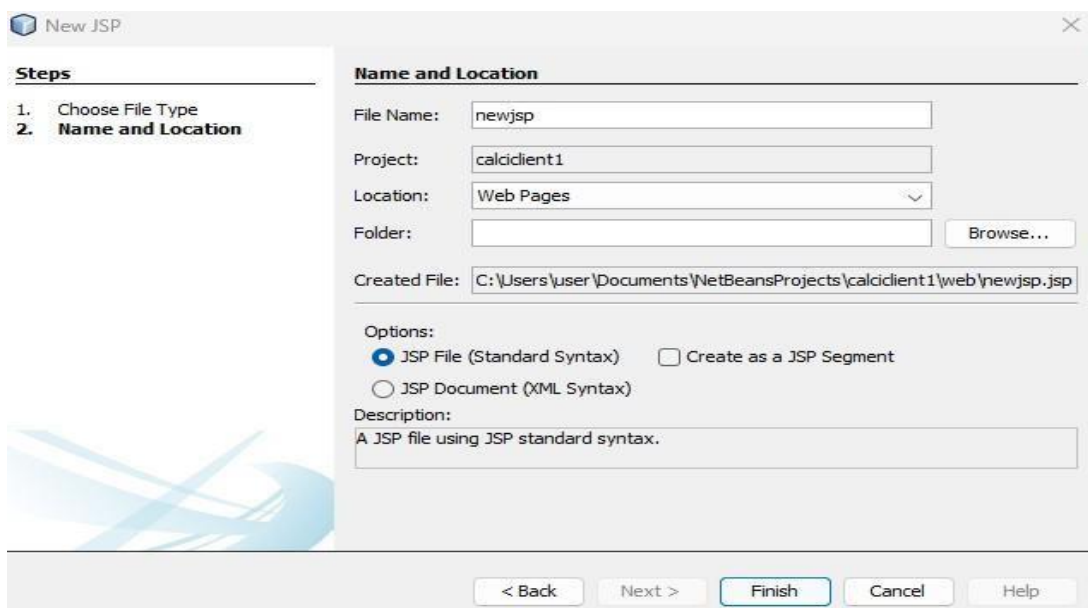
Step 6: Right click the project, Clean and build it.



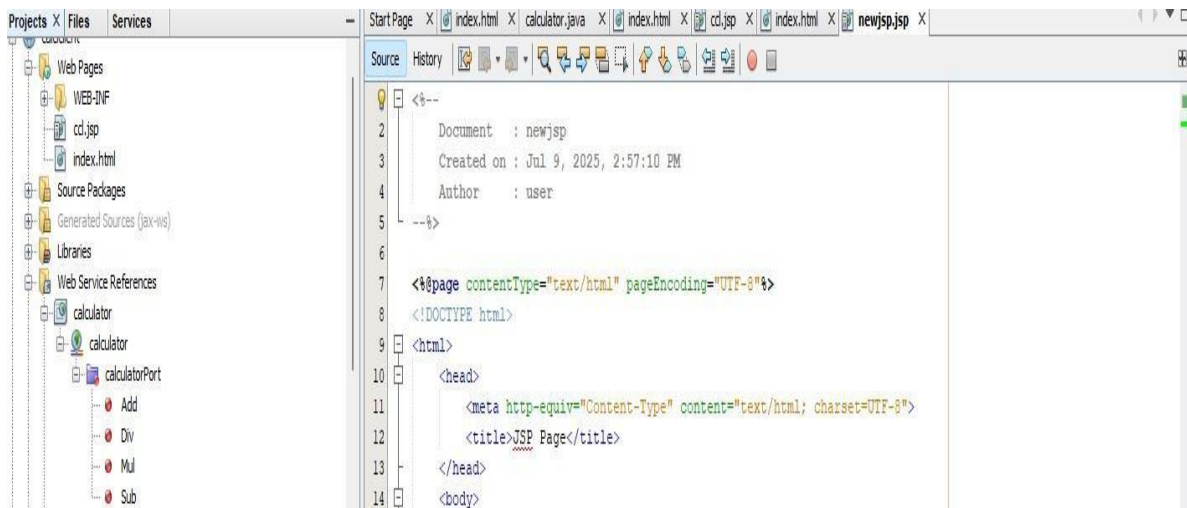
Step 7: Expand the client project, choose web pages, right click and select JSP.



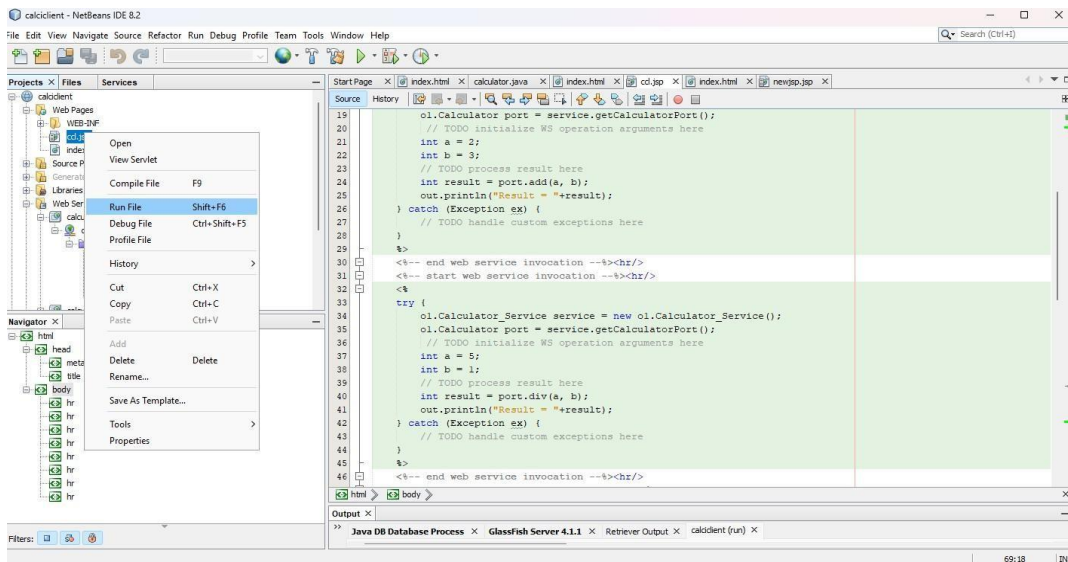
Step 8: Assign a name for your JSP file and click finish.



Step 9: Delete the content within the body tag in JSP file.



Step 10 : Expand the Web Service References, drag and drop the arithmetic operation interfaces inside the <body> tag, change the values of the parameters, then right-click the JSP file and select Run.



Hello World!

Result = 11

Result = 3

Result = Hello John !

Result = 6

Result = 1

RESULT : Thus, development of a new webservice calculator for designing a calculator using NetBeans IDE is implemented successfully.

EXP NO: 1 B
DATE:

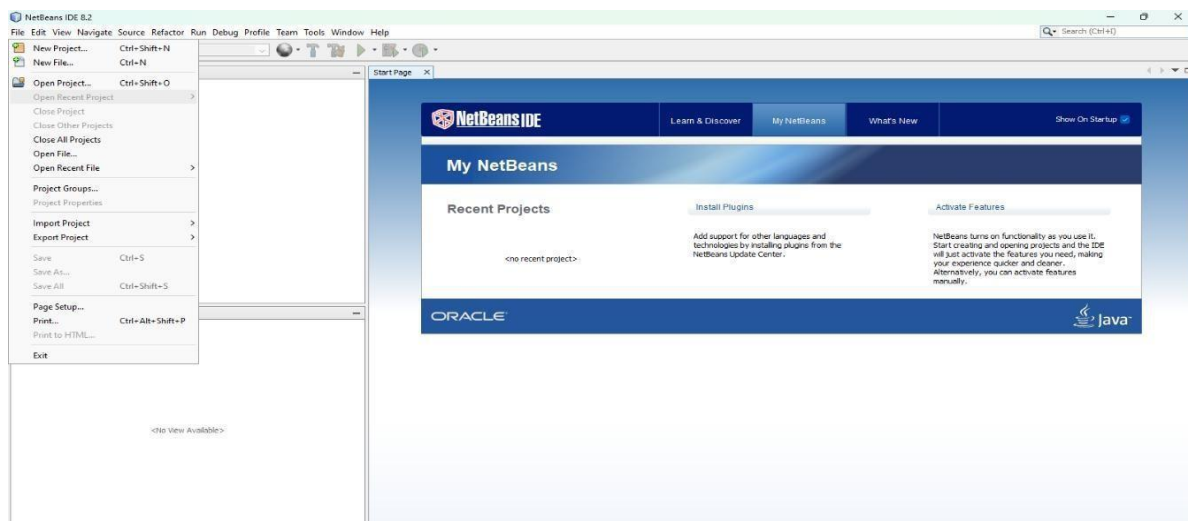
DEVELOP A NEW WEBSERVICE FOR CONVERTING TEMPERATURE FROM FAHRENHEIT TO CELSIUS

AIM: To develop a new webservice for converting temperature from Fahrenheit to Celsius using NetBeans IDE.

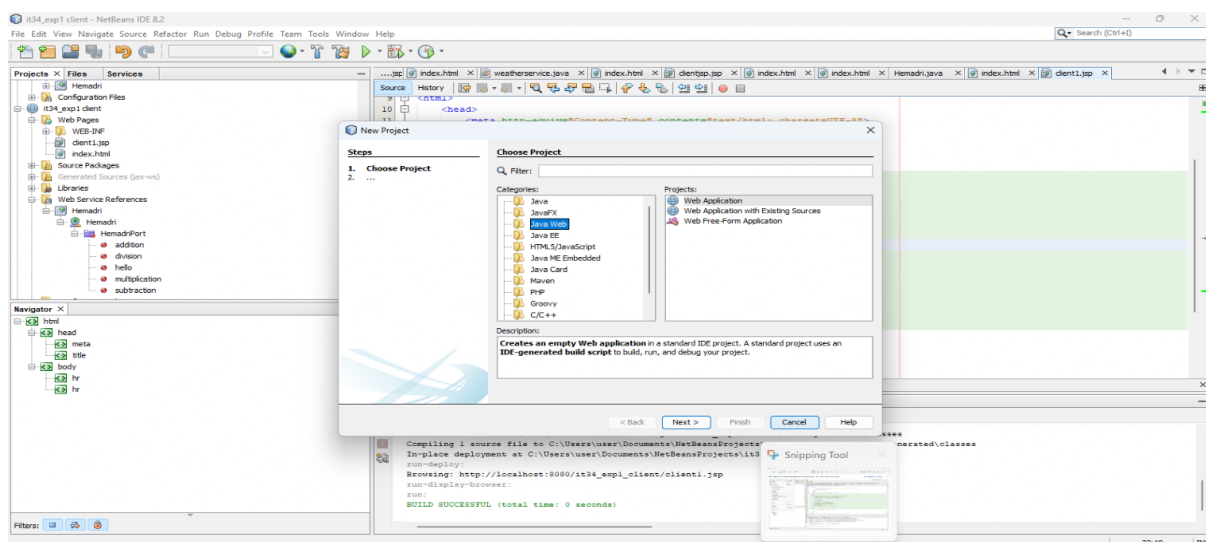
PROCEDURE:

Creating Web Service – (Fahrenheit to Celsius).

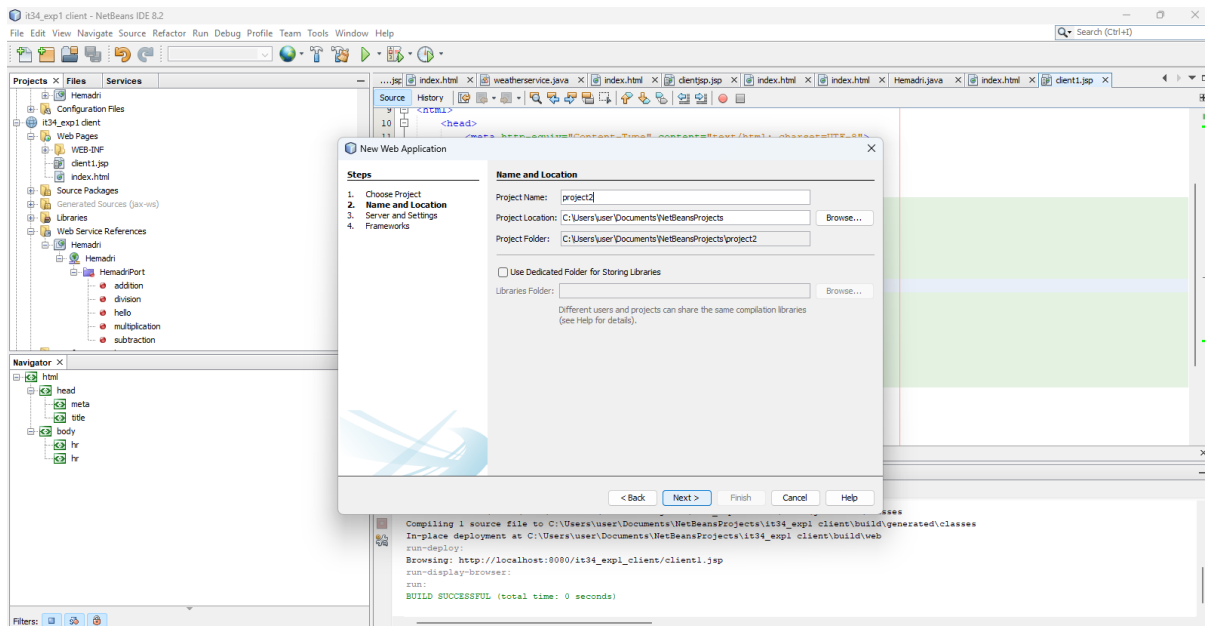
Step 1: Install NetBeans IDE , now go to file and click new project.



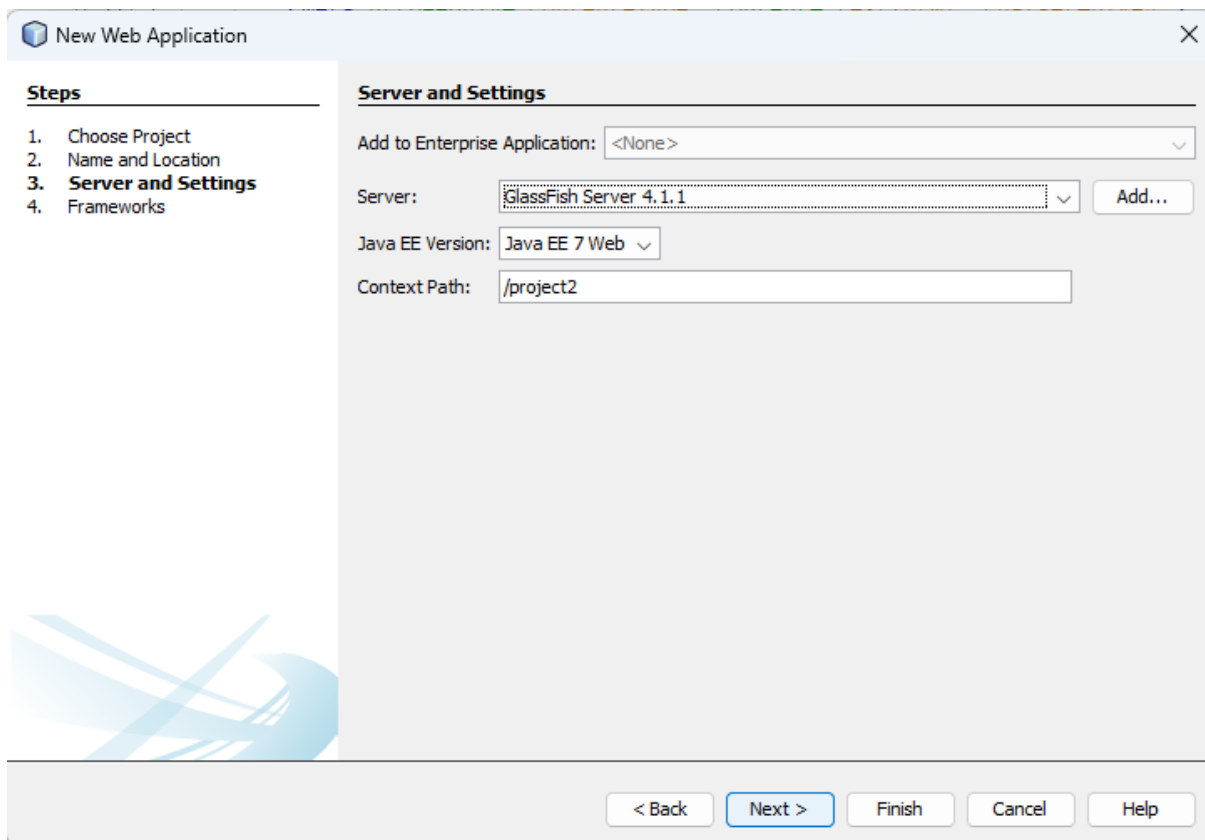
Step 2: Select Java Web Application , click finish and next.

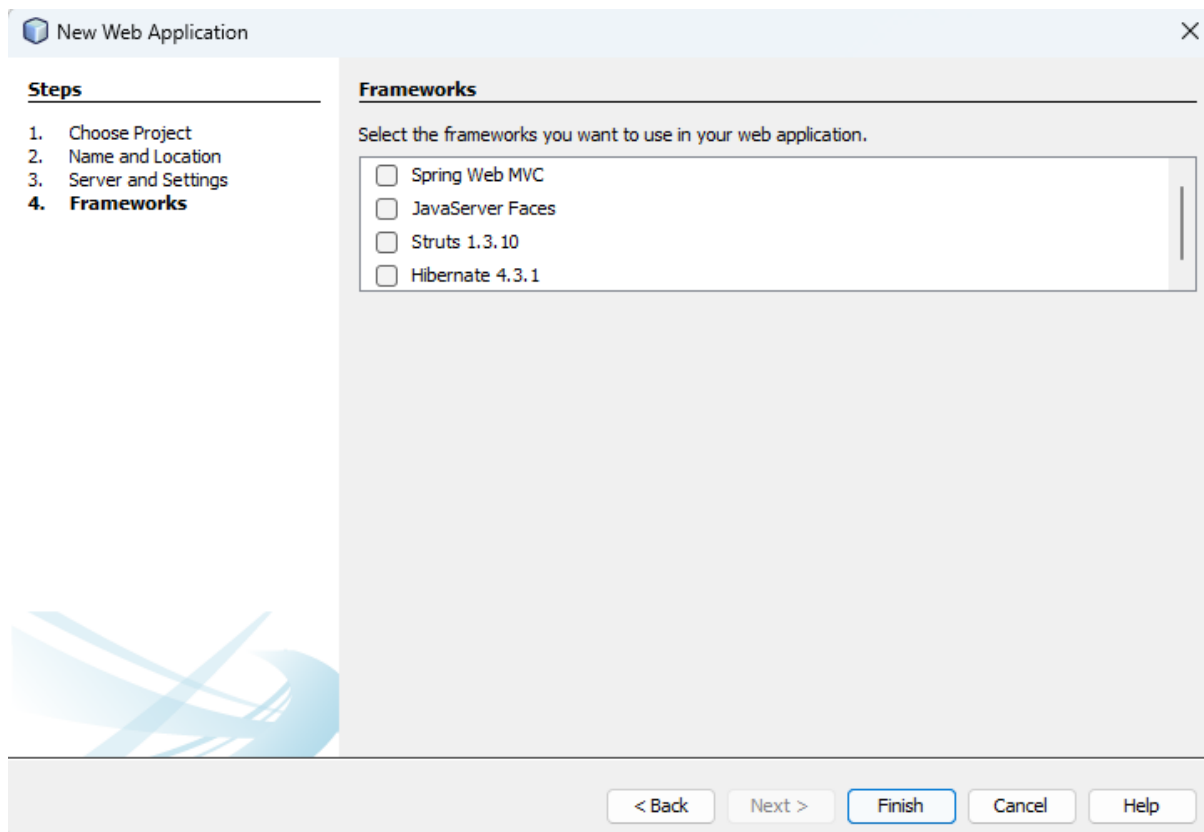


Step 3: Give the project name as TemperatureConverterService and click Next. Then, right-click the TemperatureConverterService project, select New → Web Service, and create a web service.

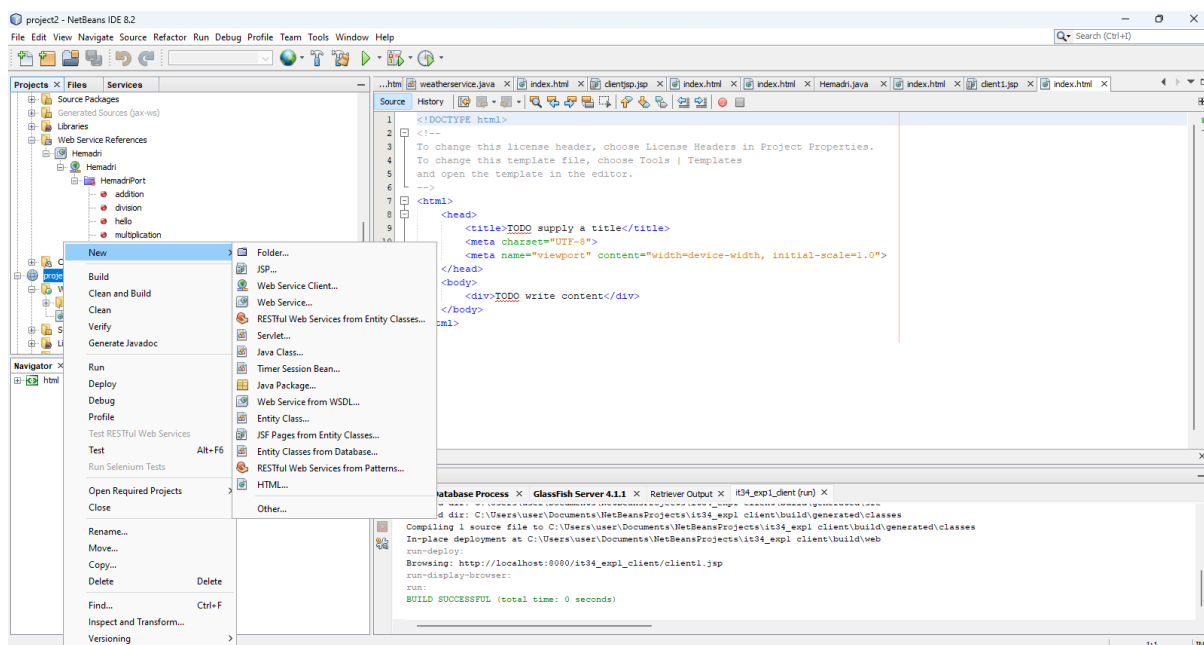


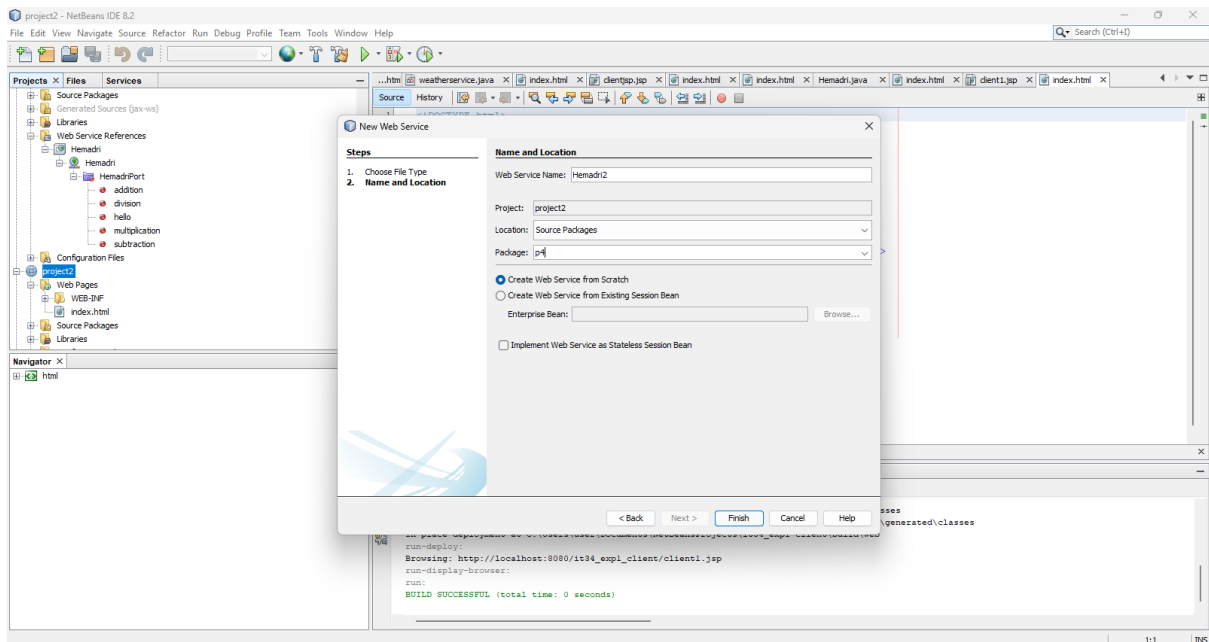
Step 4: In the Server and Settings window, ensure that the Server Name is set to GlassFish Server and the Java EE Version is Java EE 7 Web. Then click Finish.



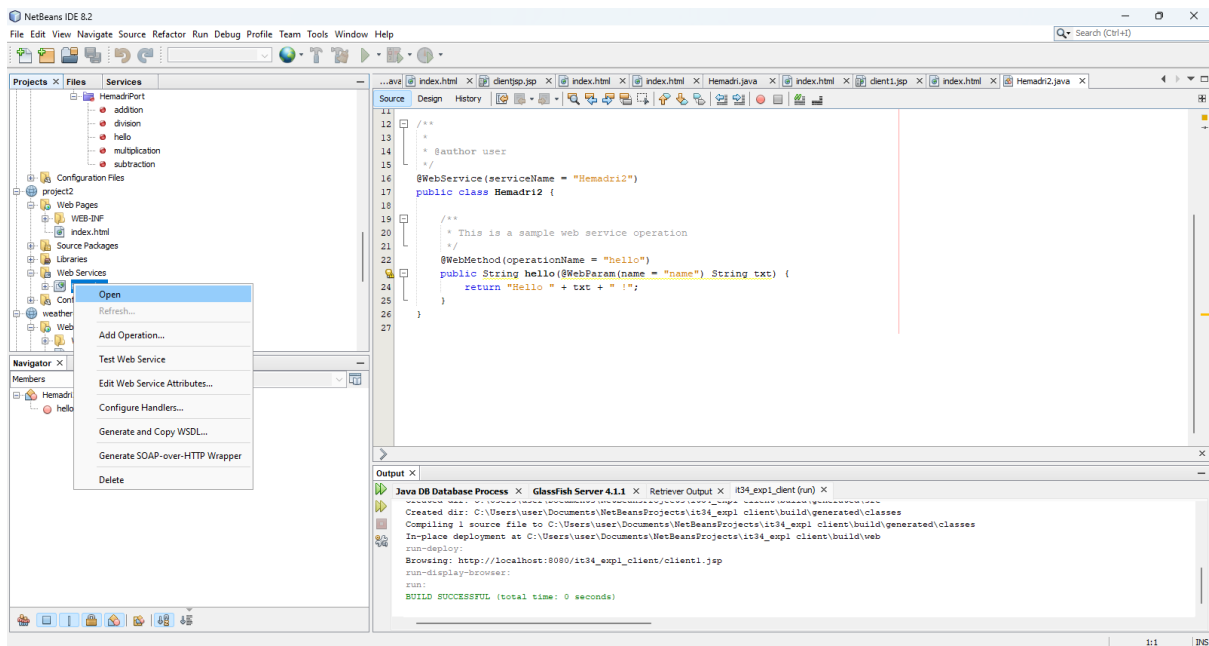


Step 5 : Right-click the TemperatureConverterService project, select New → Web Service. Enter the Web Service Name as TemperatureService, assign the package name (e.g., com.temperature), and click Finish.

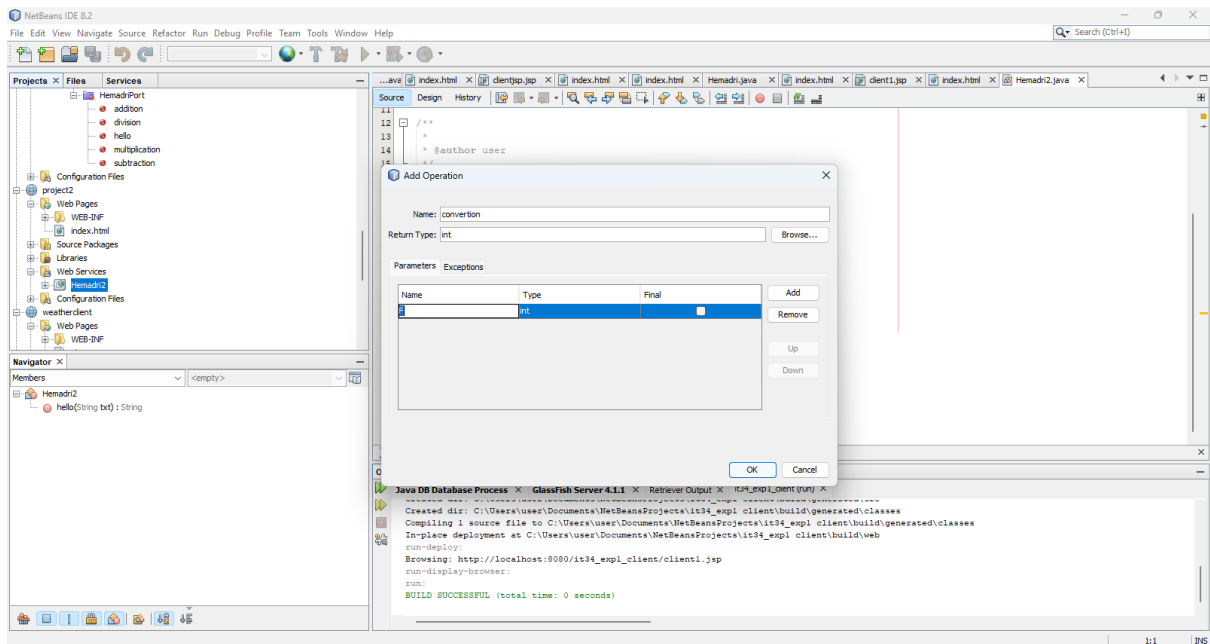




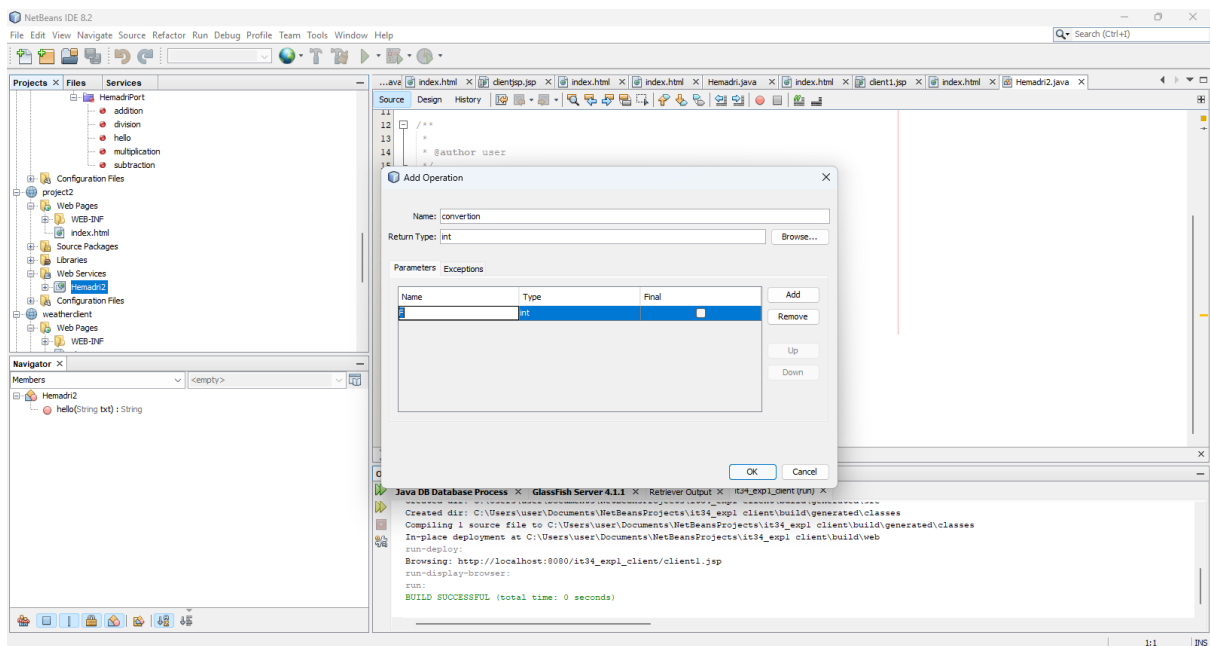
Step 6: Expand the Web Service folder, right-click the TemperatureService web service, and select Add Operation.



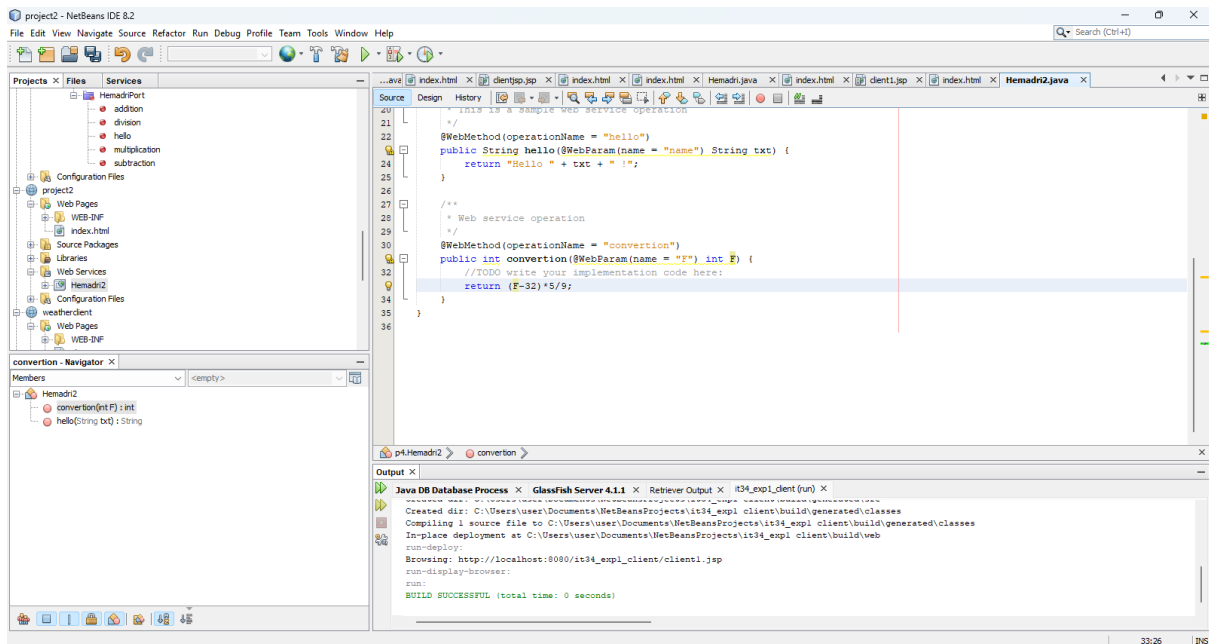
Step 7: Give the operation name as fahrenheitToCelsius and choose the Return Type as Double (double).



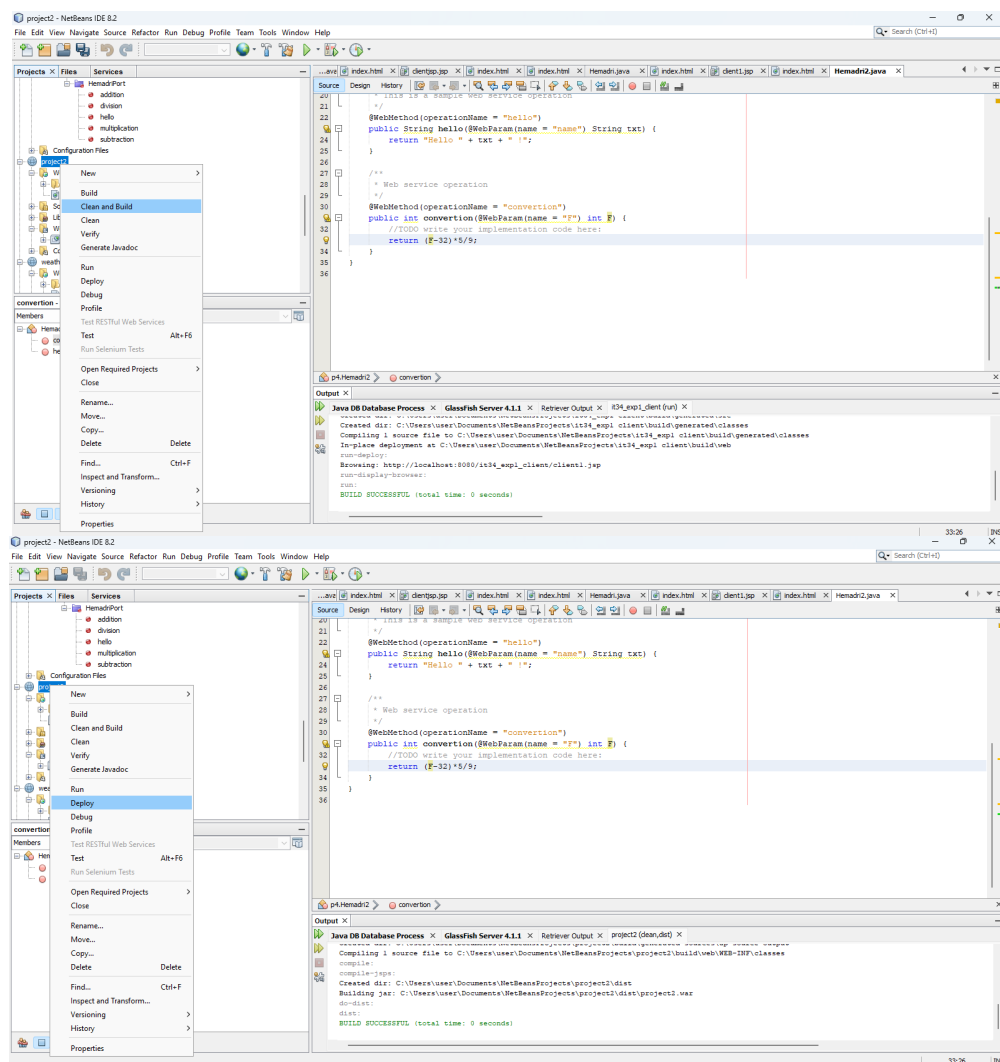
Step 8 : Click on the Add button, enter the parameter name as fahrenheit, choose the Data Type as Double (double), and click OK.



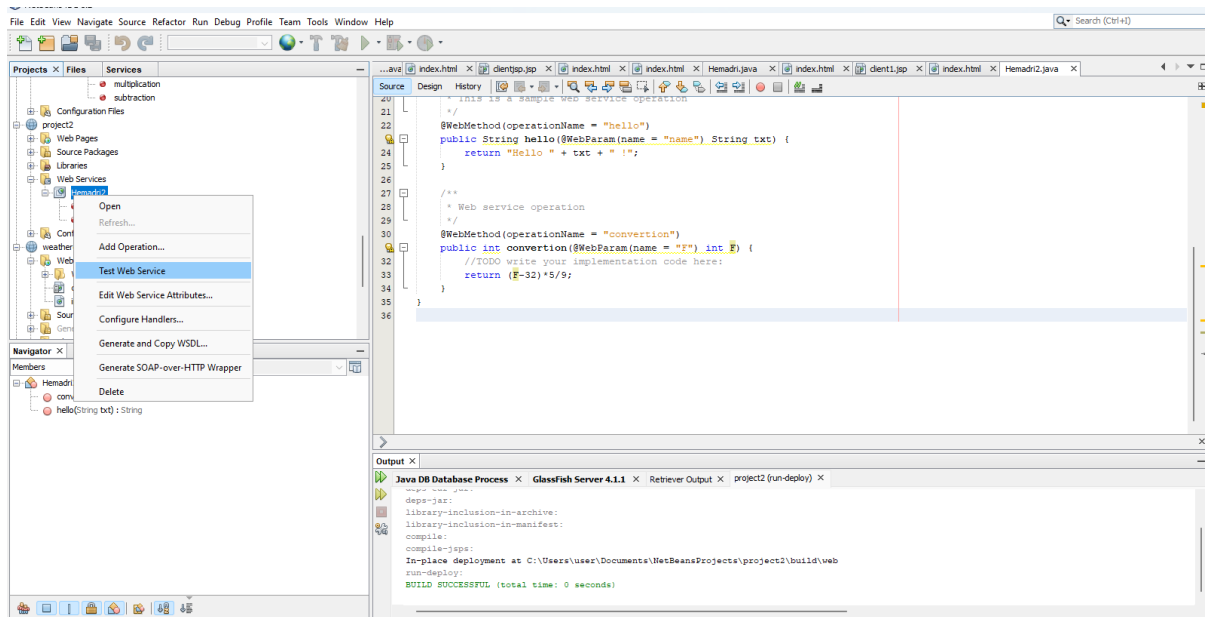
Step 9 : In the web method code, replace return 0; with the Fahrenheit to Celsius conversion formula: $\text{return (fahrenheit - 32) * 5 / 9;}$



Step 10: Right-click the TemperatureConverterService project and select Clean and Build. After the build is completed successfully, right-click the project again and select Deploy.



Step 11 : Right-click the TemperatureService web service and select Test Web Service to test the Fahrenheit to Celsius conversion.



ftoC Method invocation

Method parameter(s)

Type	Value
int	65

Method returned

float : "18.0"

SOAP Request

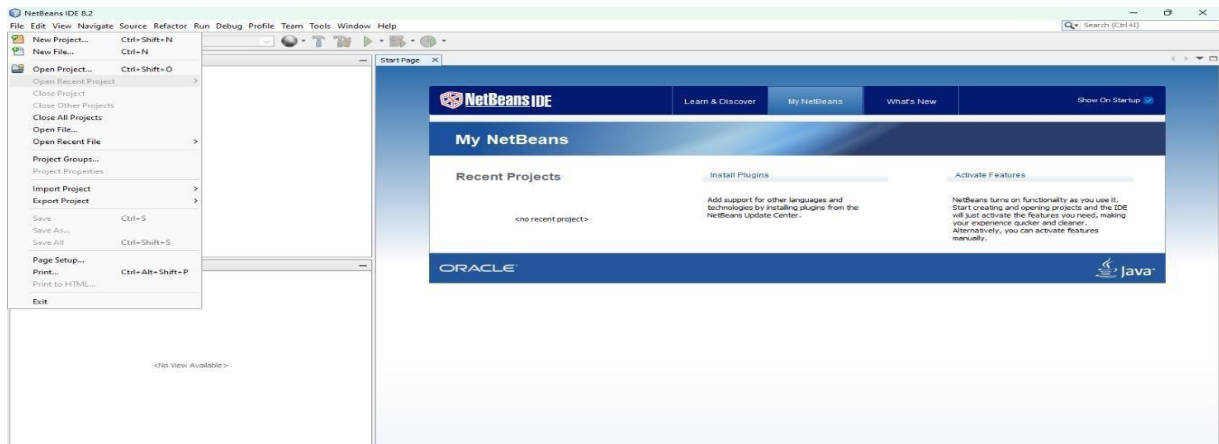
```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body>
    <ns2:FtoC xmlns:ns2="http://p/">
      <a>65</a>
    </ns2:FtoC>
  </S:Body>
</S:Envelope>
```

SOAP Response

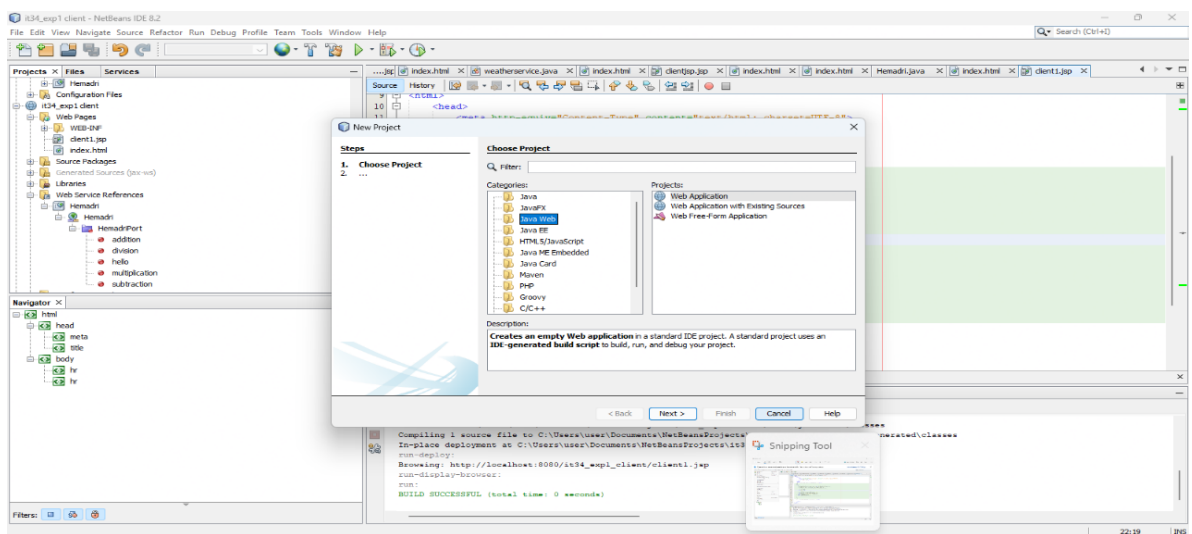
```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body>
    <ns2:FtoC xmlns:ns2="http://p/">
      <a>18.0</a>
    </ns2:FtoC>
  </S:Body>
</S:Envelope>
```

Creating a Client Application Fahrenheit to Celsius

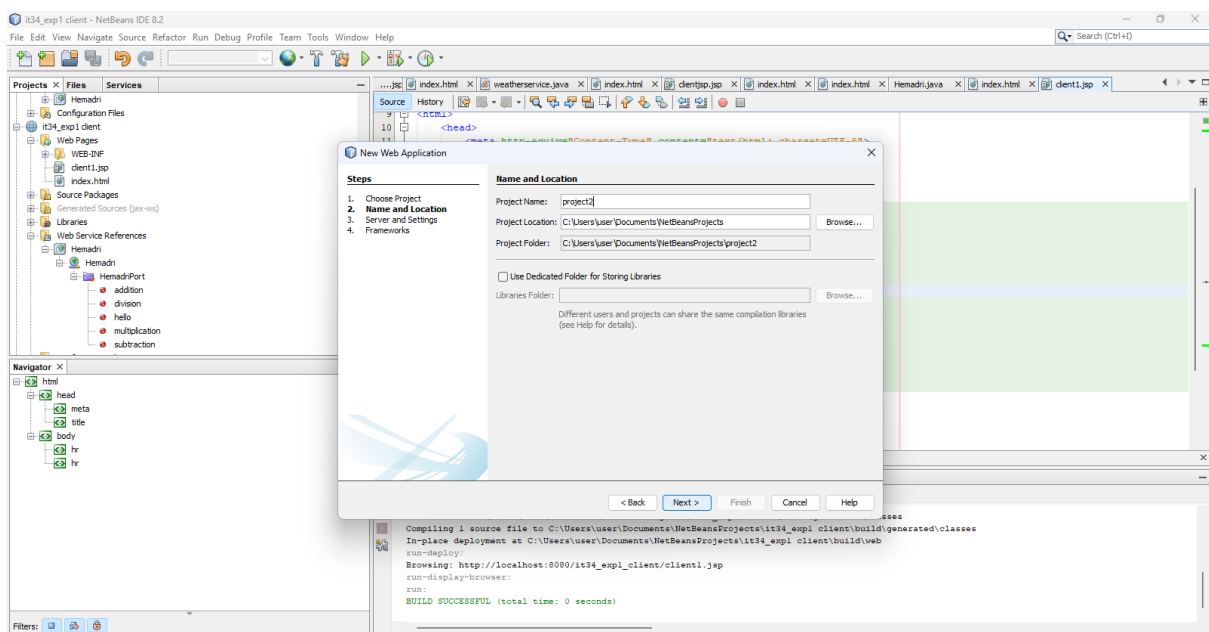
Step 1 : Install NetBeans IDE , now go to file and click new project.



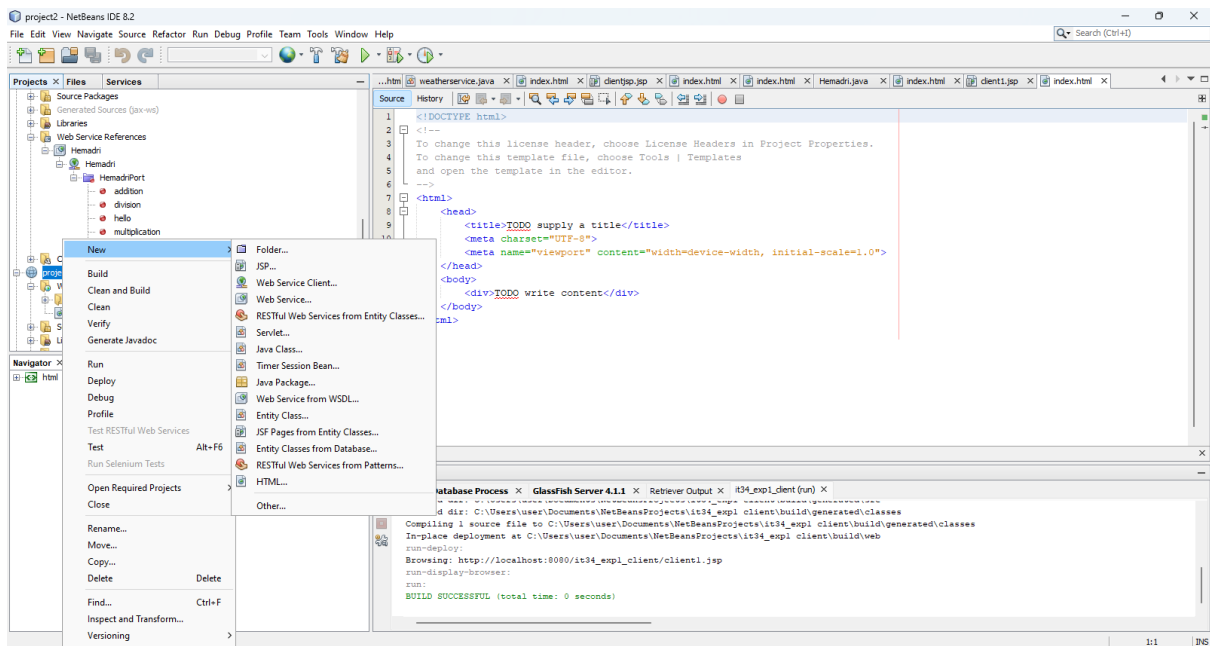
Step 2 : Select Java Web Application , click finish and next.



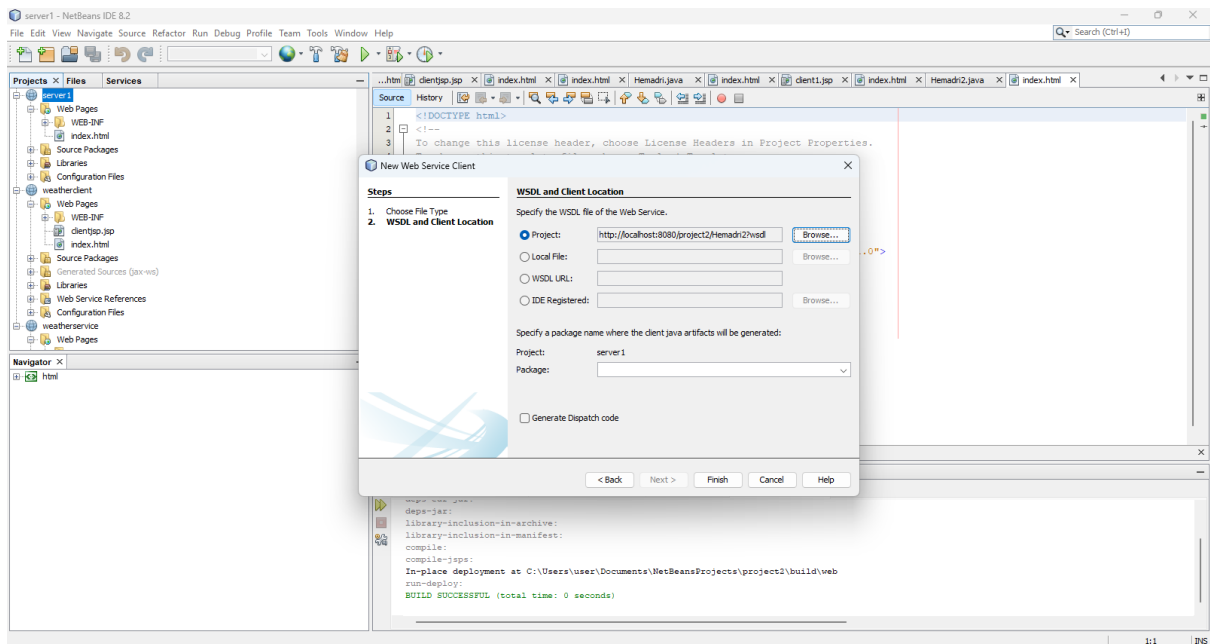
Step 3 : Assign a name for your client project as TemperatureConverterClient and click Next.



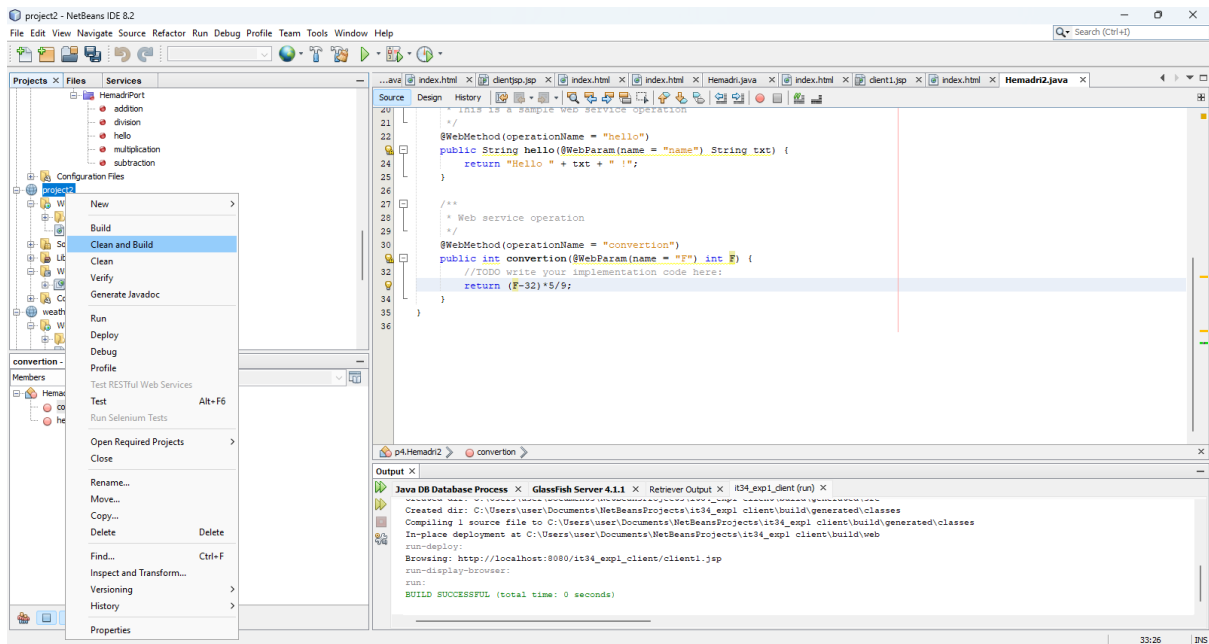
Step 4 : Right-click the TemperatureConverterClient project, select New → Web Service Client. If Web Service Client is not listed, click Other → Web Services → Web Service Client, then click Next.



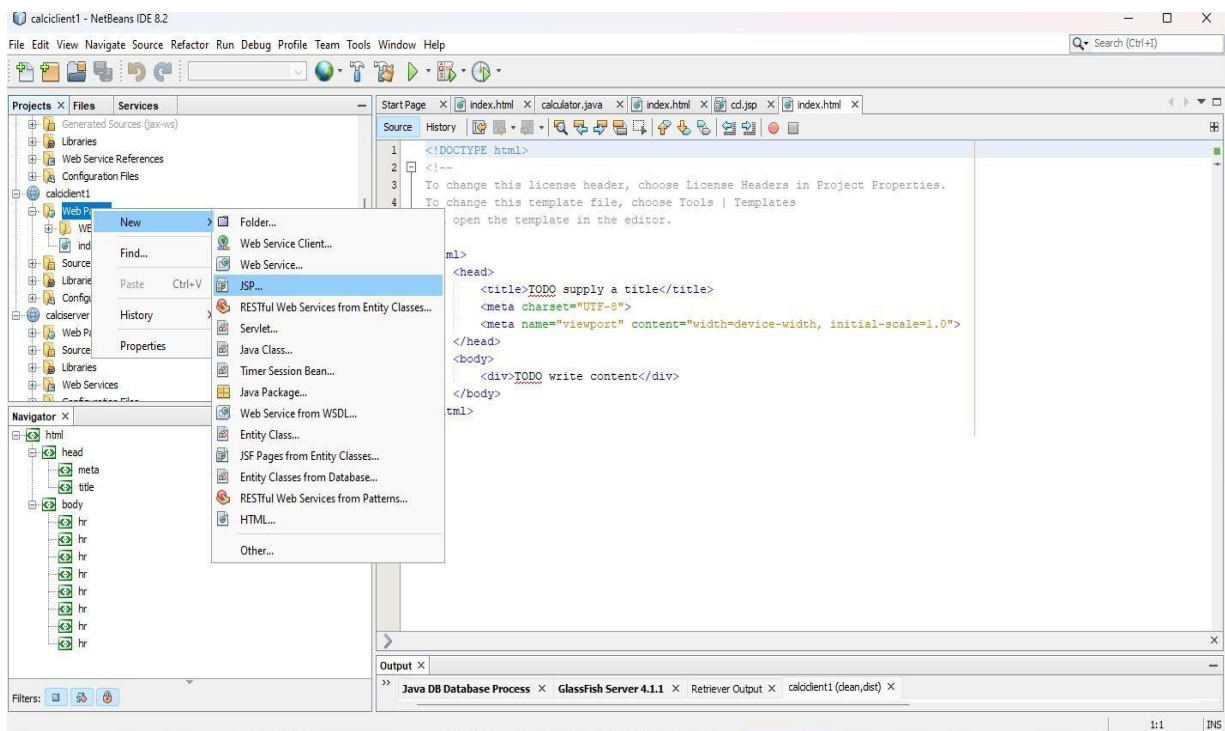
Step 5 : Browse and select the TemperatureService web service (WSDL) created in the service project. Click OK, then click Finish to create the Web Service Client.



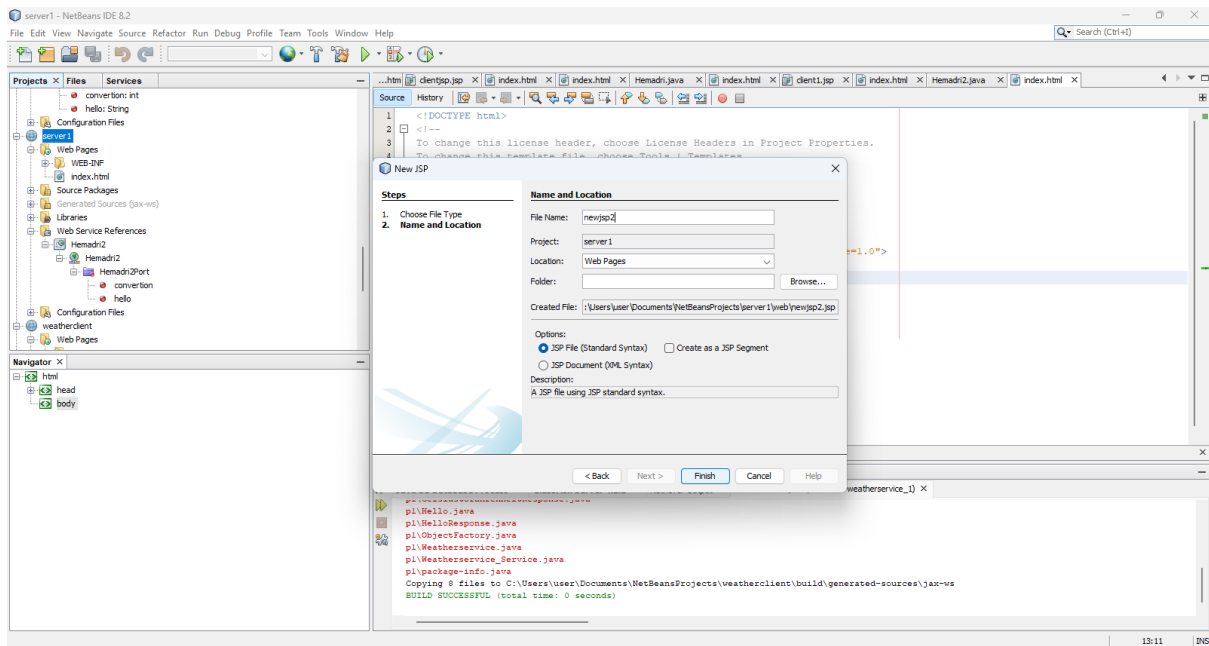
Step 6 : Right-click the TemperatureConverterClient project and select Clean and Build. After the build is completed successfully, run the client project.



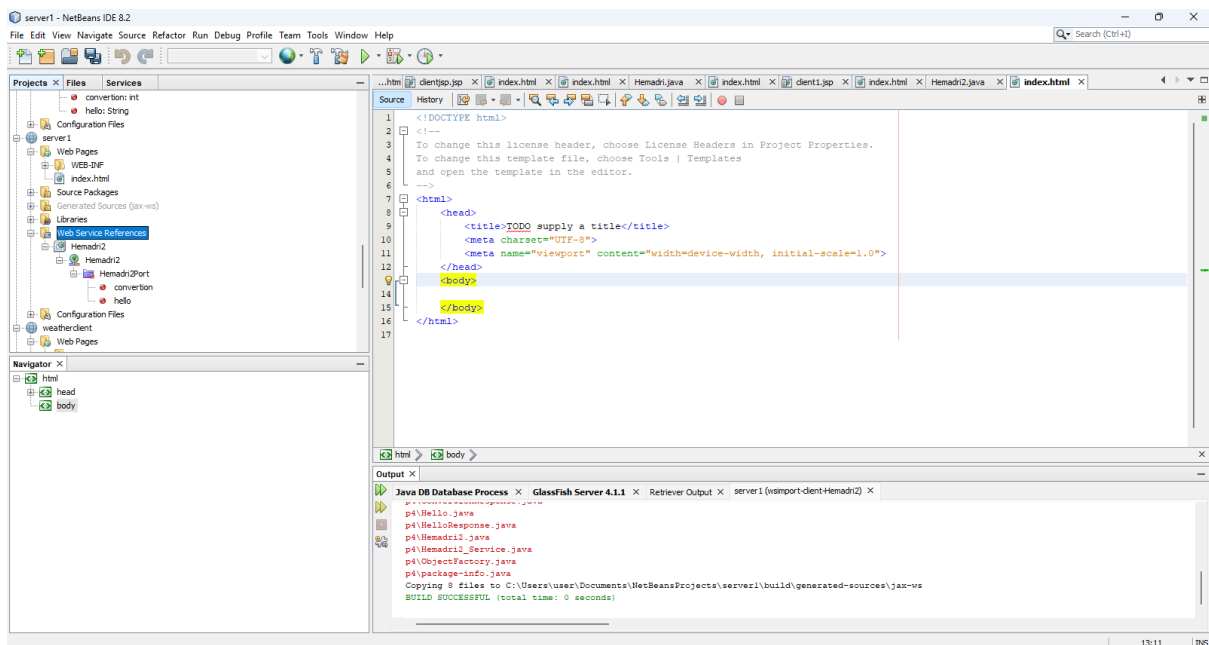
Step 7 : Expand the TemperatureConverterClient project, right-click Web Pages, select New → JSP, and create a new JSP file.



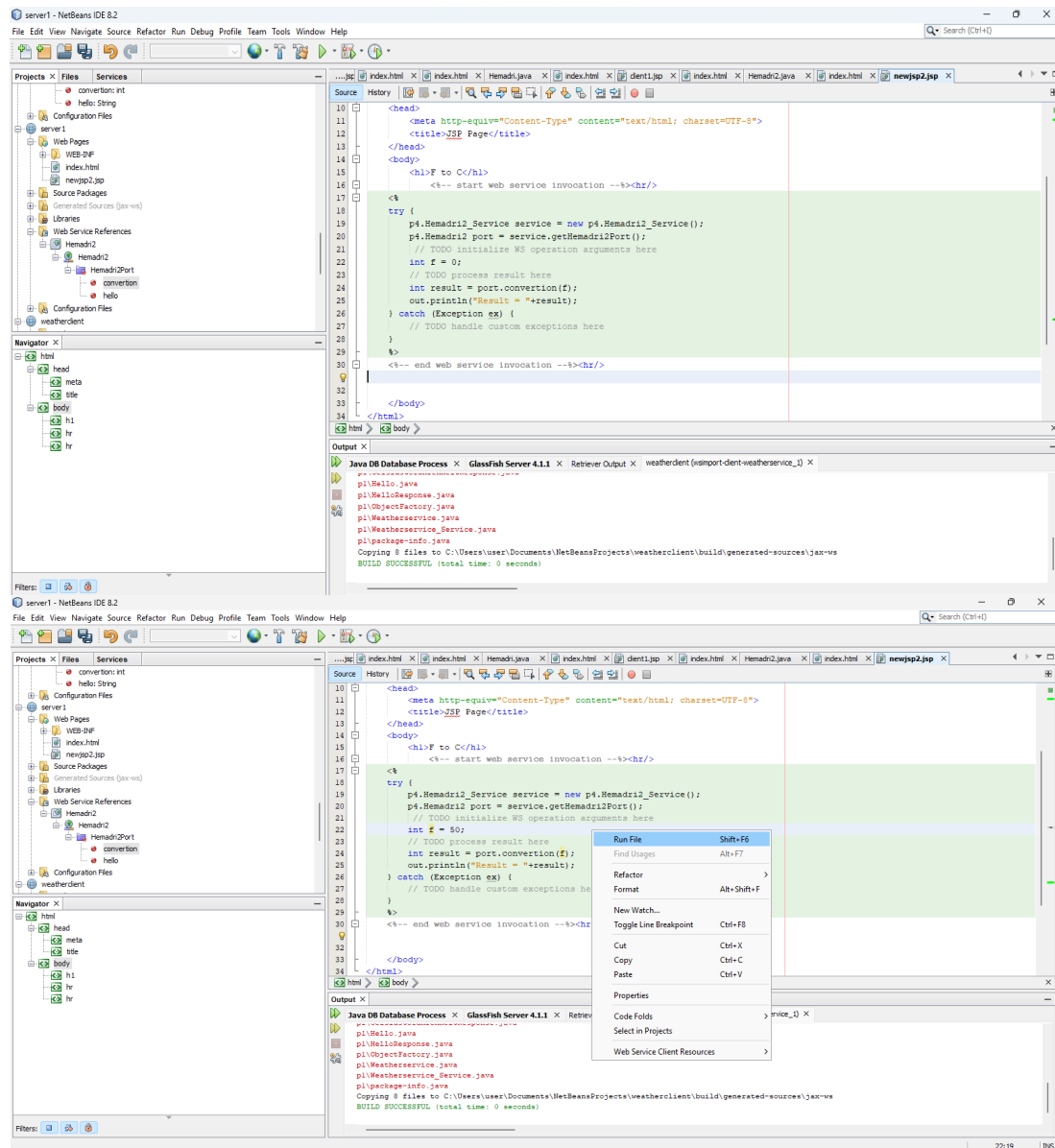
Step 8 : Assign a name for the JSP file (for example, index.jsp) and click Finish.



Step 9 : Open the JSP file and delete the default content inside the <body> tag.



Step 10 : Expand Web Service References, drag and drop the fahrenheitToCelsius operation into the <body> tag of the JSP file. Enter the Fahrenheit value, then right-click the JSP file and select Run to display the Celsius result.



Hello World!

Result = 11

Result = 3

Result = Hello John !

Result = 6

Result = 1

Result = 18.0

RESULT : Thus, the web service for converting temperature from Fahrenheit to Celsius using NetBeans IDE was designed, implemented, tested, and executed successfully.

EXP NO: 2 SIMULATION OF CLOUD ENVIRONMENT USING CLOUDSIM

DATE:

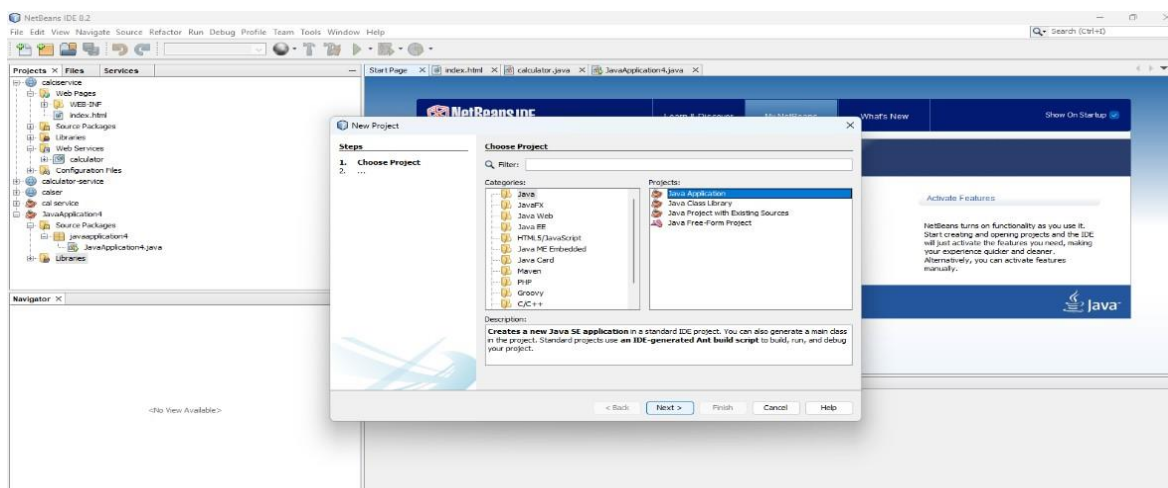
AIM: To simulate various cloud computing scenarios using the CloudSim toolkit by creating datacentres, virtual machines (VMs), and cloudlets, and to observe their execution in a cloud environment.

DESCRIPTION:

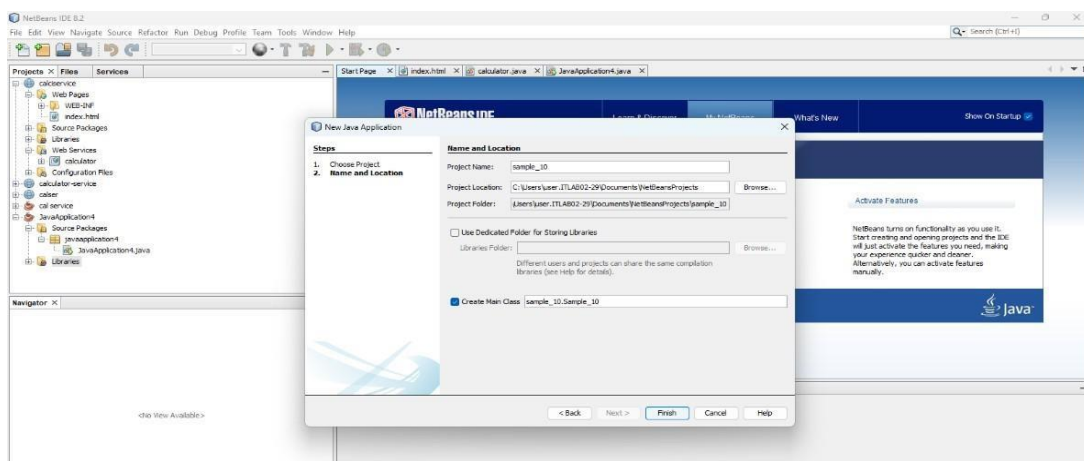
CloudSim is an open-source simulation toolkit used for modelling and simulating cloud computing infrastructures and services. It enables users to create and test cloud environments consisting of datacentres, hosts, virtual machines (VMs), brokers, and cloudlets without requiring actual cloud hardware. CloudSim is widely used for evaluating resource allocation policies, scheduling algorithms, VM provisioning, and cloud performance in a cost-effective and flexible manner.

PROCEDURE:

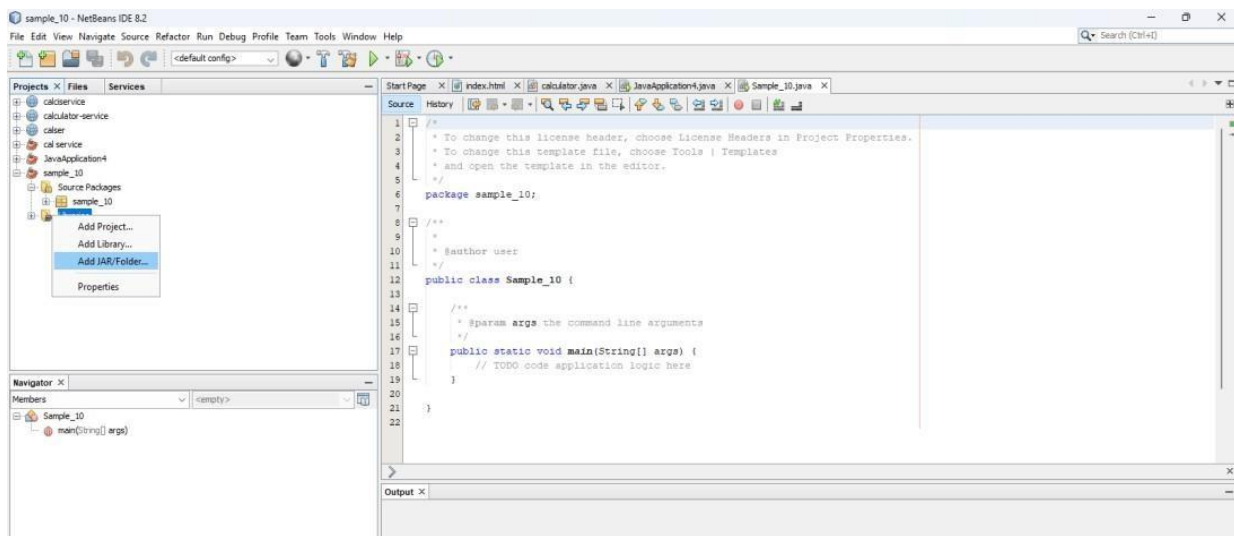
1. Open NetBeans IDE.
2. Select File → New Project → Java → Java Application.



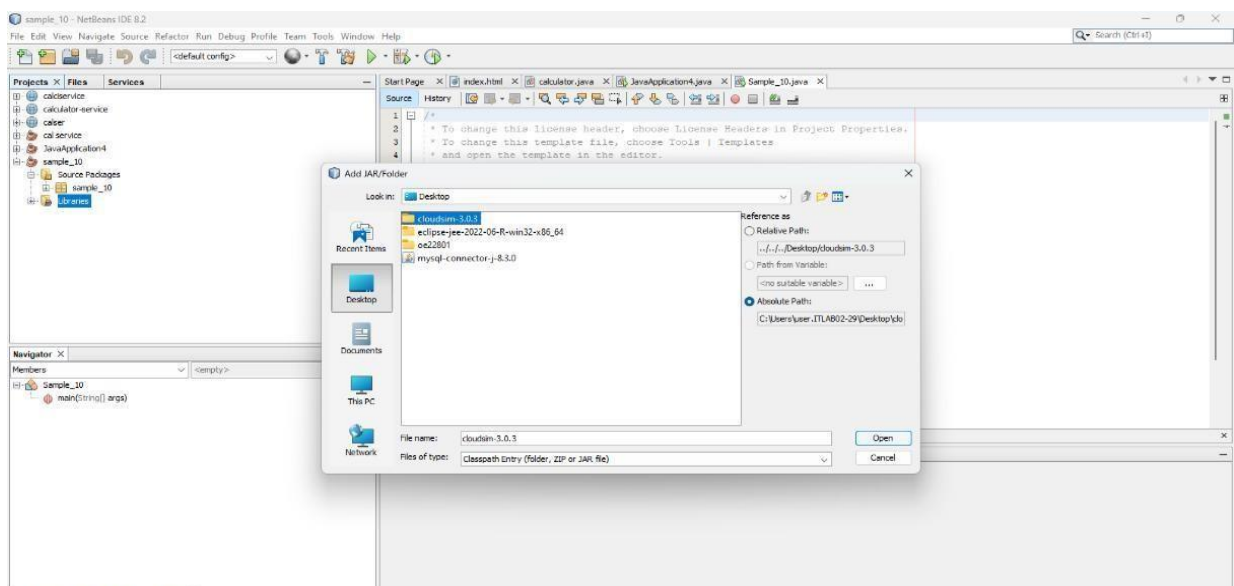
3. Enter the project name and click Finish.



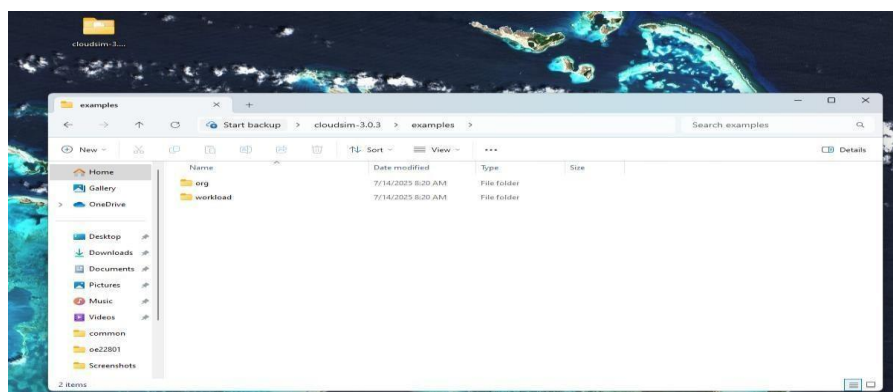
- Expand the created project, right-click Libraries, and select Add JAR/Folder.

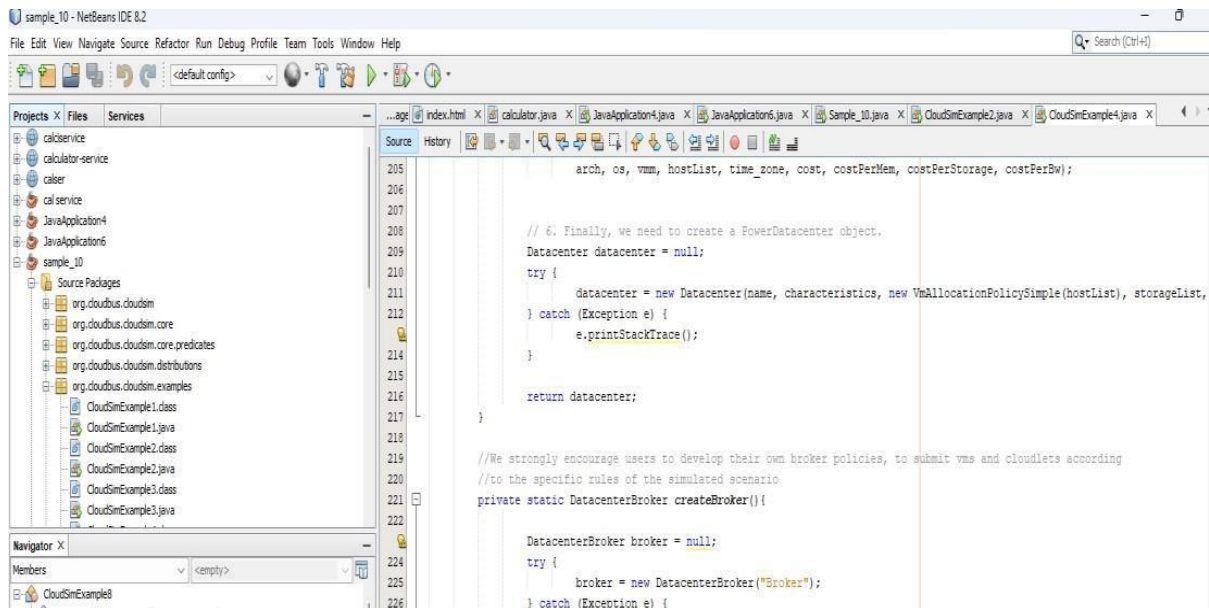


- Browse to Desktop → CloudSim 3.0 → jars, select cloudsim-3.0.jar, and click Open.

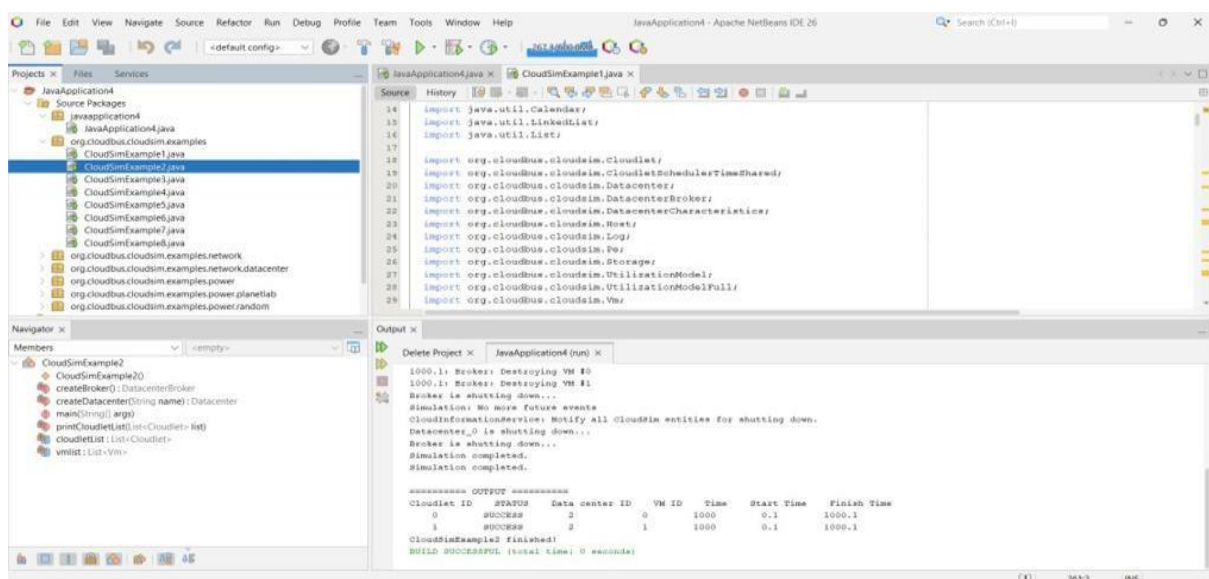


- Open Desktop → CloudSim 3.0 → examples → org → cloudbus → cloudsim → examples.
- Copy the required example program.





8. Return to NetBeans, right-click Source Packages, and paste the example package, or create a new Java class using New → Java Class.
9. Copy and paste the required program into the Java class.
10. Save the program.
11. Right-click the Java class and select Run File.
12. Observe the simulation output in the Output window.



13. Repeat the above steps for each of the following programs.

QUESTIONS:**1. Create a basic simulation using CloudSim.****CODE:**

```

import java.util.ArrayList;
import java.util.Calendar;
import java.util.LinkedList;
import java.util.List;
import org.cloudbus.cloudsim.Datacenter;
import org.cloudbus.cloudsim.DatacenterCharacteristics;
import org.cloudbus.cloudsim.Host;
import org.cloudbus.cloudsim.Log;
import org.cloudbus.cloudsim.Pe;
import org.cloudbus.cloudsim.Storage;
import org.cloudbus.cloudsim.VmAllocationPolicySimple;
import org.cloudbus.cloudsim.VmSchedulerTimeShared;
import org.cloudbus.cloudsim.core.CloudSim;
import org.cloudbus.cloudsim.provisioners.BwProvisionerSimple;
import org.cloudbus.cloudsim.provisioners.PeProvisionerSimple;
import org.cloudbus.cloudsim.provisioners.RamProvisionerSimple;

public class Q1 {

    public static void main(String[] args) {
        Log.println("Starting bare minimum CloudSim simulation...");

        try {
            // 1. Initialize CloudSim
            int numUser = 1;
            Calendar calendar = Calendar.getInstance();
            boolean traceFlag = false;
            CloudSim.init(numUser, calendar, traceFlag);

            // 2. Create the Datacenter (which hosts the physical infrastructure)
            Datacenter datacenter0 = createDatacenter("Datacenter_0");

            // 3. Start the Simulation
            CloudSim.startSimulation();

            // 4. Stop the Simulation
            CloudSim.stopSimulation();

            Log.println("Simulation finished successfully!");
        } catch (Exception e) {
            e.printStackTrace();
            Log.println("Uncaught error occurred");
        }
    }
}

```

```

private static Datacenter createDatacenter(String name) {
// 1. Create a list to hold our physical machines (Hosts)
List<Host> hostList = new ArrayList<Host>();

// 2. Create Processing Elements (CPUs/Cores)
List<Pe> peList = new ArrayList<Pe>();
int mips = 1000;
peList.add(new Pe(0, new PeProvisionerSimple(mips)));

// 3. Create a Host with its configurations
int hostId = 0;
int ram = 2048; // host memory (MB)
long storage = 1000000; // host storage
int bw = 10000;

hostList.add(
new Host(
hostId,
new RamProvisionerSimple(ram),
new BwProvisionerSimple(bw),
storage,
peList,
new VmSchedulerTimeShared(peList)
)
);

// 4. Set Datacenter characteristics
String arch = "x86";
String os = "Linux";
String vmm = "Xen";
double time_zone = 10.0;
double cost = 3.0;
double costPerMem = 0.05;
double costPerStorage = 0.001;
double costPerBw = 0.0;

LinkedList<Storage> storageList = new LinkedList<Storage>();

DatacenterCharacteristics characteristics = new DatacenterCharacteristics(
arch, os, vmm, hostList, time_zone, cost, costPerMem, costPerStorage, costPerBw);

// 5. Create and return the Datacenter object
Datacenter datacenter = null;
try {
datacenter = new Datacenter(name, characteristics, new VmAllocationPolicySimple(hostList),
storageList, 0);
} catch (Exception e) {
e.printStackTrace();
}
}

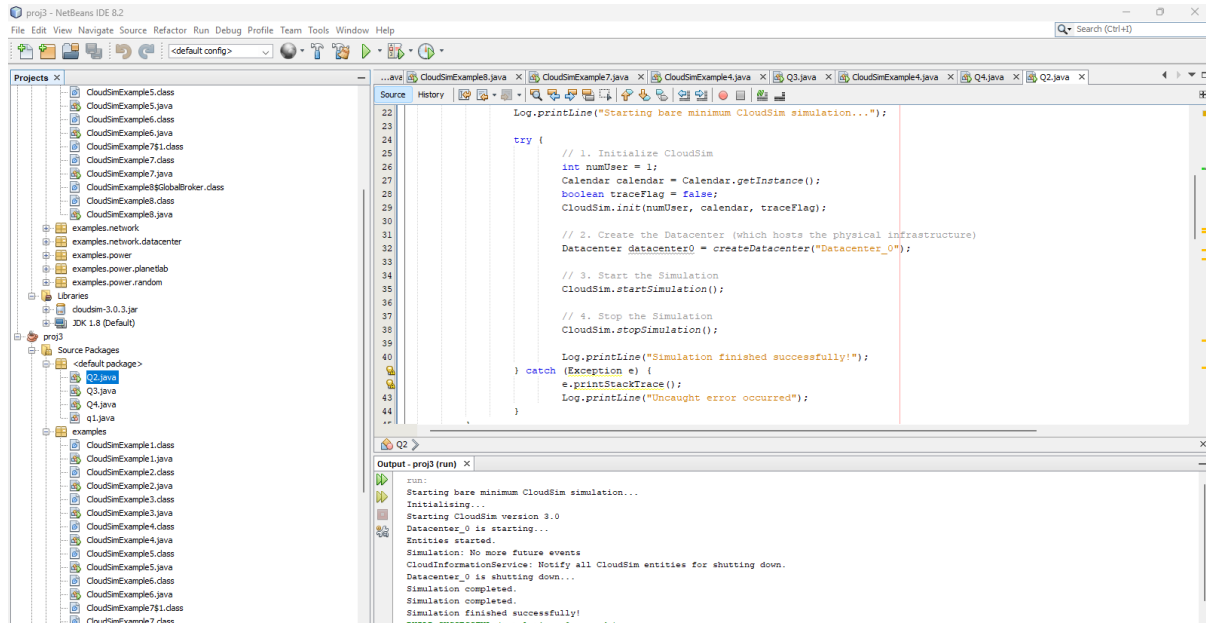
```

```

return datacenter;
}
}

```

OUTPUT:



2. Create a simple datacenter using CloudSim.

CODE:

```

/*
 * Title:      CloudSim Toolkit
 * Description: CloudSim (Cloud Simulation) Toolkit for Modeling and Simulation
 *              of Clouds
 * Licence:    GPL - http://www.gnu.org/copyleft/gpl.html
 *
 * Copyright (c) 2009, The University of Melbourne, Australia
 */

```

```

import java.text.DecimalFormat;
import java.util.ArrayList;
import java.util.Calendar;
import java.util.LinkedList;
import java.util.List;

```

```

import org.cloudbus.cloudsim.Cloudlet;
import org.cloudbus.cloudsim.CloudletSchedulerTimeShared;
import org.cloudbus.cloudsim.Datacenter;
import org.cloudbus.cloudsim.DatacenterBroker;
import org.cloudbus.cloudsim.DatacenterCharacteristics;
import org.cloudbus.cloudsim.Host;
import org.cloudbus.cloudsim.Log;

```

```

import org.cloudbus.cloudsim.Pe;
import org.cloudbus.cloudsim.Storage;
import org.cloudbus.cloudsim.UtilizationModel;
import org.cloudbus.cloudsim.UtilizationModelFull;
import org.cloudbus.cloudsim.Vm;
import org.cloudbus.cloudsim.VmAllocationPolicySimple;
import org.cloudbus.cloudsim.VmSchedulerTimeShared;
import org.cloudbus.cloudsim.core.CloudSim;
import org.cloudbus.cloudsim.provisioners.BwProvisionerSimple;
import org.cloudbus.cloudsim.provisioners.PeProvisionerSimple;
import org.cloudbus.cloudsim.provisioners.RamProvisionerSimple;

/**
 * A simple example showing how to create a datacenter with one host and run one
 * cloudlet on it.
 */
public class q2 {

    /** The cloudlet list. */
    private static List<Cloudlet> cloudletList;

    /** The vm list. */
    private static List<Vm> vmList;

    /**
     * Creates main() to run this example.
     *
     * @param args the args
     */
    @SuppressWarnings("unused")
    public static void main(String[] args) {

        Log.println("Starting CloudSimExample1...");

        try {
            // First step: Initialize the CloudSim package. It should be called
            // before creating any entities.
            int num_user = 1; // number of cloud users
            Calendar calendar = Calendar.getInstance();
            boolean trace_flag = false; // mean trace events

            // Initialize the CloudSim library
            CloudSim.init(num_user, calendar, trace_flag);

            // Second step: Create Datacenters
            // Datacenters are the resource providers in CloudSim. We need at
            // list one of them to run a CloudSim simulation
            Datacenter datacenter0 = createDatacenter("Datacenter_0");

            // Third step: Create Broker

```

```
DatacenterBroker broker = createBroker();
int brokerId = broker.getId();

// Fourth step: Create one virtual machine
vmList = new ArrayList<Vm>();

// VM description
int vmid = 0;
int mips = 1000;
long size = 10000; // image size (MB)
int ram = 512; // vm memory (MB)
long bw = 1000;
int pesNumber = 1; // number of cpus
String vmm = "Xen"; // VMM name

// create VM
Vm vm = new Vm(vmid, brokerId, mips, pesNumber, ram, bw, size, vmm, new
CloudletSchedulerTimeShared());

// add the VM to the vmList
vmList.add(vm);

// submit vm list to the broker
broker.submitVmList(vmList);

// Fifth step: Create one Cloudlet
cloudletList = new ArrayList<Cloudlet>();

// Cloudlet properties
int id = 0;
long length = 400000;
long fileSize = 300;
long outputSize = 300;
UtilizationModel utilizationModel = new UtilizationModelFull();

Cloudlet cloudlet = new Cloudlet(id, length, pesNumber, fileSize, outputSize,
utilizationModel, utilizationModel, utilizationModel);
cloudlet.setUserId(brokerId);
cloudlet.setVmId(vmid);

// add the cloudlet to the list
cloudletList.add(cloudlet);

// submit cloudlet list to the broker
broker.submitCloudletList(cloudletList);

// Sixth step: Starts the simulation
CloudSim.startSimulation();

CloudSim.stopSimulation();
```

```

//Final step: Print results when simulation is over
List<Cloudlet> newList = broker.getCloudletReceivedList();
printCloudletList(newList);

Log.println("CloudSimExample1 finished!");
} catch (Exception e) {
e.printStackTrace();
Log.println("Unwanted errors happen");
}
}

/**
 * Creates the datacenter.
 *
 * @param name the name
 *
 * @return the datacenter
 */
private static Datacenter createDatacenter(String name) {

// Here are the steps needed to create a PowerDatacenter:
// 1. We need to create a list to store
// our machine
List<Host> hostList = new ArrayList<Host>();

// 2. A Machine contains one or more PEs or CPUs/Cores.
// In this example, it will have only one core.
List<Pe> peList = new ArrayList<Pe>();

int mips = 1000;

// 3. Create PEs and add these into a list.
peList.add(new Pe(0, new PeProvisionerSimple(mips))); // need to store Pe id and MIPS Rating

// 4. Create Host with its id and list of PEs and add them to the list
// of machines
int hostId = 0;
int ram = 2048; // host memory (MB)
long storage = 1000000; // host storage
int bw = 10000;

hostList.add(
new Host(
hostId,
new RamProvisionerSimple(ram),
new BwProvisionerSimple(bw),
storage,
peList,
new VmSchedulerTimeShared(peList)

```

```

)
); // This is our machine

// 5. Create a DatacenterCharacteristics object that stores the
// properties of a data center: architecture, OS, list of
// Machines, allocation policy: time- or space-shared, time zone
// and its price (G$/Pe time unit).
String arch = "x86"; // system architecture
String os = "Linux"; // operating system
String vmm = "Xen";
double time_zone = 10.0; // time zone this resource located
double cost = 3.0; // the cost of using processing in this resource
double costPerMem = 0.05; // the cost of using memory in this resource
double costPerStorage = 0.001; // the cost of using storage in this
// resource
double costPerBw = 0.0; // the cost of using bw in this resource
LinkedList<Storage> storageList = new LinkedList<Storage>(); // we are not adding SAN
// devices by now

DatacenterCharacteristics characteristics = new DatacenterCharacteristics(
arch, os, vmm, hostList, time_zone, cost, costPerMem,
costPerStorage, costPerBw);

// 6. Finally, we need to create a PowerDatacenter object.
Datacenter datacenter = null;
try {
datacenter = new Datacenter(name, characteristics, new VmAllocationPolicySimple(hostList),
storageList, 0);
} catch (Exception e) {
e.printStackTrace();
}

return datacenter;
}

// We strongly encourage users to develop their own broker policies, to
// submit vms and cloudlets according
// to the specific rules of the simulated scenario
/**
 * Creates the broker.
 *
 * @return the datacenter broker
 */
private static DatacenterBroker createBroker() {
DatacenterBroker broker = null;
try {
broker = new DatacenterBroker("Broker");
} catch (Exception e) {
e.printStackTrace();
}
return null;
}

```

```

    }
    return broker;
}

/**
 * Prints the Cloudlet objects.
 *
 * @param list list of Cloudlets
 */
private static void printCloudletList(List<Cloudlet> list) {
    int size = list.size();
    Cloudlet cloudlet;

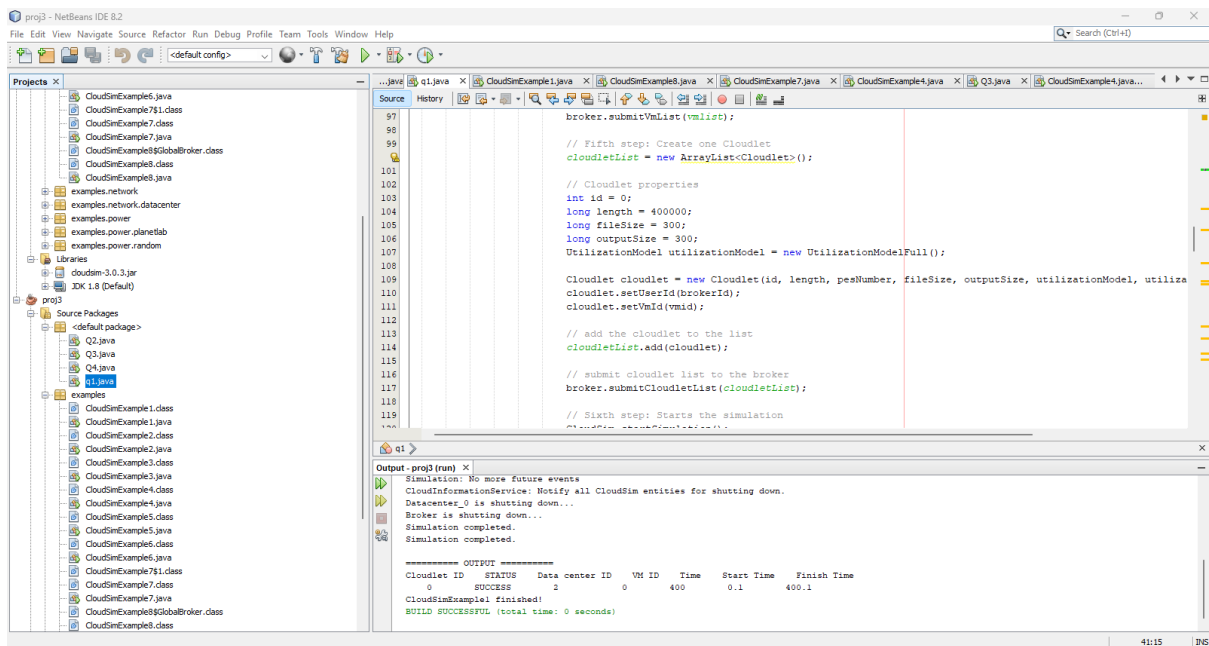
    String indent = "  ";
    Log.println();
    Log.println("===== OUTPUT =====");
    Log.println("Cloudlet ID" + indent + "STATUS" + indent
        + "Data center ID" + indent + "VM ID" + indent + "Time" + indent
        + "Start Time" + indent + "Finish Time");

    DecimalFormat dft = new DecimalFormat("###.##");
    for (int i = 0; i < size; i++) {
        cloudlet = list.get(i);
        Log.print(indent + cloudlet.getCloudletId() + indent + indent);

        if (cloudlet.getCloudletStatus() == Cloudlet.SUCCESS) {
            Log.print("SUCCESS");

            Log.println(indent + indent + cloudlet.getResourceId()
                + indent + indent + cloudlet.getVmId()
                + indent + indent
                + dft.format(cloudlet.getActualCPUTime()) + indent
                + indent + dft.format(cloudlet.getExecStartTime())
                + indent + indent
                + dft.format(cloudlet.getFinishTime()));
        }
    }
}

```

OUTPUT:**3. Create a single virtual machine and cloudlet using CloudSim.****CODE:**

```

/*
 * Title:      CloudSim Toolkit
 * Description: CloudSim (Cloud Simulation) Toolkit for Modeling and Simulation
 *              of Clouds
 * Licence:    GPL - http://www.gnu.org/copyleft/gpl.html
 *
 * Copyright (c) 2009, The University of Melbourne, Australia
 */

```

```
package org.cloudbus.cloudsim.examples;
```

```

import java.text.DecimalFormat;
import java.util.ArrayList;
import java.util.Calendar;
import java.util.LinkedList;
import java.util.List;
import org.cloudbus.cloudsim.Cloudlet;
import org.cloudbus.cloudsim.CloudletSchedulerTimeShared;
import org.cloudbus.cloudsim.Datacenter;
import org.cloudbus.cloudsim.DatacenterBroker;
import org.cloudbus.cloudsim.DatacenterCharacteristics;
import org.cloudbus.cloudsim.Host;
import org.cloudbus.cloudsim.Log;
import org.cloudbus.cloudsim.Pe;
import org.cloudbus.cloudsim.Storage;
import org.cloudbus.cloudsim.UtilizationModel;
import org.cloudbus.cloudsim.UtilizationModelFull;

```

```

import org.cloudbus.cloudsim.Vm;
import org.cloudbus.cloudsim.VmAllocationPolicySimple;
import org.cloudbus.cloudsim.VmSchedulerSpaceShared;
import org.cloudbus.cloudsim.core.CloudSim;
import org.cloudbus.cloudsim.provisioners.BwProvisionerSimple;
import org.cloudbus.cloudsim.provisioners.PeProvisionerSimple;
import org.cloudbus.cloudsim.provisioners.RamProvisionerSimple;

/**
 * A simple example showing how to create
 * two datacenters with one host each and
 * run two cloudlets on them.
 */
public class Q3 {

    /** The cloudlet list. */
    private static List<Cloudlet> cloudletList;

    /** The vm list. */
    private static List<Vm> vmList;

    /**
     * Creates main() to run this example
     */
    public static void main(String[] args) {

        Log.println("Starting CloudSimExample4...");

        try {
            // First step: Initialize the CloudSim package. It should be called
            // before creating any entities.
            int num_user = 1; // number of cloud users
            Calendar calendar = Calendar.getInstance();
            boolean trace_flag = false; // mean trace events

            // Initialize the GridSim library
            CloudSim.init(num_user, calendar, trace_flag);

            // Second step: Create Datacenters
            // Datacenters are the resource providers in CloudSim. We need at list one of them to run a
            // CloudSim simulation
            @SuppressWarnings("unused")
            Datacenter datacenter0 = createDatacenter("Datacenter_0");
            @SuppressWarnings("unused")
            Datacenter datacenter1 = createDatacenter("Datacenter_1");

            // Third step: Create Broker
            DatacenterBroker broker = createBroker();
            int brokerId = broker.getId();

```

```
//Fourth step: Create one virtual machine  
vmList = new ArrayList<Vm>();
```

```
//VM description  
int vmid = 0;  
int mips = 250;  
long size = 10000; //image size (MB)  
int ram = 512; //vm memory (MB)  
long bw = 1000;  
int pesNumber = 1; //number of cpus  
String vmm = "Xen"; //VMM name
```

```
//create two VMs  
Vm vm1 = new Vm(vmid, brokerId, mips, pesNumber, ram, bw, size, vmm, new  
CloudletSchedulerTimeShared());
```

```
//add the VMs to the vmList  
vmList.add(vm1);
```

```
//submit vm list to the broker  
broker.submitVmList(vmList);
```

```
//Fifth step: Create two Cloudlets  
cloudletList = new ArrayList<Cloudlet>();
```

```
//Cloudlet properties  
int id = 0;  
long length = 40000;  
long fileSize = 300;  
long outputSize = 300;  
UtilizationModel utilizationModel = new UtilizationModelFull();
```

```
Cloudlet cloudlet1 = new Cloudlet(id, length, pesNumber, fileSize, outputSize,  
utilizationModel, utilizationModel, utilizationModel);  
cloudlet1.setUserId(brokerId);
```

```
//add the cloudlets to the list  
cloudletList.add(cloudlet1);
```

```
//submit cloudlet list to the broker  
broker.submitCloudletList(cloudletList);
```

```
//bind the cloudlets to the vms. This way, the broker  
// will submit the bound cloudlets only to the specific VM  
broker.bindCloudletToVm(cloudlet1.getCloudletId(), vm1.getId());
```

```

// Sixth step: Starts the simulation
CloudSim.startSimulation();

// Final step: Print results when simulation is over
List<Cloudlet> newList = broker.getCloudletReceivedList();

CloudSim.stopSimulation();

    printCloudletList(newList);

Log.println("CloudSimExample4 finished!");
}
catch (Exception e) {
e.printStackTrace();
Log.println("The simulation has been terminated due to an unexpected error");
}
}

private static Datacenter createDatacenter(String name){

// Here are the steps needed to create a PowerDatacenter:
// 1. We need to create a list to store
//    our machine
List<Host> hostList = new ArrayList<Host>();

// 2. A Machine contains one or more PEs or CPUs/Cores.
// In this example, it will have only one core.
List<Pe> peList = new ArrayList<Pe>();

int mips = 1000;

// 3. Create PEs and add these into a list.
peList.add(new Pe(0, new PeProvisionerSimple(mips))); // need to store Pe id and MIPS
Rating

//4. Create Host with its id and list of PEs and add them to the list of machines
int hostId=0;
int ram = 2048; //host memory (MB)
long storage = 1000000; //host storage
int bw = 10000;

//in this example, the VMAllocationPolicy in use is SpaceShared. It means that only one VM
//is allowed to run on each Pe. As each Host has only one Pe, only one VM can run on each
Host.
hostList.add(
    new Host(
        hostId,

```

```

    new RamProvisionerSimple(ram),
    new BwProvisionerSimple(bw),
    storage,
    peList,
    new VmSchedulerSpaceShared(peList)
)
); // This is our first machine

// 5. Create a DatacenterCharacteristics object that stores the
//   properties of a data center: architecture, OS, list of
//   Machines, allocation policy: time- or space-shared, time zone
//   and its price (G$/Pe time unit).
String arch = "x86";    // system architecture
String os = "Linux";    // operating system
String vmm = "Xen";
double time_zone = 10.0;    // time zone this resource located
double cost = 3.0;    // the cost of using processing in this resource
double costPerMem = 0.05; // the cost of using memory in this resource
double costPerStorage = 0.001; // the cost of using storage in this resource
double costPerBw = 0.0; // the cost of using bw in this resource
LinkedList<Storage> storageList = new LinkedList<Storage>(); //we are not adding SAN
devices by now

    DatacenterCharacteristics characteristics = new DatacenterCharacteristics(
        arch, os, vmm, hostList, time_zone, cost, costPerMem, costPerStorage, costPerBw);

// 6. Finally, we need to create a PowerDatacenter object.
Datacenter datacenter = null;
try {
    datacenter = new Datacenter(name, characteristics, new
VmAllocationPolicySimple(hostList), storageList, 0);
} catch (Exception e) {
    e.printStackTrace();
}

return datacenter;
}

//We strongly encourage users to develop their own broker policies, to submit vms and
cloudlets according
//to the specific rules of the simulated scenario
private static DatacenterBroker createBroker(){

    DatacenterBroker broker = null;
    try {
        broker = new DatacenterBroker("Broker");
    } catch (Exception e) {
        e.printStackTrace();
    }
    return null;
}

```

```

    }
    return broker;
}

/**
 * Prints the Cloudlet objects
 * @param list list of Cloudlets
 */
private static void printCloudletList(List<Cloudlet> list) {
    int size = list.size();
    Cloudlet cloudlet;

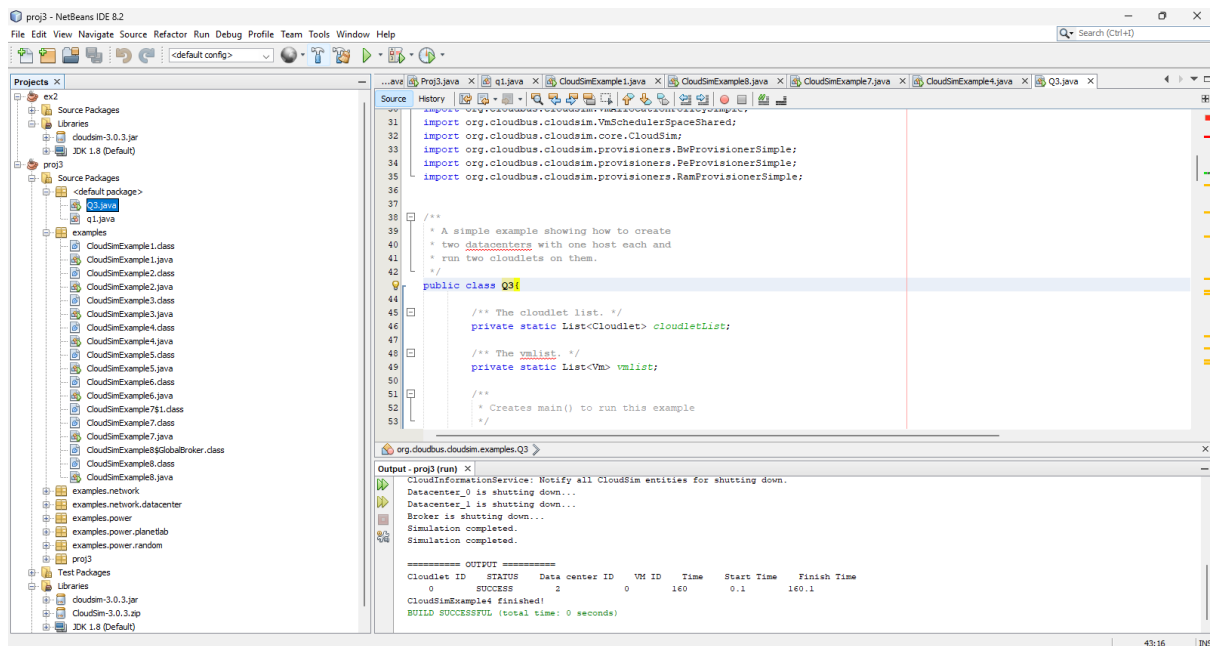
    String indent = "  ";
    Log.println();
    Log.println("===== OUTPUT =====");
    Log.println("Cloudlet ID" + indent + "STATUS" + indent +
        "Data center ID" + indent + "VM ID" + indent + "Time" + indent + "Start Time" + indent +
        "Finish Time");

    DecimalFormat dft = new DecimalFormat("###.##");
    for (int i = 0; i < size; i++) {
        cloudlet = list.get(i);
        Log.print(indent + cloudlet.getCloudletId() + indent + indent);

        if (cloudlet.getCloudletStatus() == Cloudlet.SUCCESS){
            Log.print("SUCCESS");

            Log.println( indent + indent + cloudlet.getResourceId() + indent + indent + indent +
                cloudlet.getVmId() +
                indent + indent + dft.format(cloudlet.getActualCPUTime()) + indent + indent +
                dft.format(cloudlet.getExecStartTime())+
                indent + indent + dft.format(cloudlet.getFinishTime()));
        }
    }
}

```

OUTPUT:**4. Create multiple virtual machines and multiple cloudlets using CloudSim.****CODE:**

```

/*
 * Title:      CloudSim Toolkit
 * Description: CloudSim (Cloud Simulation) Toolkit for Modeling and Simulation
 *              of Clouds
 * Licence:    GPL - http://www.gnu.org/copyleft/gpl.html
 *
 * Copyright (c) 2009, The University of Melbourne, Australia
 */

```

```
package org.cloudbus.cloudsim.examples;
```

```

import java.text.DecimalFormat;
import java.util.ArrayList;
import java.util.Calendar;
import java.util.LinkedList;
import java.util.List;
import org.cloudbus.cloudsim.Cloudlet;
import org.cloudbus.cloudsim.CloudletSchedulerTimeShared;
import org.cloudbus.cloudsim.Datacenter;
import org.cloudbus.cloudsim.DatacenterBroker;
import org.cloudbus.cloudsim.DatacenterCharacteristics;
import org.cloudbus.cloudsim.Host;
import org.cloudbus.cloudsim.Log;
import org.cloudbus.cloudsim.Pe;
import org.cloudbus.cloudsim.Storage;
import org.cloudbus.cloudsim.UtilizationModel;

```

```

import org.cloudbus.cloudsim.UtilizationModelFull;
import org.cloudbus.cloudsim.Vm;
import org.cloudbus.cloudsim.VmAllocationPolicySimple;
import org.cloudbus.cloudsim.VmSchedulerSpaceShared;
import org.cloudbus.cloudsim.core.CloudSim;
import org.cloudbus.cloudsim.provisioners.BwProvisionerSimple;
import org.cloudbus.cloudsim.provisioners.PeProvisionerSimple;
import org.cloudbus.cloudsim.provisioners.RamProvisionerSimple;

/**
 * A simple example showing how to create
 * two datacenters with one host each and
 * run two cloudlets on them.
 */
public class Q4{

    /** The cloudlet list. */
    private static List<Cloudlet> cloudletList;

    /** The vm list. */
    private static List<Vm> vmList;

    /**
     * Creates main() to run this example
     */
    public static void main(String[] args) {

        Log.println("Starting CloudSimExample4...");

        try {
            // First step: Initialize the CloudSim package. It should be called
            // before creating any entities.
            int num_user = 1; // number of cloud users
            Calendar calendar = Calendar.getInstance();
            boolean trace_flag = false; // mean trace events

            // Initialize the GridSim library
            CloudSim.init(num_user, calendar, trace_flag);

            // Second step: Create Datacenters
            //Datacenters are the resource providers in CloudSim. We need at list one of them to run a
            //CloudSim simulation
            @SuppressWarnings("unused")
            Datacenter datacenter0 = createDatacenter("Datacenter_0");
            @SuppressWarnings("unused")
            Datacenter datacenter1 = createDatacenter("Datacenter_1");

            //Third step: Create Broker
            DatacenterBroker broker = createBroker();

```

```
int brokerId = broker.getId();

//Fourth step: Create one virtual machine
vmList = new ArrayList<Vm>();

//VM description
int vmid = 0;
int mips = 250;
long size = 10000; //image size (MB)
int ram = 512; //vm memory (MB)
long bw = 1000;
int pesNumber = 1; //number of cpus
String vmm = "Xen"; //VMM name

//create two VMs
Vm vm1 = new Vm(vmid, brokerId, mips, pesNumber, ram, bw, size, vmm, new
CloudletSchedulerTimeShared());

vmid++;
Vm vm2 = new Vm(vmid, brokerId, mips, pesNumber, ram, bw, size, vmm, new
CloudletSchedulerTimeShared());

//add the VMs to the vmList
vmList.add(vm1);
vmList.add(vm2);

//submit vm list to the broker
broker.submitVmList(vmList);

//Fifth step: Create two Cloudlets
cloudletList = new ArrayList<Cloudlet>();

//Cloudlet properties
int id = 0;
long length = 40000;
long fileSize = 300;
long outputSize = 300;
UtilizationModel utilizationModel = new UtilizationModelFull();

Cloudlet cloudlet1 = new Cloudlet(id, length, pesNumber, fileSize, outputSize,
utilizationModel, utilizationModel, utilizationModel);
cloudlet1.setUserId(brokerId);

id++;
Cloudlet cloudlet2 = new Cloudlet(id, length, pesNumber, fileSize, outputSize,
utilizationModel, utilizationModel, utilizationModel);
cloudlet2.setUserId(brokerId);

//add the cloudlets to the list
```

```

cloudletList.add(cloudlet1);
cloudletList.add(cloudlet2);

//submit cloudlet list to the broker
broker.submitCloudletList(cloudletList);

//bind the cloudlets to the vms. This way, the broker
// will submit the bound cloudlets only to the specific VM
broker.bindCloudletToVm(cloudlet1.getCloudletId(),vm1.getId());
broker.bindCloudletToVm(cloudlet2.getCloudletId(),vm2.getId());

// Sixth step: Starts the simulation
CloudSim.startSimulation();

// Final step: Print results when simulation is over
List<Cloudlet> newList = broker.getCloudletReceivedList();

CloudSim.stopSimulation();

    printCloudletList(newList);

Log.println("CloudSimExample4 finished!");
}
catch (Exception e) {
e.printStackTrace();
Log.println("The simulation has been terminated due to an unexpected error");
}
}

private static Datacenter createDatacenter(String name){

// Here are the steps needed to create a PowerDatacenter:
// 1. We need to create a list to store
//    our machine
List<Host> hostList = new ArrayList<Host>();

// 2. A Machine contains one or more PEs or CPUs/Cores.
// In this example, it will have only one core.
List<Pe> peList = new ArrayList<Pe>();

int mips = 1000;

// 3. Create PEs and add these into a list.
peList.add(new Pe(0, new PeProvisionerSimple(mips))); // need to store Pe id and MIPS
Rating

//4. Create Host with its id and list of PEs and add them to the list of machines
int hostId=0;

```

```
int ram = 2048; //host memory (MB)
long storage = 1000000; //host storage
int bw = 10000;
```

//in this example, the VMAllocationPolicy in use is SpaceShared. It means that only one VM //is allowed to run on each Pe. As each Host has only one Pe, only one VM can run on each Host.

```
hostList.add(
    new Host(
        hostId,
        new RamProvisionerSimple(ram),
        new BwProvisionerSimple(bw),
        storage,
        peList,
        new VmSchedulerSpaceShared(peList)
    )
); // This is our first machine
```

// 5. Create a DatacenterCharacteristics object that stores the
// properties of a data center: architecture, OS, list of
// Machines, allocation policy: time- or space-shared, time zone
// and its price (G\$/Pe time unit).

```
String arch = "x86"; // system architecture
String os = "Linux"; // operating system
String vmm = "Xen";
double time_zone = 10.0; // time zone this resource located
double cost = 3.0; // the cost of using processing in this resource
double costPerMem = 0.05; // the cost of using memory in this resource
double costPerStorage = 0.001; // the cost of using storage in this resource
double costPerBw = 0.0; // the cost of using bw in this resource
LinkedList<Storage> storageList = new LinkedList<Storage>(); //we are not adding SAN
devices by now
```

```
DatacenterCharacteristics characteristics = new DatacenterCharacteristics(
    arch, os, vmm, hostList, time_zone, cost, costPerMem, costPerStorage, costPerBw);
```

// 6. Finally, we need to create a PowerDatacenter object.

```
Datacenter datacenter = null;
try {
    datacenter = new Datacenter(name, characteristics, new
VmAllocationPolicySimple(hostList), storageList, 0);
} catch (Exception e) {
    e.printStackTrace();
}

return datacenter;
}
```

```
//We strongly encourage users to develop their own broker policies, to submit vms and
cloudlets according
//to the specific rules of the simulated scenario
private static DatacenterBroker createBroker(){

    DatacenterBroker broker = null;
    try {
        broker = new DatacenterBroker("Broker");
    } catch (Exception e) {
        e.printStackTrace();
        return null;
    }
    return broker;
}

/**
 * Prints the Cloudlet objects
 * @param list list of Cloudlets
 */
private static void printCloudletList(List<Cloudlet> list) {
    int size = list.size();
    Cloudlet cloudlet;

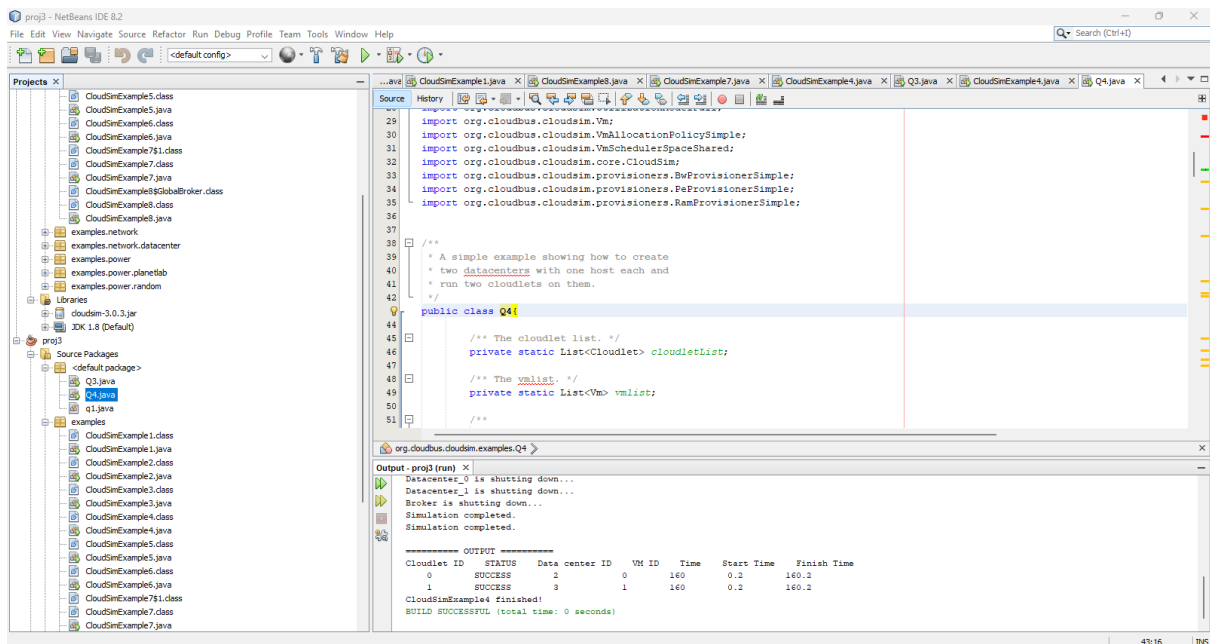
    String indent = "  ";
    Log.println();
    Log.println("===== OUTPUT =====");
    Log.println("Cloudlet ID" + indent + "STATUS" + indent +
        "Data center ID" + indent + "VM ID" + indent + "Time" + indent + "Start Time" + indent +
        "Finish Time");

    DecimalFormat dft = new DecimalFormat("###.##");
    for (int i = 0; i < size; i++) {
        cloudlet = list.get(i);
        Log.print(indent + cloudlet.getCloudletId() + indent + indent);

        if (cloudlet.getCloudletStatus() == Cloudlet.SUCCESS){
            Log.print("SUCCESS");

            Log.println( indent + indent + cloudlet.getResourceId() + indent + indent + indent +
                cloudlet.getVmId() +
                indent + indent + dft.format(cloudlet.getActualCPUTime()) + indent + indent +
                dft.format(cloudlet.getExecStartTime())+
                indent + indent + dft.format(cloudlet.getFinishTime()));
        }
    }

}
}
```

OUTPUT:

RESULT: The CloudSim toolkit was successfully configured in NetBeans, and various cloud computing scenarios were simulated.

EXP NO: 3
DATE:

INSTALLATION OF EUCALYPTUS

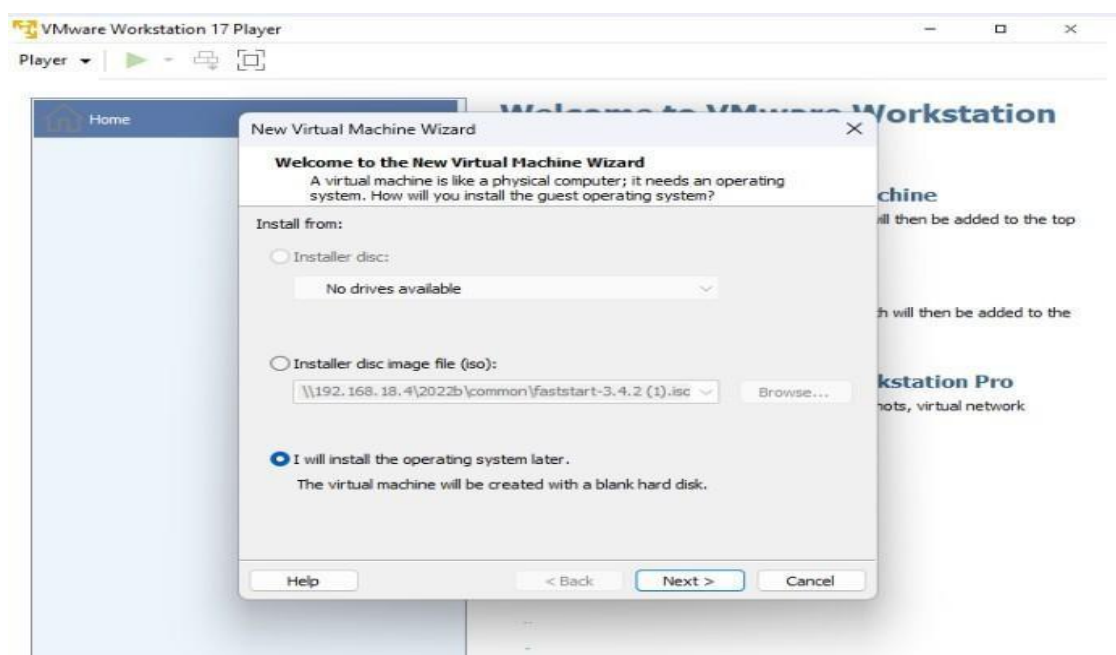
AIM: To preform installation of eucalyptus.

PROCEDURE:

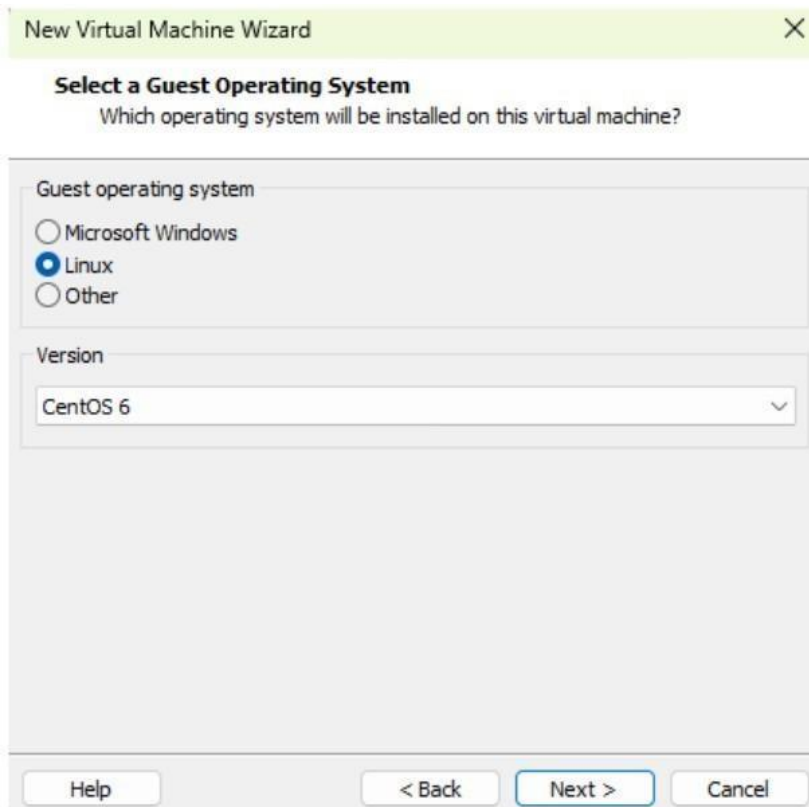
1. Click on VMWare workstation in your desktop and click on Create a New Virtual Machine.



2. Choose the 'I will install the operating system later' option and click Next.

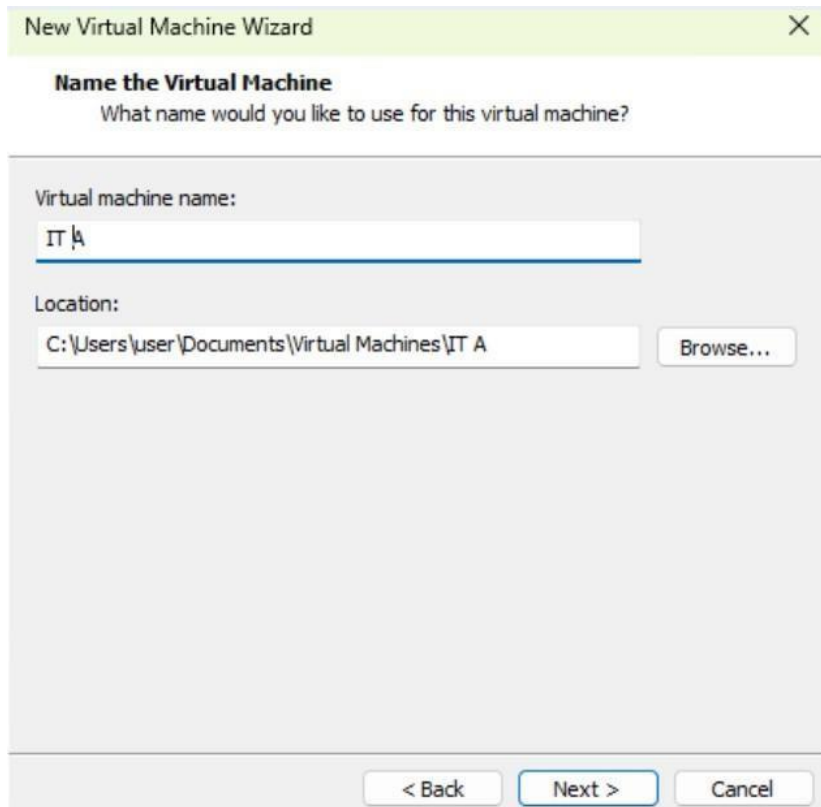


3. Choose Linux as the Guest Operating system, select CentOS 6 as version and click Next.



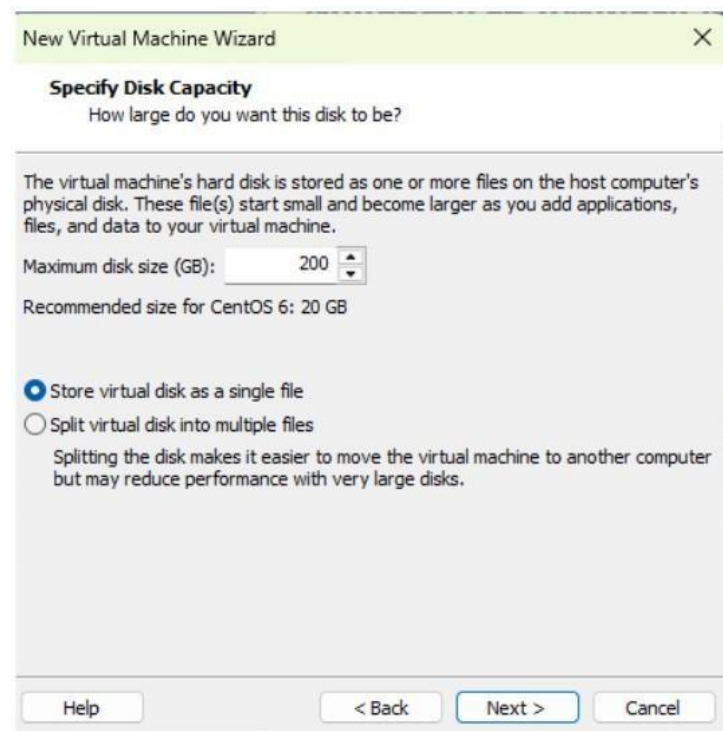
The screenshot shows the 'New Virtual Machine Wizard' dialog box. The title bar is green with a close button. The main heading is 'Select a Guest Operating System' with the subtitle 'Which operating system will be installed on this virtual machine?'. Below this, there are two sections: 'Guest operating system' with three radio buttons ('Microsoft Windows', 'Linux' (selected), and 'Other'), and 'Version' with a dropdown menu showing 'CentOS 6'. At the bottom, there are four buttons: 'Help', '< Back', 'Next >' (highlighted with a blue border), and 'Cancel'.

4. Name the Virtual Machine as John, click Next.

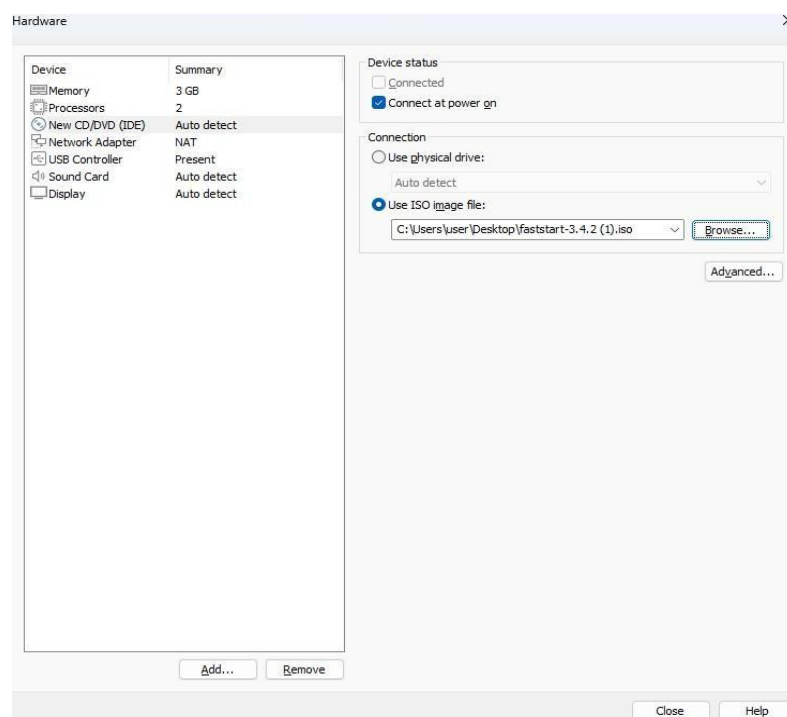


The screenshot shows the 'New Virtual Machine Wizard' dialog box. The title bar is green with a close button. The main heading is 'Name the Virtual Machine' with the subtitle 'What name would you like to use for this virtual machine?'. Below this, there are two sections: 'Virtual machine name:' with a text box containing 'IT A', and 'Location:' with a text box containing 'C:\Users\user\Documents\Virtual Machines\IT A' and a 'Browse...' button. At the bottom, there are three buttons: '< Back', 'Next >' (highlighted with a blue border), and 'Cancel'.

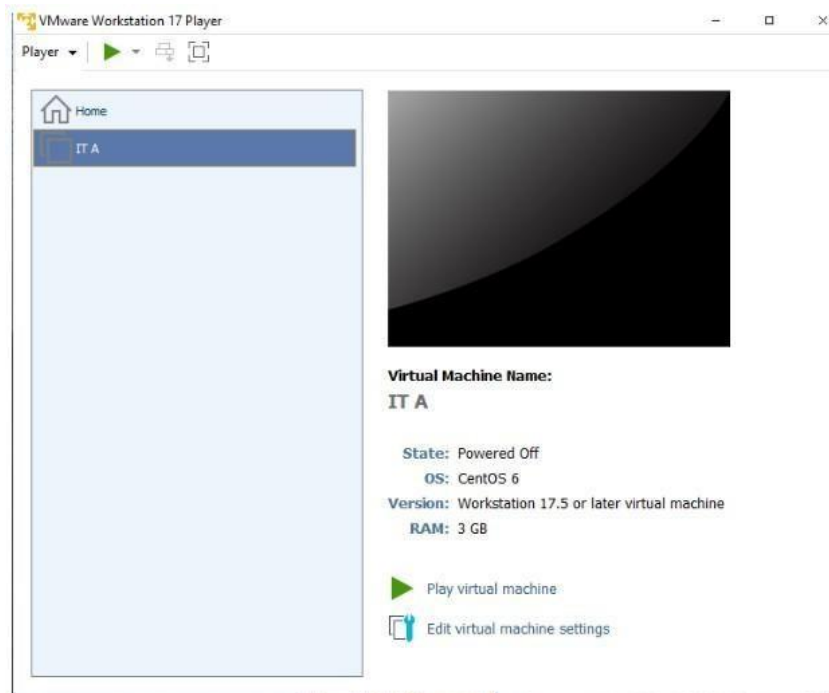
5. Enter the maximum disk size as 200GB and choose the option. Store virtual disk as a single file.



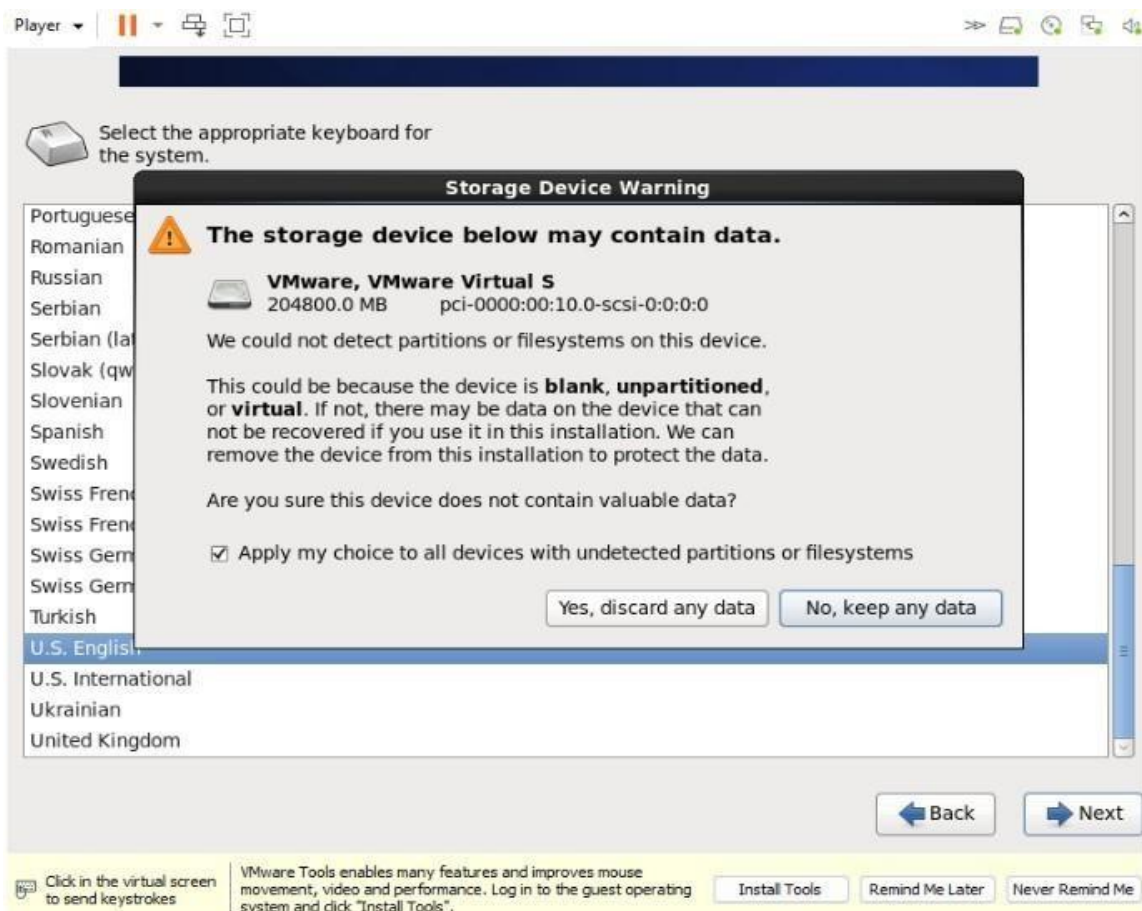
6. Click on Customize Hardware and configure the Virtual Machine as below:
 1. Choose the memory as 3GB(3072 MB)
 2. Enter the number of processors as 2.
 3. Choose ISO image file, click Browse, select faststart-3.4.2.iso, click close and then finish.



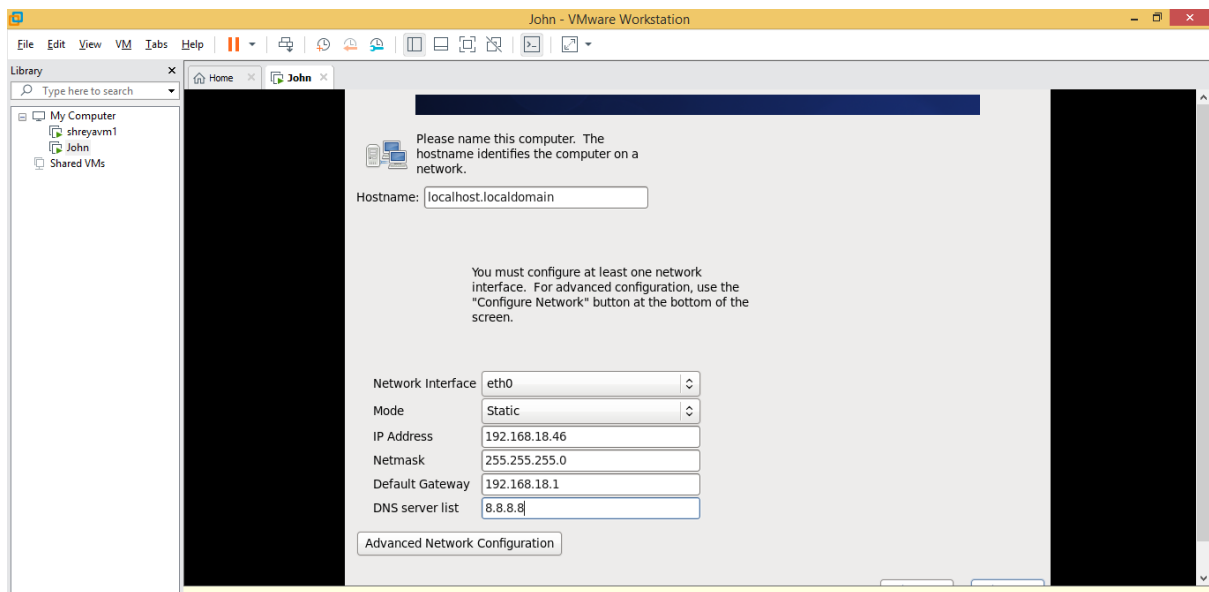
7. Click Play Virtual Machine, choose Install CentOS 6 single system cloud in 60x., then press skip and then ok.



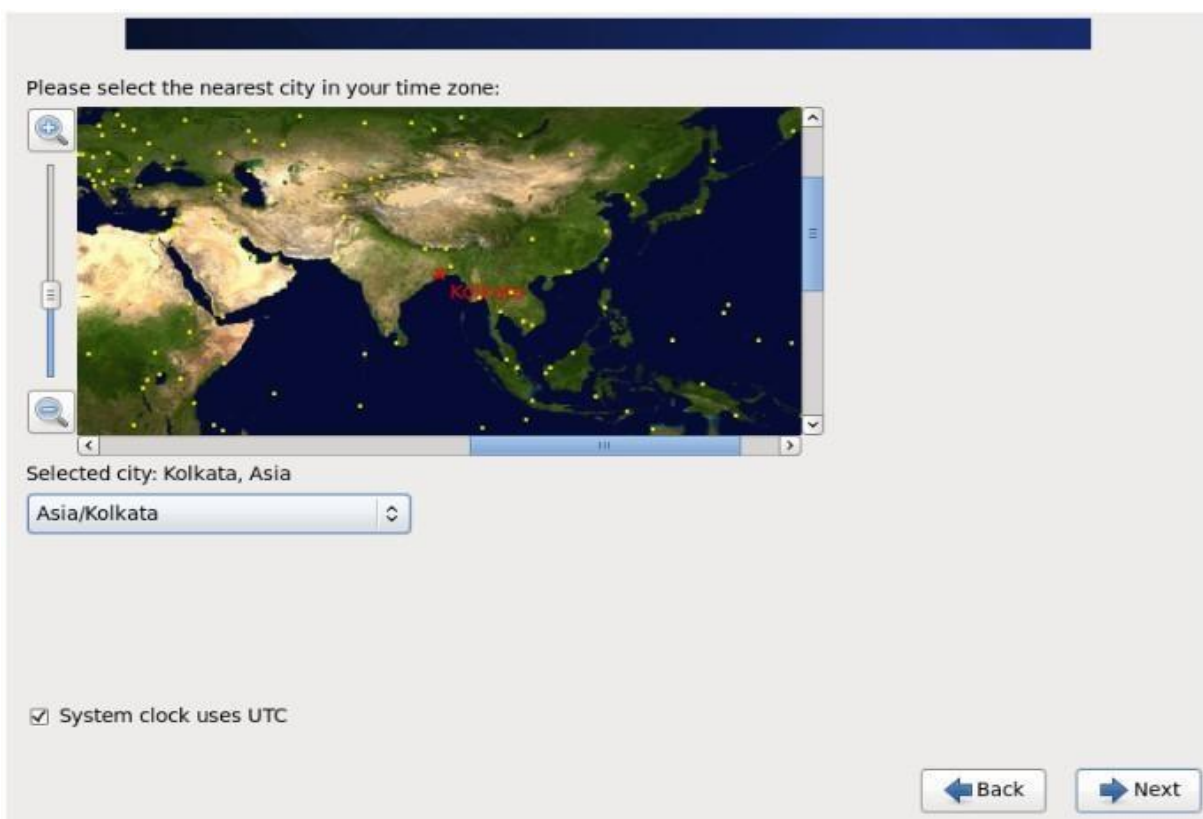
8. Select language as English and click Next. And Click Yes, discard my data.



9. Set hostname as “root”, set ip= 192.168.18.35, netmask= 255.255.254.0, gateway= 192.168.18.1, mask=8.8.8.8.



10. Locate Asia/Kolkata and click Next.



11. Enter root password : 123456 and enter it again to confirm. Press Next, and click Use anyway.

Player ▾ | [Icons] | [Icons]

The root account is used for administering the system. Enter a password for the root user.

Root Password: [.....]

Confirm: [.....]

[Back] [Next]

Click in the virtual screen to send keystrokes

VMware Tools enables many features and improves mouse movement, video and performance. Log in to the guest operating system and click "Install Tools".

[Install Tools] [Remind Me Later] [Never Remind Me]

12. Set public IP range/list: 192.168.18.36-192.168.18.41 (Your system number- your system number+5).

Please enter the required information to configure your Eucalyptus Cloud.

Public IP range/list: [192.168.18.36-192.168.18.41]

Below are some advanced options which you will generally not need to modify.

Public Network Interface: [eth0 ▾]

Private Network Interface: [br0 ▾]

Private Subnet: [172.31.254.0]

Private Netmask: [255.255.254.0]

Addresses per security group: [32 ▾]

DNS server: [8.8.8.8]

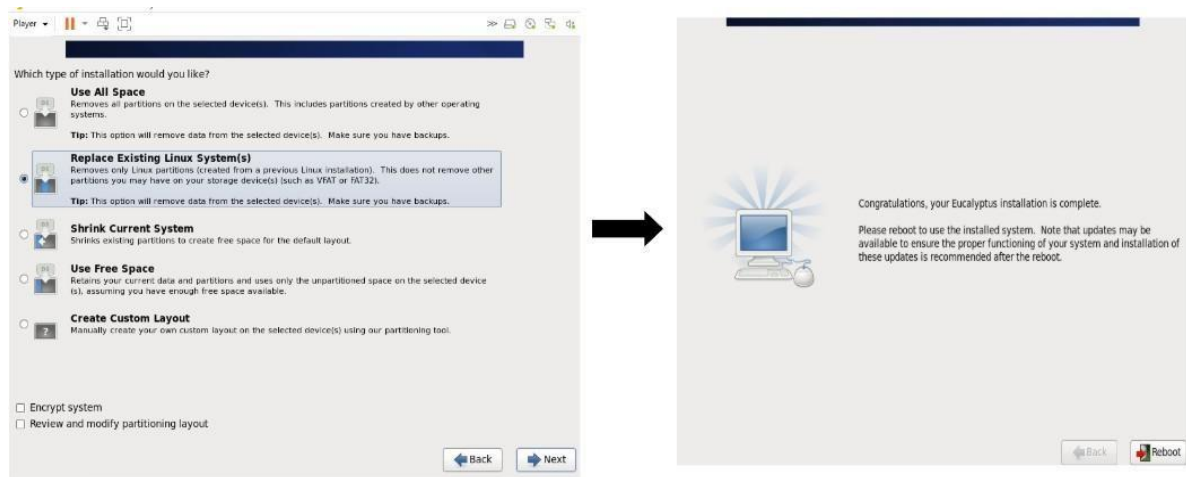
NC Bridge IP: [172.31.252.1]

Below is some brief information about the configuration settings on this screen. For more detailed explanations, please see the Eucalyptus Administrators' Guide.

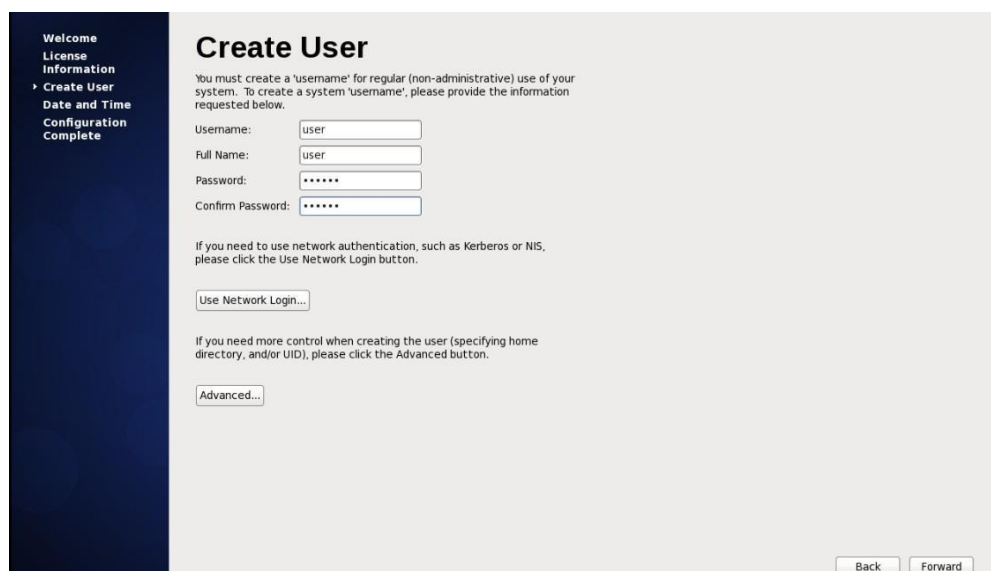
Public IP range - This is the range of available public IP addresses which can be mapped to instances. It should be two IPs separated by a dash (e.g. 192.168.1.100-192.168.1.200). This range should have enough IP addresses for the maximum number of instances which are intended to run concurrently in the cloud. These addresses must be on the same subnet as your public network interface, and must not be in use by other systems on the network.

[Back] [Next]

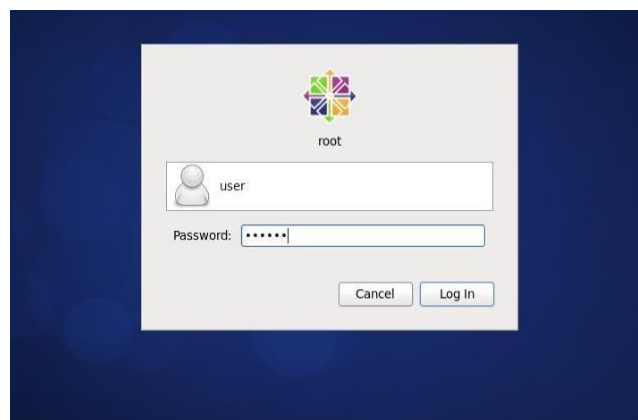
13. Click Replace existing Linux system and press Write changes; press reboot- it will take a few minutes.



14. Click Forward twice, then enter username and Full name as “user” and password as : 123456. Then click forward



15. Click finish. Virtual Machine is created with Linux system. And Enter password: 123456 and log in.



RESULT: Thus, setting up cloud using Eucalyptus has been completed successfully.

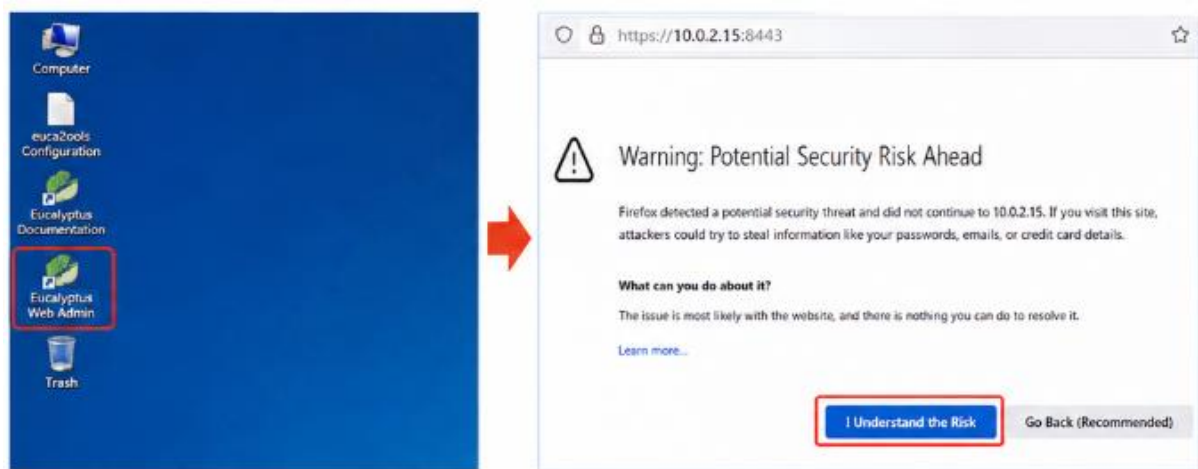
EXP NO: 4 RUNNING VIRTUAL MACHINE OF DIFFERENT CONFIGURATIONS

DATE:

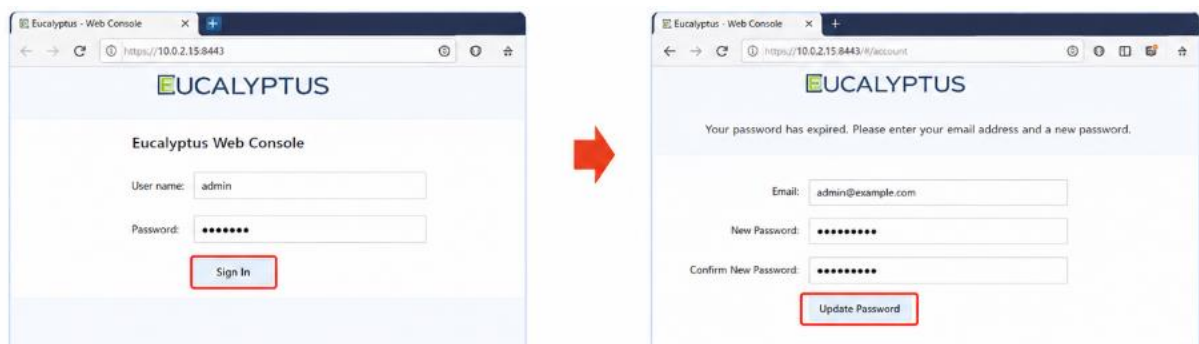
AIM: To run Virtual Machines of different configurations.

PROCEDURE:

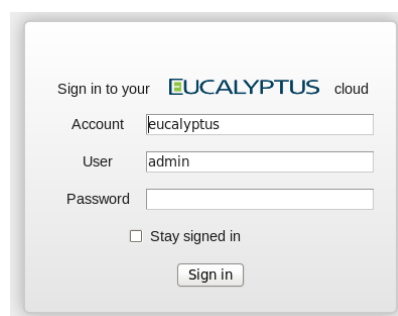
1. Open the Virtual Machine (VM) and click on Eucalyptus Web Admin from the desktop. In the browser, click "I Understand the Risk", then click "Add Exception" and "Confirm Security Exception."



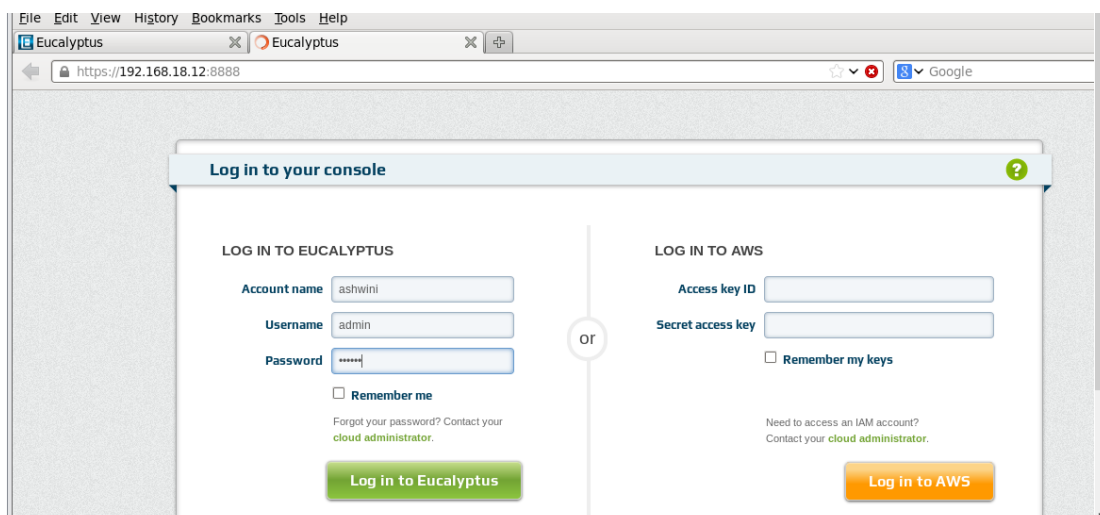
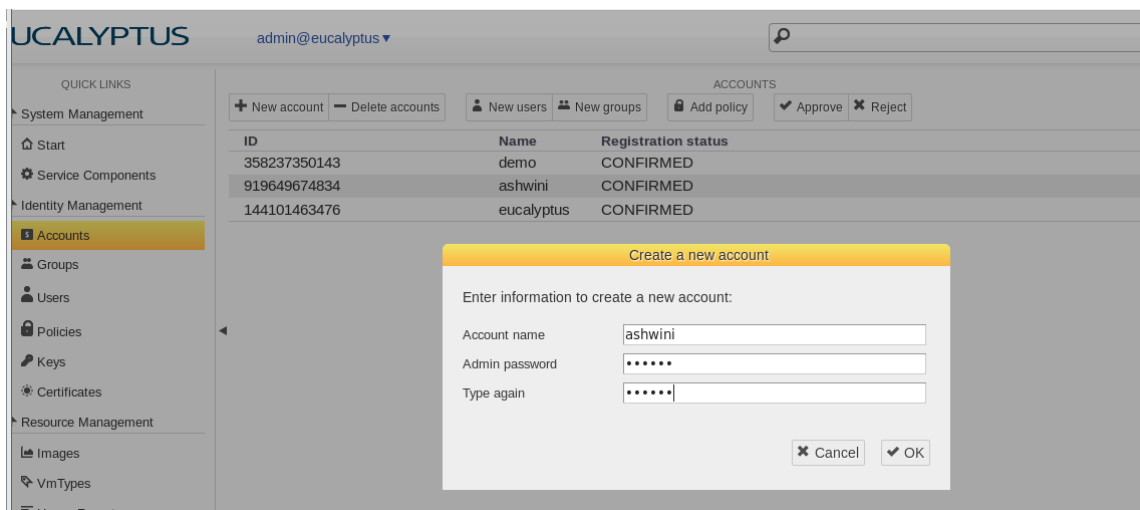
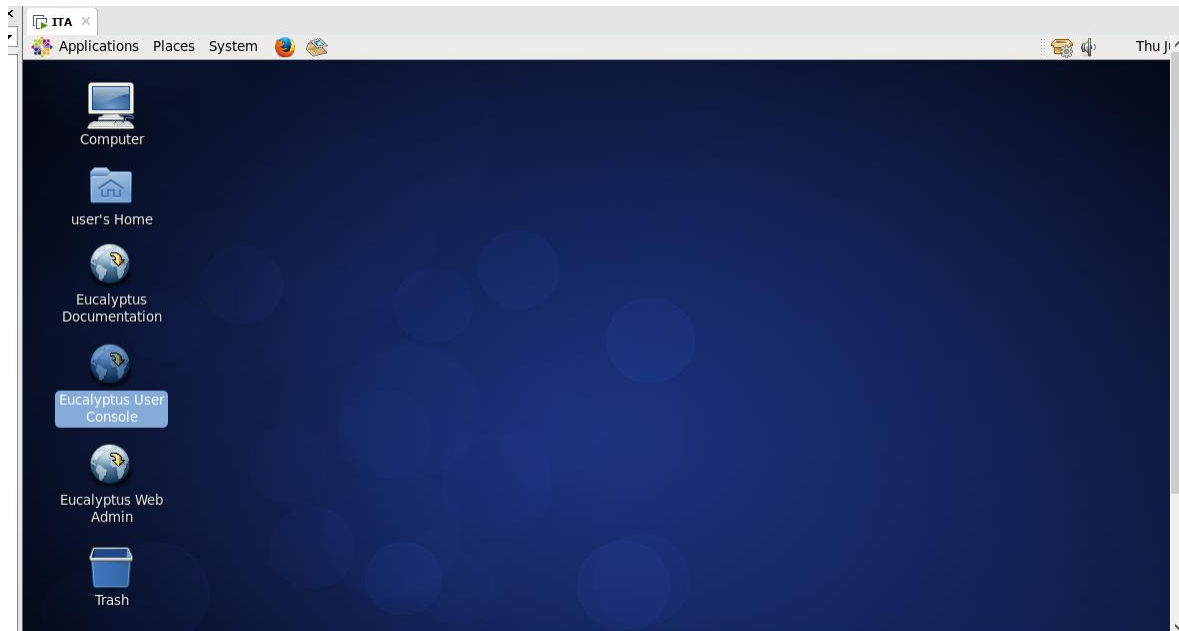
2. Log in as the administrator. Enter the username admin and password admin, then click Sign In. When prompted, enter your email ID and reset/set the new password as admin1



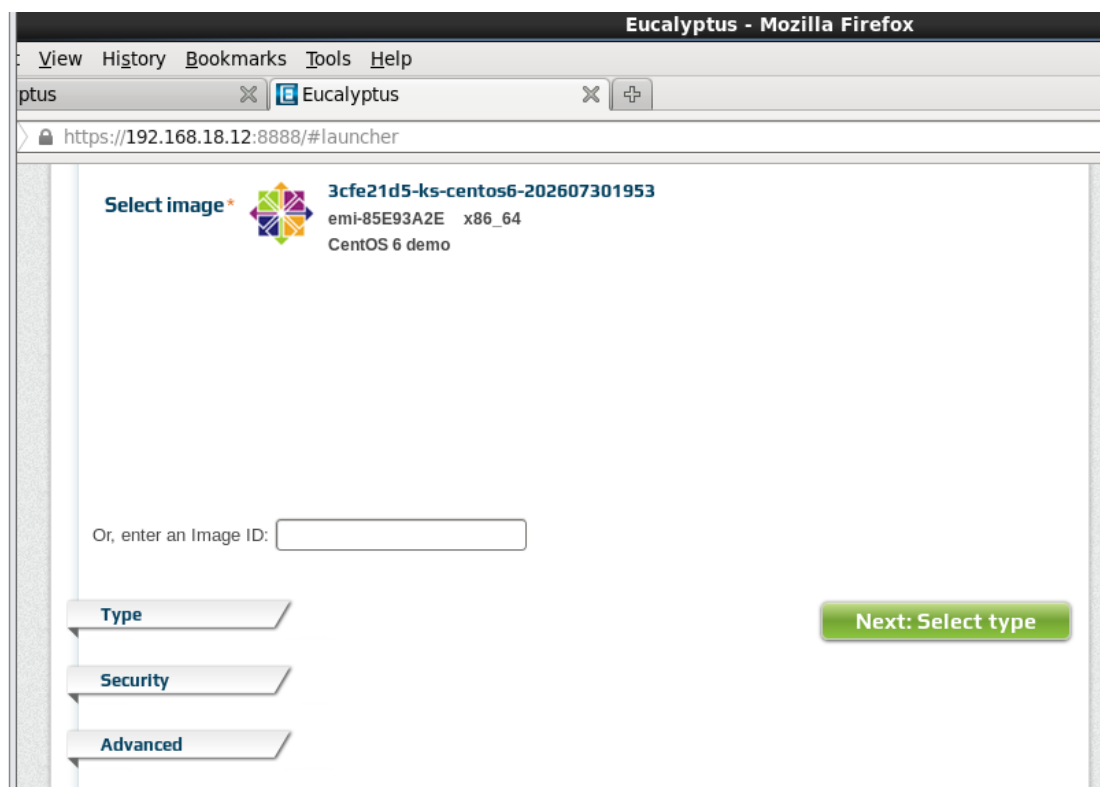
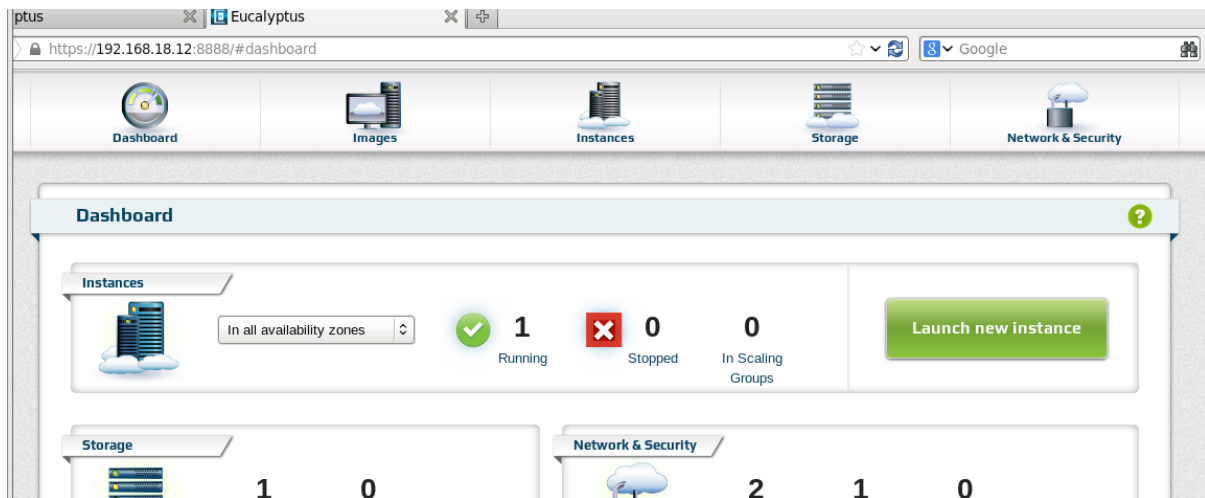
3. Log in again using the username admin and password admin1, then click OK.



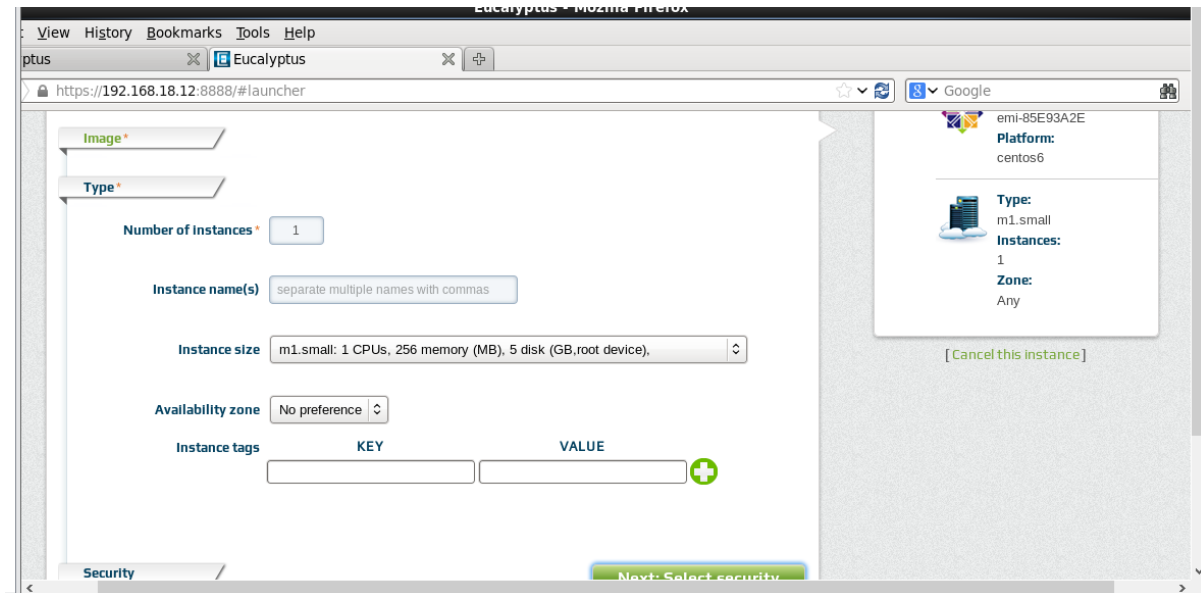
4. Open the Eucalyptus User Console. Enter the Account Name as your username, Username as admin, and Password as admin1. Click Login.



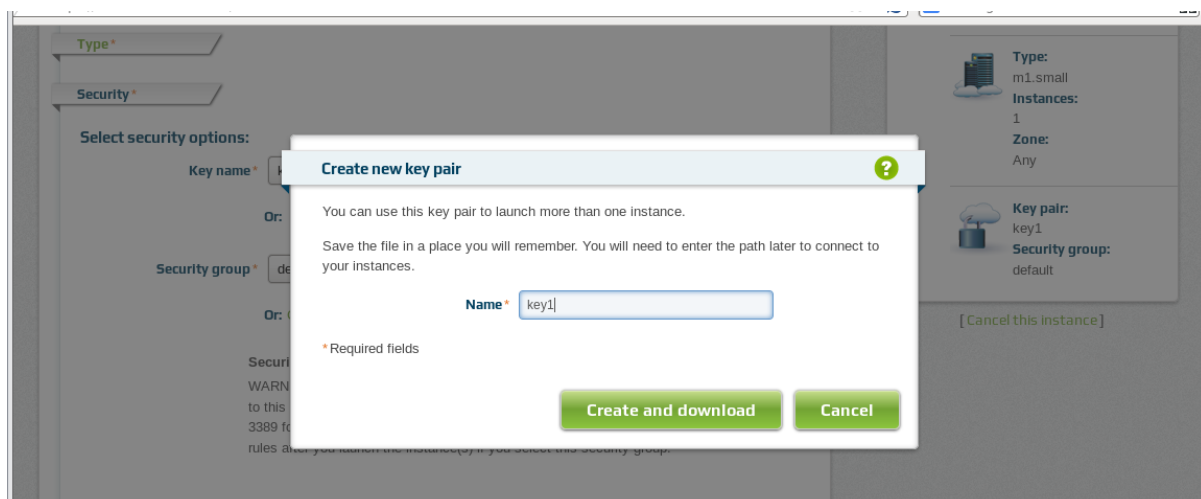
- Click Launch New Instance. In the Select Image page, choose the required image and select the instance type.



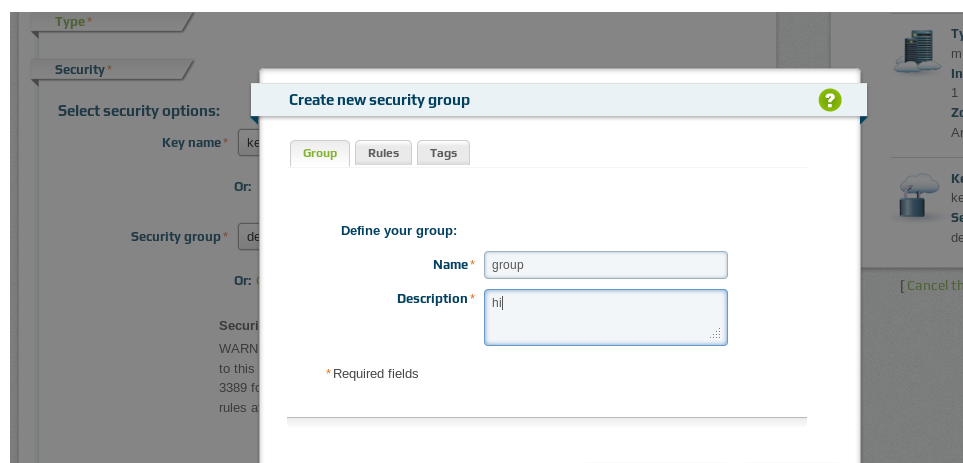
- Set the Number of Instances as 1. Enter the Instance Name as your username and click Select Security.



- Click Create New Key Pair, enter the key pair name as key1, click Create, then Download the key pair and click OK.



- Click Create New Security Group, enter the group name as group1, provide a description (e.g., "Hi User"), and click Create Group.



9. Click Launch Instance.

Select security options:

Key name *

Or: [Create new key pair](#)

Security group *

Or: [Create new security group](#)

Security group group1 incoming traffic rules

WARNING, NO RULES DEFINED! Your instance(s) will not be accessible until you add rules to this security group (commonly, open port 22 for SSH access to Linux instances and port 3389 for RDP access to Windows instances). You can use the Manage Rules dialog to add rules after you launch the instance(s) if you select this security group.

Advanced

Launch instance(s)

Or: [Select advanced options](#)

10. Wait until the instance status changes to Running. This indicates that the virtual machine instance has been successfully created and launched.

EUCALYPTUS admin@ashwini

Dashboard Images Instances Storage Network & Security

Manage instances

Launch new instance More actions

1 instances found. Showing: 10 | 25 | 50 | 100

INSTANCE	STATUS	IMAGE ID	AVAILABILITY ZONE	PUBLIC ADDRESS	PRIVATE ADDRESS	KEY NAME	SECURITY GROUP	LAUNCH TIME
☐ i-FCED3CC2	☒	emi-85E93A2E	CLUSTER01	192.168.18.13	172.31.254.239	key1	group1	08:21:51 PM Jul 30th 2026

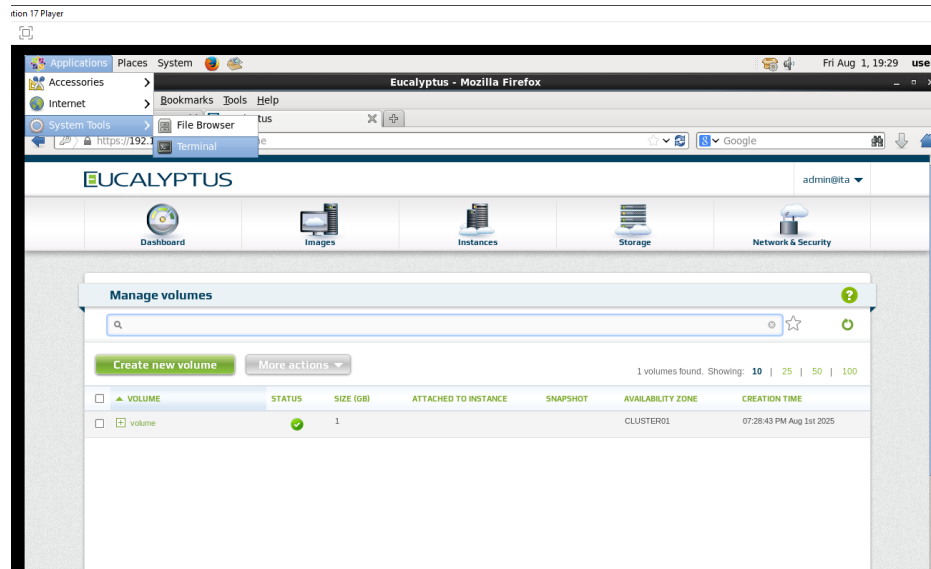
RESULT: Therefore, running the VM of different configurations is done successfully.

EXP NO: 5
DATE:

AIM:

PROCEDURE:

1. Create an instance in the Eucalyptus User Console.
2. In the desktop window, click on Applications, select system tool, then select Terminal.



3. Type the su root command in the terminal, enter 123456 as password, then click enter.
4. List out the files and directories in the root using ls command.
5. Change the directory to Downloads using the cd Downloads command.
6. Type the command chmod 6000 key1.pem to change mode to 6000.
7. Then type the ssh -i key1.pem cloud-user@192.168.18.16 command.
8. Then enter password as 123456.

```

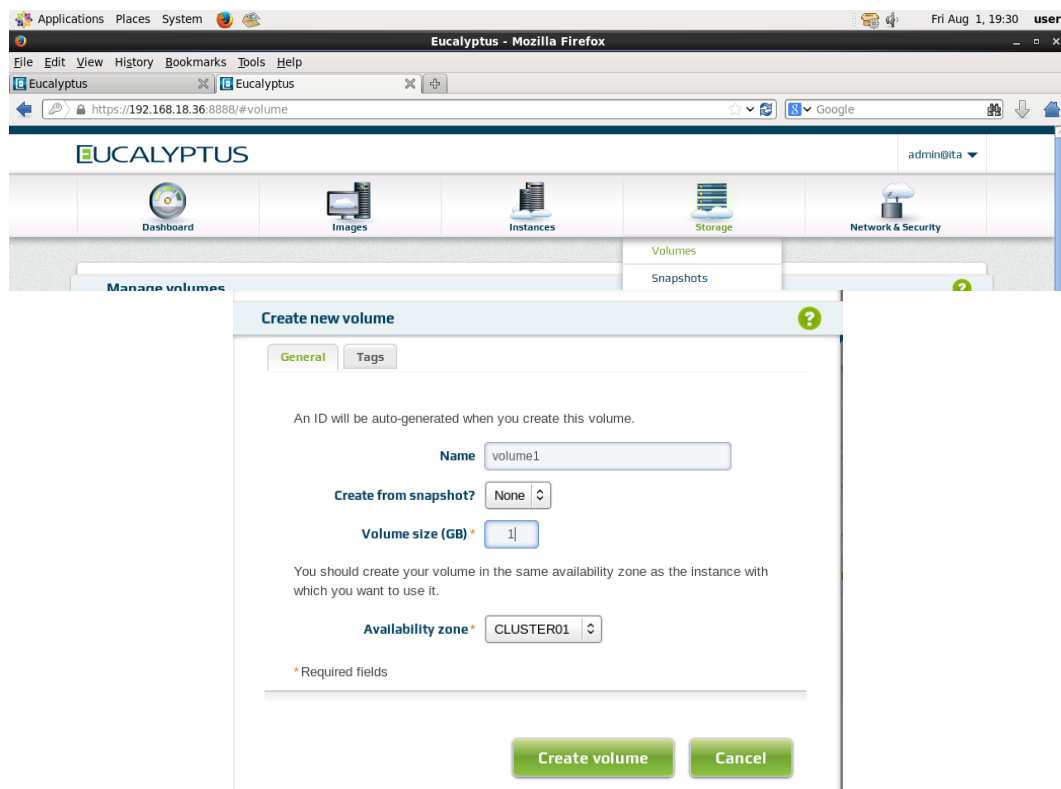
user@root: /home/user/Downloads
File Edit View Search Terminal Help
[user@root ~]$ su root
Password:
[root@root user]# ls
credentials  Documents  Music      Public      Videos
Desktop     Downloads  Pictures   Templates
[root@root user]# cd Downloads
[root@root Downloads]# Chmod 600 key1.pem
bash: Chmod: command not found
[root@root Downloads]# chmod 600 key1.pem
[root@root Downloads]# ssh -i key1.pem cloud-user@192.168.18.16
cloud-user@192.168.18.16's password:
Permission denied, please try again.
cloud-user@192.168.18.16's password:
Permission denied, please try again.
cloud-user@192.168.18.16's password:
Permission denied (publickey,gssapi-keyex,gssapi-with-mic,password).
[root@root Downloads]#

```

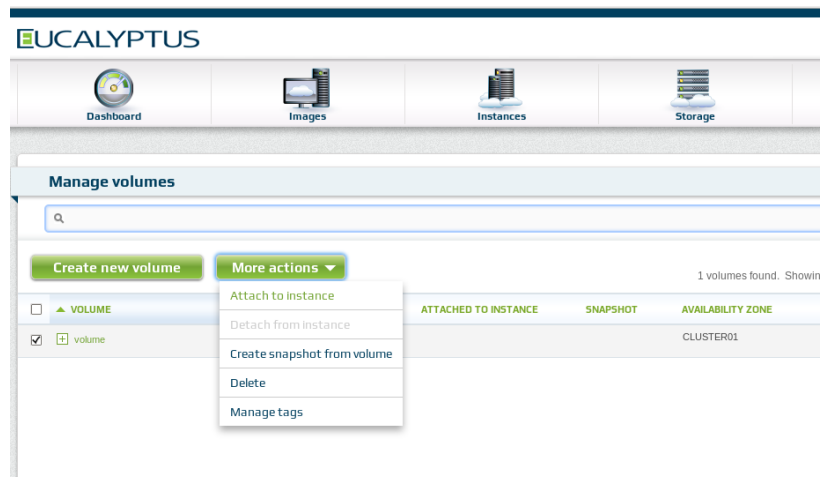
9. Then list out the blocks using lsblk.

```
[root@root Downloads]# lsblk
NAME
MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
loop0
 7:0      0  1.3G  0 loop
└─euca-AID3WCL9FF0096KGHXX0K-i-A73642F5-emi-C47D3787-25e43c3b-p0-snap (dm-1)
253:1     0  1.3G  0 dm
└─euca-AID3WCL9FF0096KGHXX0K-i-A73642F5-emi-C47D3787-25e43c3b (dm-2)
253:2     0  1.3G  0 dm
└─euca-AID3WCL9FF0096KGHXX0K-i-A73642F5-dsk-C47D3787-76e491e1 (dm-12)
253:12    0    5G  0 dm
└─euca-AID3WCL9FF0096KGHXX0K-i-A73642F5-dsk-C47D3787-76e491e1p1 (dm-13)
253:13    0  1.3G  0 dm
└─euca-AID3WCL9FF0096KGHXX0K-i-A73642F5-dsk-C47D3787-76e491e1p2 (dm-14)
253:14    0  3.3G  0 dm
└─euca-AID3WCL9FF0096KGHXX0K-i-A73642F5-dsk-C47D3787-76e491e1p3 (dm-15)
253:15    0  512M  0 dm
loop1
 7:1      0  1.3G  0 loop
└─euca-AID3WCL9FF0096KGHXX0K-i-A73642F5-emi-C47D3787-25e43c3b-p0-back (dm-0)
253:0     0  1.3G  0 dm
└─euca-AID3WCL9FF0096KGHXX0K-i-A73642F5-emi-C47D3787-25e43c3b-p0-snap (dm-1)
253:1     0  1.3G  0 dm
└─euca-AID3WCL9FF0096KGHXX0K-i-A73642F5-emi-C47D3787-25e43c3b (dm-2)
253:2     0  1.3G  0 dm
└─euca-AID3WCL9FF0096KGHXX0K-i-A73642F5-dsk-C47D3787-76e491e1 (dm-12)
253:12    0    5G  0 dm
└─euca-AID3WCL9FF0096KGHXX0K-i-A73642F5-dsk-C47D3787-76e491e1p1 (dm-13)
253:13    0  1.3G  0 dm
└─euca-AID3WCL9FF0096KGHXX0K-i-A73642F5-dsk-C47D3787-76e491e1p2 (dm-14)
253:14    0  3.3G  0 dm
└─euca-AID3WCL9FF0096KGHXX0K-i-A73642F5-dsk-C47D3787-76e491e1p3 (dm-15)
253:15    0  512M  0 dm
loop2
 7:2      0  3.3G  0 loop
└─euca-AID3WCL9FF0096KGHXX0K-i-A73642F5-prt-03328ext3-57f32f76-p0-snap (dm-4)
253:4     0  3.3G  0 dm
└─euca-AID3WCL9FF0096KGHXX0K-i-A73642F5-prt-03328ext3-57f32f76 (dm-5)
253:5     0  3.3G  0 dm
```

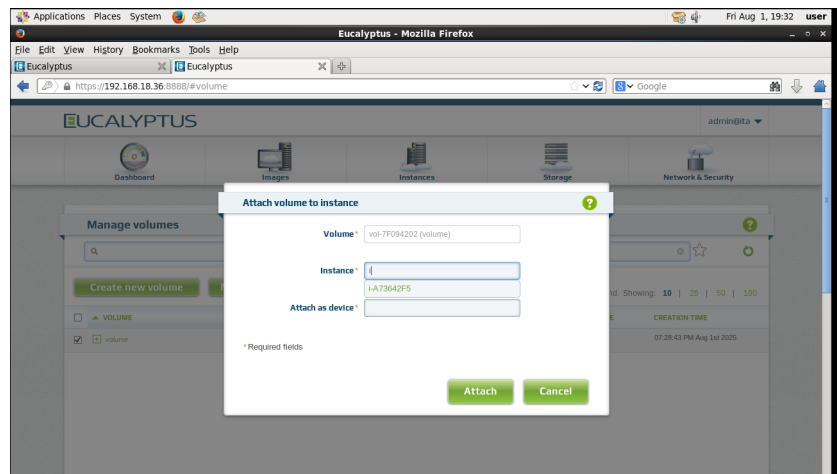
10. Now minimise the terminal, in the eucalyptus console, click on Storage, select volume, give name as volume1 and size as 1 GB and click on create volume.



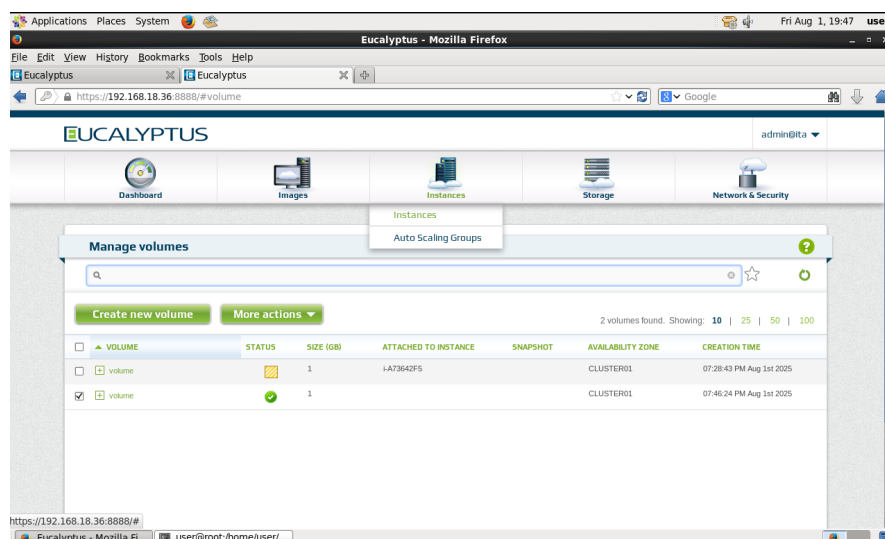
11. Select the volume, then click on more actions, then select attach instances.

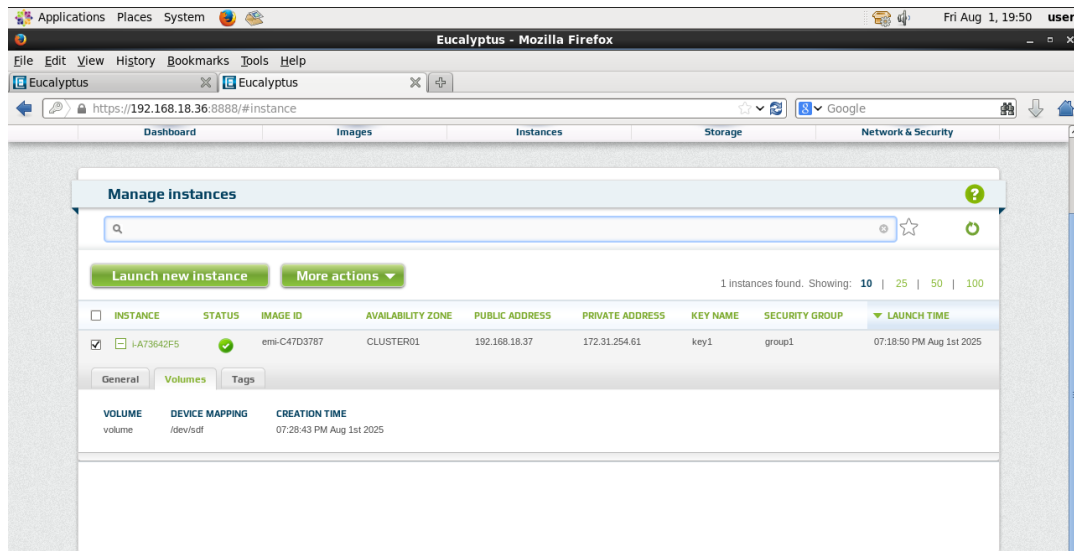


12. A screen with Default volume appears, then type the instance name, choose the instance number and then click attach option.



13. Now click on instances, expand your instance, click volume and check if the volume we created is attached to the instance we created.





14. To check if the volume is attached, enter `lsblk` command in the terminal again and see if the size is 1 GB.

```

└─euca--ebs--storage--vg--vol--7F094202-euca--vol--7F094202 (dm-16)
253:16  0      1G  0 lvm
sr0
11:0    1  1024M  0 rom
sda
8:0     0   200G  0 disk
└─sda1
8:1     0    500M  0 part /boot
└─sda2
8:2     0  196.6G  0 part /
└─sda3
8:3     0     3G  0 part [SWAP]
└─sda4
8:4     0     1K  0 part
└─sda5
8:5     0    18M  0 part
euca-zero (dm-9)
253:9   0     1P  0 dm
└─euca-AID3WCL9FF0096KGHXXOK-i-A73642F5-dsk-C47D3787-76e491e1-p0-snap (dm-11)
253:11  0   31.5K  0 dm
└─euca-AID3WCL9FF0096KGHXXOK-i-A73642F5-dsk-C47D3787-76e491e1 (dm-12)
253:12  0     5G  0 dm
└─euca-AID3WCL9FF0096KGHXXOK-i-A73642F5-dsk-C47D3787-76e491e1p1 (dm-13)
253:13  0    1.3G  0 dm
└─euca-AID3WCL9FF0096KGHXXOK-i-A73642F5-dsk-C47D3787-76e491e1p2 (dm-14)
253:14  0    3.3G  0 dm
└─euca-AID3WCL9FF0096KGHXXOK-i-A73642F5-dsk-C47D3787-76e491e1p3 (dm-15)
253:15  0    512M  0 dm
sdb
8:16   0     1G  0 disk
[root@root Downloads]#

```

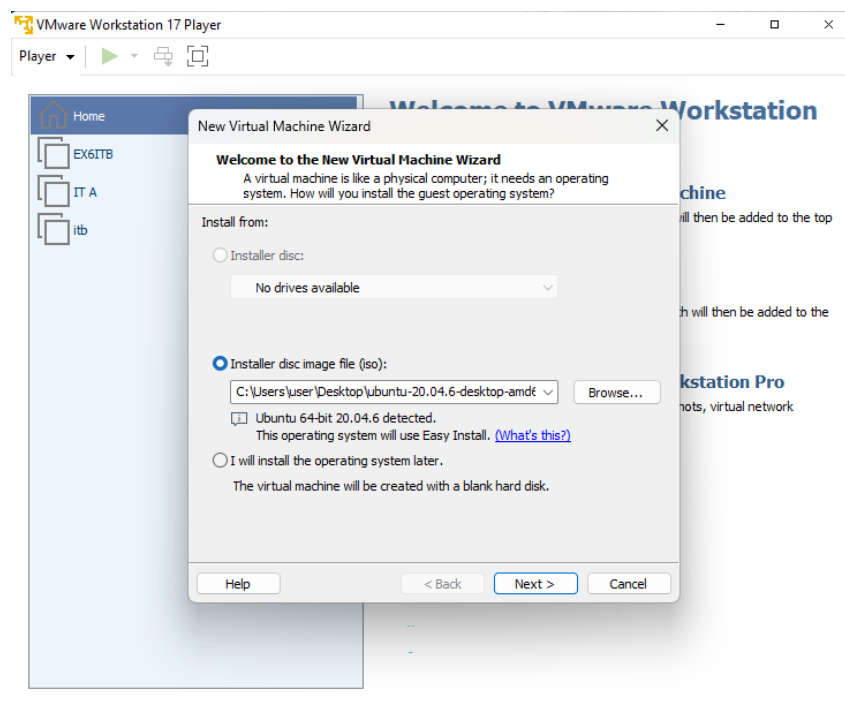
RESULT:

**EXP NO: 6 INSTALL A C COMPILER IN THE VIRTUAL MACHINE AND
DATE: EXECUTE A SAMPLE C PROGRAM**

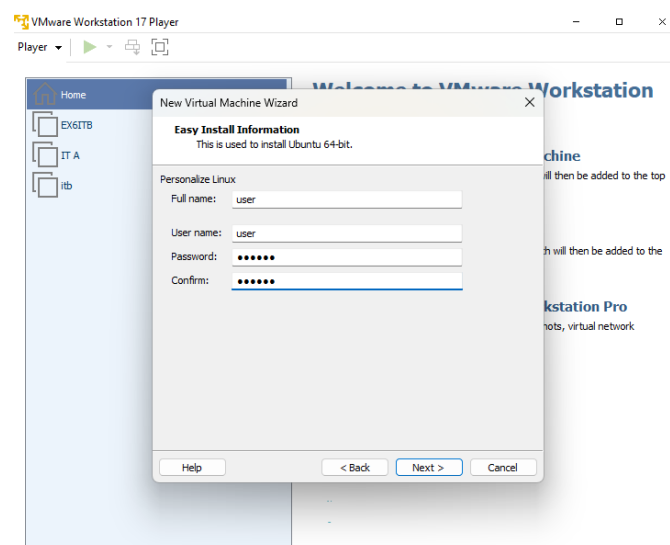
AIM: To install a C compiler in the virtual machine and execute a sample program.

PROCEDURE:

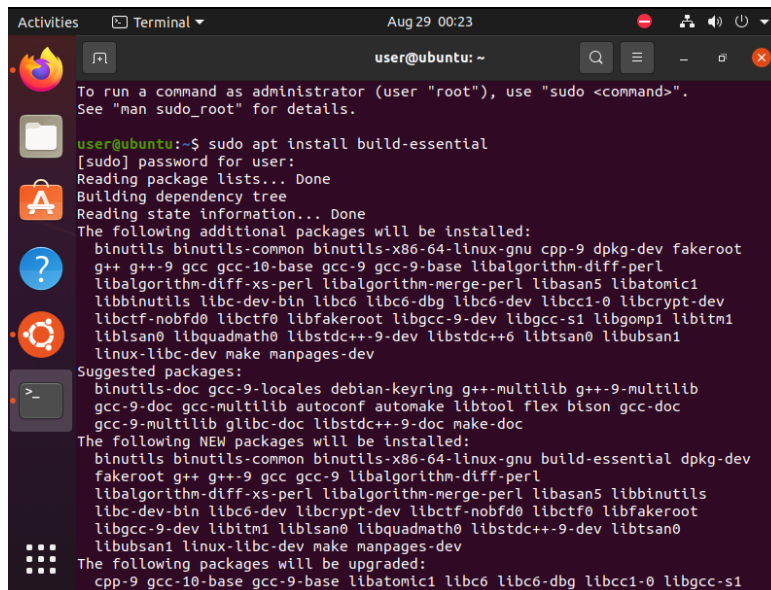
1. Open VMWare workstation and install UNIX OS by choosing unix AMD64 image file from the desktop.



2. Enter Full name as user, user name as user and set password as 123456, confirm the password and click next.



3. Name your VM, follow the same steps as exercise 5. Finish once created and open the hotspot login.
4. Open the terminal in Ubuntu OS, install the C compiler using the command: `sudo apt install build-essential` and enter password as 1233456.

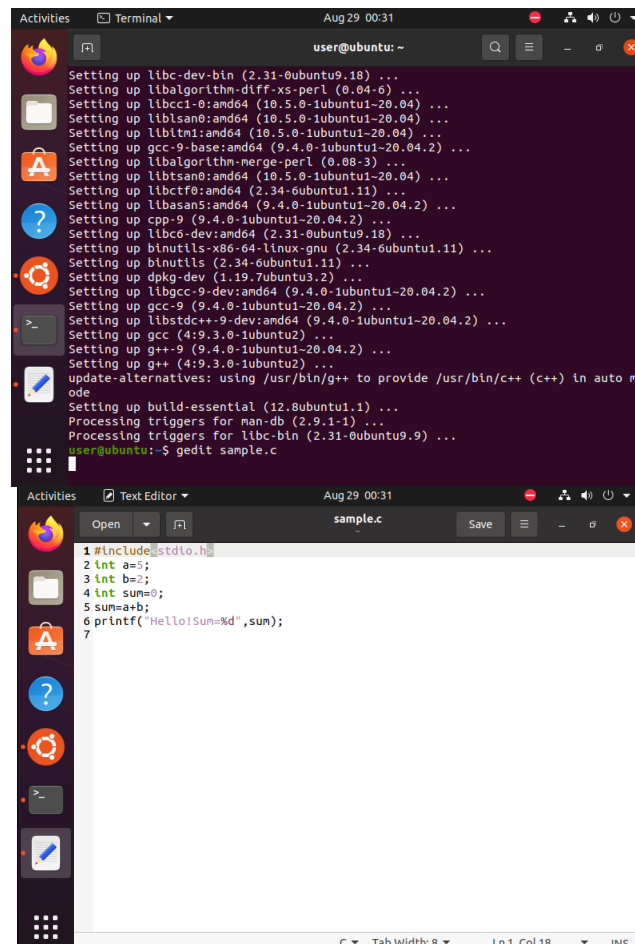


```

user@ubuntu: ~
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

user@ubuntu:~$ sudo apt install build-essential
[sudo] password for user:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu cpp-9 dpkg-dev fakeroot
  g++ g++-9 gcc gcc-10-base gcc-9 gcc-9-base libalgorithm-diff-perl
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libasan5 libatomic1
  libbinutils libc-dev-bin libc6 libc6-dbg libc6-dev libcc1-0 libcrypt-dev
  libctf-nobfd0 libctf0 libfakeroot libgcc-9-dev libgcc-s1 libgomp1 libitm1
  liblsan0 libquadmath0 libstdc++-9-dev libstdc++6 libtsan0 libubsan1
  linux-libc-dev make manpages-dev
Suggested packages:
  binutils-doc gcc-9-locales debian-keyring g++-multilib g++-9-multilib
  gcc-9-doc gcc-multilib autoconf automake libtool flex bison gcc-doc
  gcc-9-multilib glibc-doc libstdc++-9-doc make-doc
The following NEW packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu build-essential dpkg-dev
  fakeroot g++ g++-9 gcc gcc-9 libalgorithm-diff-perl
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libasan5 libbinutils
  libc-dev-bin libc6-dev libcrypt-dev libctf-nobfd0 libctf0 libfakeroot
  libgcc-9-dev libitm1 liblsan0 libquadmath0 libstdc++-9-dev libtsan0
  libubsan1 linux-libc-dev make manpages-dev
The following packages will be upgraded:
  cpp-9 gcc-10-base gcc-9-base libatomic1 libc6 libc6-dbg libcc1-0 libgcc-s1
  
```

5. Create a sample C program using the command `gedit sample.c`



```

Setting up libc-dev-bin (2.31-0ubuntu9.18) ...
Setting up libalgorithm-diff-xs-perl (0.04-6) ...
Setting up libcc1-0:amd64 (10.5.0-1ubuntu1-20.04) ...
Setting up liblsan0:amd64 (10.5.0-1ubuntu1-20.04) ...
Setting up libitm1:amd64 (10.5.0-1ubuntu1-20.04) ...
Setting up gcc-9-base:amd64 (9.4.0-1ubuntu1-20.04.2) ...
Setting up libalgorithm-merge-perl (0.08-3) ...
Setting up libtsan0:amd64 (10.5.0-1ubuntu1-20.04) ...
Setting up libctf0:amd64 (2.34-6ubuntu1.11) ...
Setting up libasan5:amd64 (9.4.0-1ubuntu1-20.04.2) ...
Setting up cpp-9 (9.4.0-1ubuntu1-20.04.2) ...
Setting up libc6-dev:amd64 (2.31-0ubuntu9.18) ...
Setting up binutils-x86-64-linux-gnu (2.34-6ubuntu1.11) ...
Setting up binutils (2.34-6ubuntu1.11) ...
Setting up dpkg-dev (1.19.7ubuntu3.2) ...
Setting up libgcc-9-dev:amd64 (9.4.0-1ubuntu1-20.04.2) ...
Setting up gcc-9 (9.4.0-1ubuntu1-20.04.2) ...
Setting up libstdc++-9-dev:amd64 (9.4.0-1ubuntu1-20.04.2) ...
Setting up gcc (4:9.3.0-1ubuntu2) ...
Setting up g++-9 (9.4.0-1ubuntu1-20.04.2) ...
Setting up g++ (4:9.3.0-1ubuntu2) ...
update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto m
ode
Setting up build-essential (12.8ubuntu1.1) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for libc-bin (2.31-0ubuntu9.9) ...
user@ubuntu:~$ gedit sample.c

```

```

1 #include<stdio.h>
2 int a=5;
3 int b=2;
4 int sum=0;
5 sum=a+b;
6 printf("Hello!Sum=%d",sum);
7

```

6. Compile C program using the command gcc, and then run the program using ./a.out command.

```
user@ubuntu:~$ gedit sample.c
^Z
[1]+  Stopped                  gedit sample.c
user@ubuntu:~$ gcc sample.c
user@ubuntu:~$ ./a.out
Hello!Sum=7user@ubuntu:~$
```

RESULT: Thus, the installation of C compiler in the virtual machine and execution of a sample program has been implemented.

EXP NO: 7 SHOW THE VIRTUAL MIGRATION BASED ON CERTAIN
DATE: CONDITION FROM ONE NODE TO THE OTHER

AIM: To show the virtual machine migration based on certain condition from one node to the other.

PROCEDURE:

1. Open Browser, type localhost:9869
2. Login using username: onedadmin, password: opennebula
3. Then follow the steps to migrate VMs
 - Click on infrastructure
 - Select clusters and enter the cluster name
 - Then select host tab, and select all host
 - Then select Vnets tab, and select all vnet
 - Then select datastores tab, and select all datastores
 - And then choose host under infrastructure tab
 - Click on + symbol to add new host, name the host then click on create.
4. On instances, select VMs to migrate then follow the steps
 - Click on 8th icon, then the drop-down list will display
 - Select migrate, the popup window will display
 - Select the target host to migrate, then click on migrate

BEFORE MIGRATION:

Host: SVCE

ID	Owner	Group	Name	Status	Host	IPs
5	onedadmin	onedadmin	vm2	FAILURE	naiveenkumar	172.16.100.205
4	onedadmin	onedadmin	vm2	FAILURE	naiveenkumar	172.16.100.204
3	onedadmin	onedadmin	vm1	FAILURE	naiveenkumar	172.16.100.203
2	onedadmin	onedadmin	naiveen	FAILURE	naiveenkumar	172.16.100.202
1	onedadmin	onedadmin	naiveen	FAILURE	naiveenkumar	172.16.100.201
0	onedadmin	onedadmin	stylinu-O	FAILURE	naiveenkumar	172.16.100.200

Host: one-sandbox

The screenshot shows the OpenNebula web interface for the 'one-sandbox' host. The left sidebar contains navigation links: Dashboard, Instances, VMs, Services, Virtual Routers, Templates, Storage, Network, Infrastructure, Clusters, Hosts, Zones, System, and Settings. The main content area displays the 'Host' page for 'one-sandbox'. At the top, there are buttons for 'Select cluster', 'Enable', 'Disable', and 'Offline'. Below these are tabs for 'Info', 'Graphs', 'VMs', 'Wilds', and 'Zombies'. The 'VMs' tab is active, showing a table of VMs running on this host.

ID	Owner	Group	Name	Status	Host	IPs
7	oneadmin	oneadmin	vm8	RUNNING	one-sandbox	172.16.100.207
6	oneadmin	oneadmin	vm8	RUNNING	one-sandbox	172.16.100.206

Showing 1 to 2 of 2 entries

The screenshot shows the OpenNebula web interface for the 'VMs' page. A 'Migrate Virtual Machine' dialog box is open, displaying information about VM 6 and VM 7. The dialog box has a 'Select a Host' section with a table of available hosts.

VM 6 vm8 is currently running on Host one-sandbox
VM 7 vm8 is currently running on Host one-sandbox

Select a Host

Please select a Host from the list

ID	Name	Cluster	RVMs	Allocated CPU	Allocated MEM	Status
2	naa	default	0	0/0	0KB / -	RETRY
1	naveenkumar	rama	6	62/0	44KB / -	ERROR
0	one-sandbox	rama	2	20 / 100 (20%)	40KB / 741KB (1%)	ON

Showing 1 to 3 of 3 entries

Advanced Options

Migrate

The screenshot shows the OpenNebula web interface for the 'VMs' page. The main content area displays a table of VMs. The table has columns for ID, Owner, Group, Name, Status, Host, and IPs. The VMs are listed in descending order of ID.

ID	Owner	Group	Name	Status	Host	IPs
7	oneadmin	oneadmin	vm8	SAVE	naveenkumar	172.16.100.207
6	oneadmin	oneadmin	vm8	SAVE	naveenkumar	172.16.100.206
5	oneadmin	oneadmin	vm2	FAILURE	naveenkumar	172.16.100.205
4	oneadmin	oneadmin	vm2	FAILURE	naveenkumar	172.16.100.204
3	oneadmin	oneadmin	vm1	FAILURE	naveenkumar	172.16.100.203
2	oneadmin	oneadmin	naveen	FAILURE	naveenkumar	172.16.100.202
1	oneadmin	oneadmin	naveen	FAILURE	naveenkumar	172.16.100.201
0	oneadmin	oneadmin	ttlinux-0	FAILURE	naveenkumar	172.16.100.200

Showing 1 to 8 of 8 entries

8 TOTAL 2 ACTIVE 0 OFF 0 PENDING 6 FAILED

AFTER MIGRATION:

The screenshot shows the OpenNebula web interface at localhost:9869. The 'Hosts' page displays a table of hosts with columns: ID, Name, Cluster, RVMs, Allocated CPU, Allocated MEM, and Status. The table contains three entries:

ID	Name	Cluster	RVMs	Allocated CPU	Allocated MEM	Status
2	ras	default	0	0/0	0K/0	ERROR
1	naiveenkumar	rama	8	82/0	481K/0	ERROR
0	one-sandbox	rama	0	0/100 (0%)	0K/741K (0%)	ON

Summary statistics: 3 TOTAL, 1 ON, 0 OFF, 2 ERROR.

Host: one-sandbox

The screenshot shows the OpenNebula web interface at localhost:9869, specifically the 'Host: one-sandbox' page. The 'VMs' tab is selected, but the table is empty with the message 'There is no data available'.

Host: SVCE

The screenshot shows the OpenNebula web interface at localhost:9869, specifically the 'Host: naiveenkumar' page. The 'VMs' tab is selected, showing a table of virtual machines:

ID	Owner	Group	Name	Status	Host	IPs
7	oneadmin	oneadmin	vm0	FAILURE	naiveenkumar	172.16.100.207
6	oneadmin	oneadmin	vm0	FAILURE	naiveenkumar	172.16.100.206
5	oneadmin	oneadmin	vm2	FAILURE	naiveenkumar	172.16.100.205
4	oneadmin	oneadmin	vm2	FAILURE	naiveenkumar	172.16.100.204
3	oneadmin	oneadmin	vm1	FAILURE	naiveenkumar	172.16.100.203
2	oneadmin	oneadmin	naiveen	FAILURE	naiveenkumar	172.16.100.202
1	oneadmin	oneadmin	naiveen	FAILURE	naiveenkumar	172.16.100.201
0	oneadmin	oneadmin	ttlinux-0	FAILURE	naiveenkumar	172.16.100.200

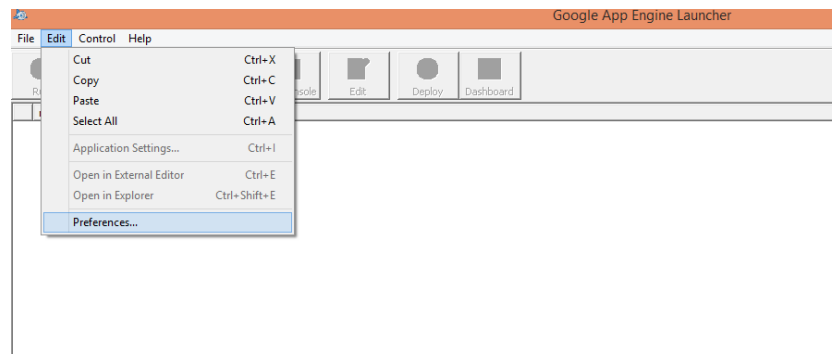
RESULT: Thus, the virtual machine migration based on the certain condition from one node to the other has been executed successfully.

**EXP NO: 8 INSTALL GOOGLE APP ENGINE AND CREATE A SIMPLE
DATE: WEB APPLICATION USING PYTHON**

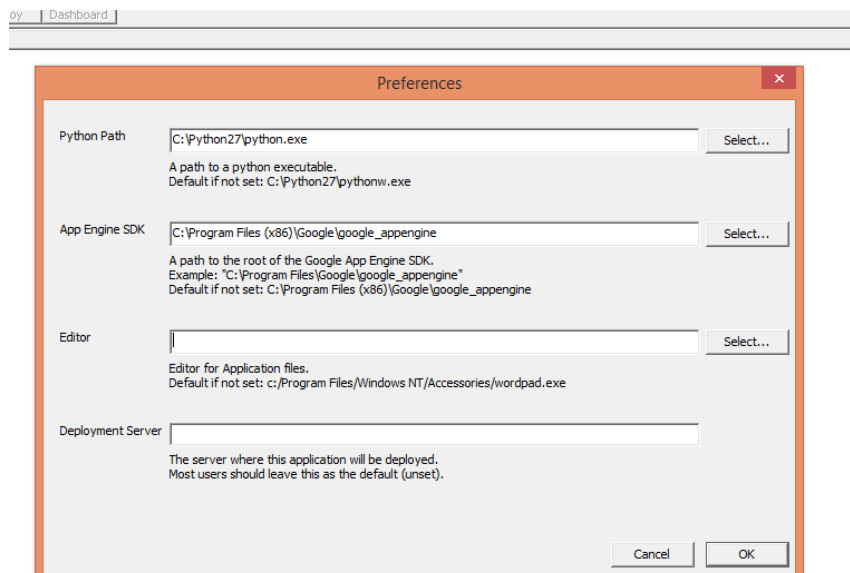
AIM: Install Google App Engine and create Hello World and other simple web applications using Java/Python.

PROCEDURE:

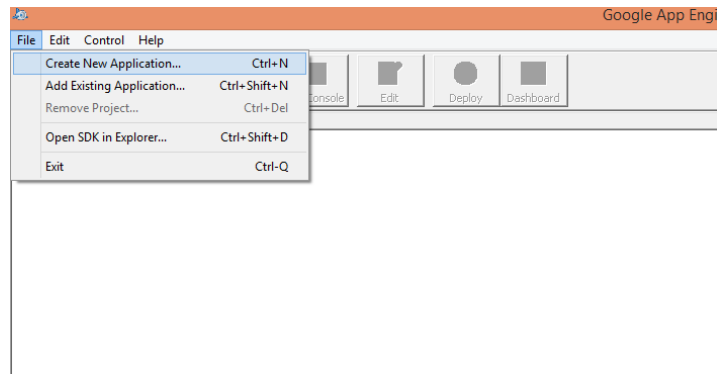
1. Open the Google App Engine Launcher and select Edit → Preferences.



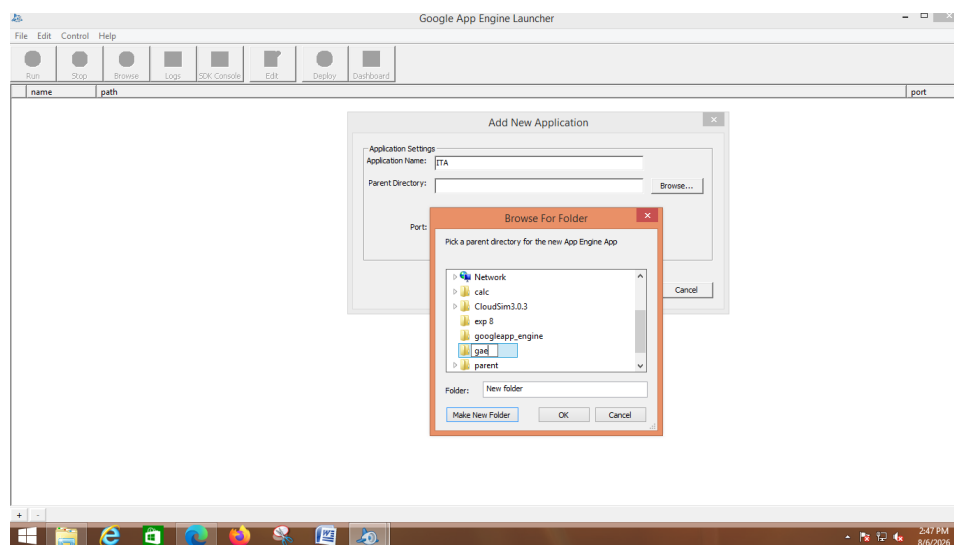
2. In the Preferences window, set the Python Path as C:\Python27\python.exe and set the App Engine SDK path as C:\Program Files (x86)\Google\google_appengine.



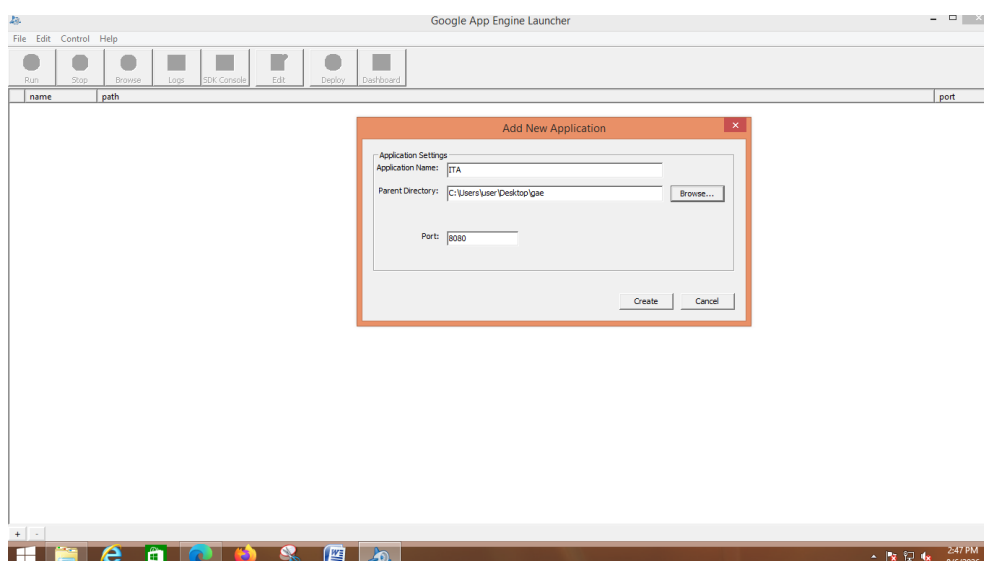
3. Open the File menu and select Create New Application.



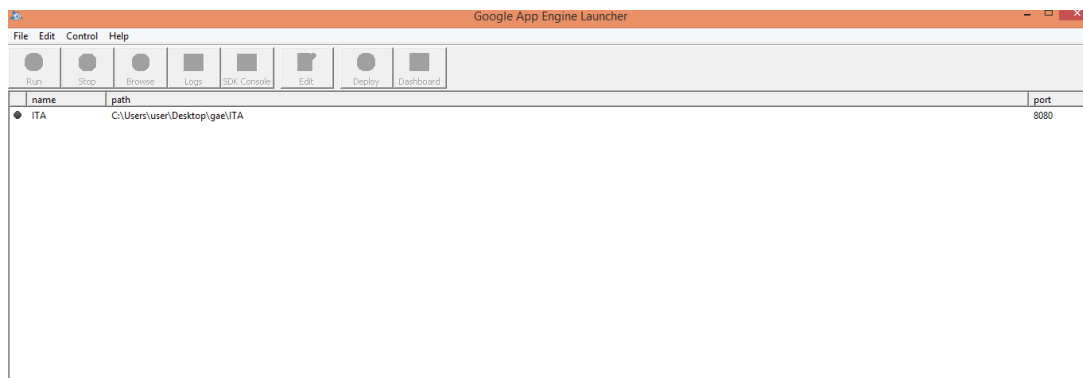
4. In the Add New Application window, enter the application name and click Browse to select the parent directory.



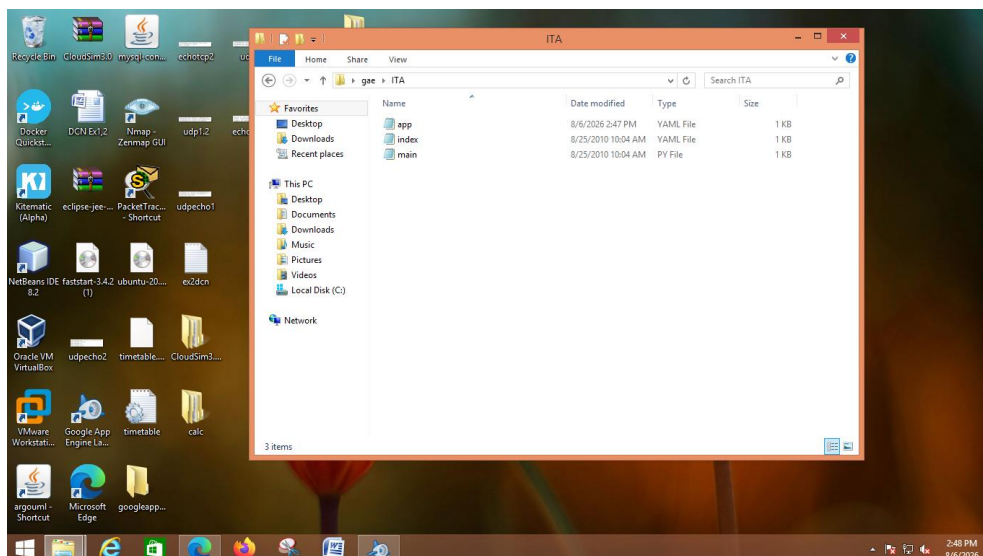
5. Enter the Application Name as ITA, select the parent directory as C:\Users\user\Desktop\gae and set the port number as 8080.



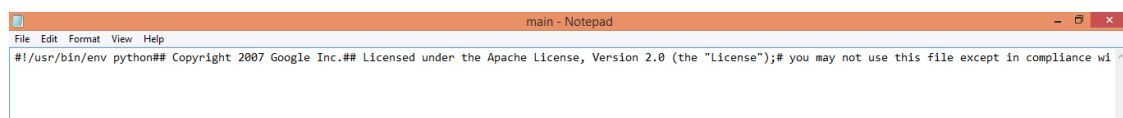
- Click Create. The application ITA is created and displayed in the Google App Engine Launcher with port 8080.



- Open the application folder. The files app.yaml, index.yaml and main.py are present in the application directory.



- Open the main.py file using a text editor and view the Python program for the web application.



- Modify the Python program to display the message “Hello IT-A Students!” and to implement a simple calculator and save the file.

CODE:

```
#!/usr/bin/env python
#
# Copyright 2007 Google Inc.
#
# Licensed under the Apache License, Version 2.0 (the "License");
# you may not use this file except in compliance with the License.
# You may obtain a copy of the License at
#
# http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.
#
from google.appengine.ext import webapp
from google.appengine.ext.webapp import util

class MainHandler(webapp.RequestHandler):
    def get(self):
        # Retrieve query parameters: ?num1=10&num2=5&op=add
        num1_str = self.request.get('num1')
        num2_str = self.request.get('num2')
        op = self.request.get('op', 'add')

        # Render a basic HTML form if no numbers are provided yet
        if not num1_str or not num2_str:
            self.response.out.write("""
<html>
<head><title>App Engine Calculator</title></head>
<body>
<h2>Simple Calculator</h2>
<form method="get">
<input type="text" name="num1" placeholder="First number" required />
<select name="op">
<option value="add">+</option>
<option value="subtract">-</option>
<option value="multiply">*</option>
<option value="divide">/</option>

```

```

        </select>
        <input type="text" name="num2" placeholder="Second number" required
/>

        <input type="submit" value="Calculate" />
    </form>
</body>
</html>
"""
)
return

try:
    num1 = float(num1_str)
    num2 = float(num2_str)
    result = 0
    error = None

    if op == 'add':
        result = num1 + num2
    elif op == 'subtract':
        result = num1 - num2
    elif op == 'multiply':
        result = num1 * num2
    elif op == 'divide':
        if num2 == 0:
            error = "Error: Division by zero is not allowed."
        else:
            result = num1 / num2
    else:
        error = "Error: Invalid operator."

    if error:
        self.response.out.write("<h3>%s</h3>" % error)
    else:
        self.response.out.write("<h3>Result: %s %s %s = %s</h3>" % (num1, op, num2,
result))

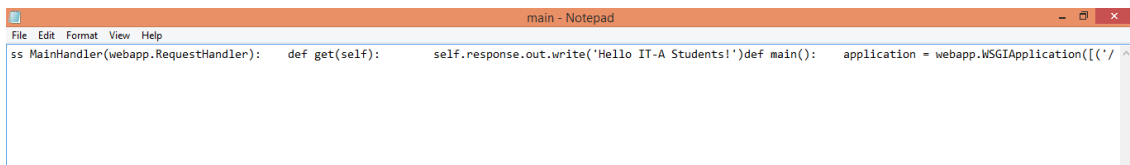
    self.response.out.write('<br><a href="/">Calculate another</a>')

except ValueError:
    self.response.out.write("<h3>Error: Please enter valid numbers.</h3><br><a
href='/'>Try again</a>")

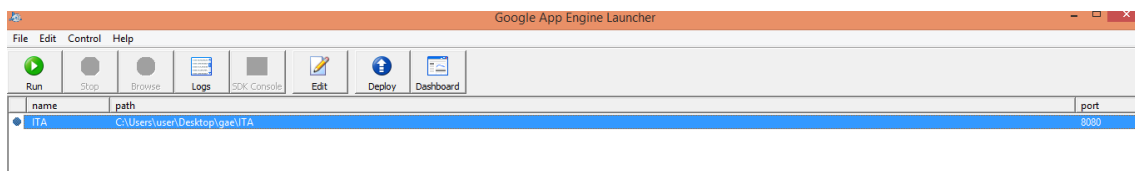
```

```
def main():
    application = webapp.WSGIApplication([('/', MainHandler)],
                                         debug=True)
    util.run_wsgi_app(application)

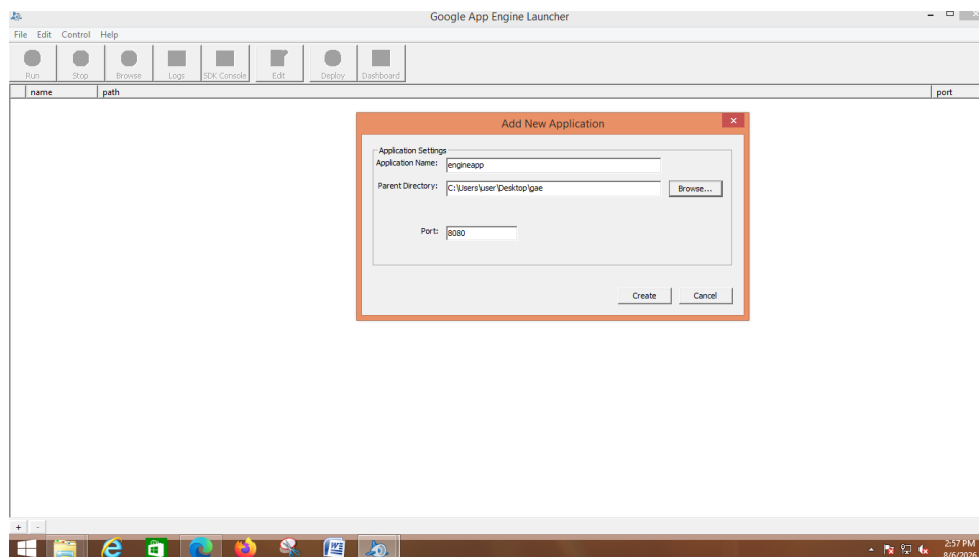
if __name__ == '__main__':
    main()
```



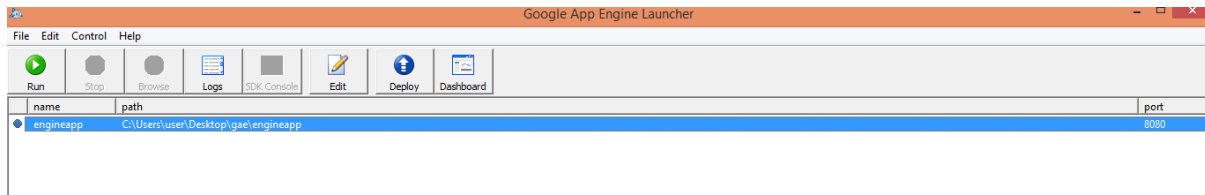
10. Select the created application ITA from the Google App Engine Launcher.



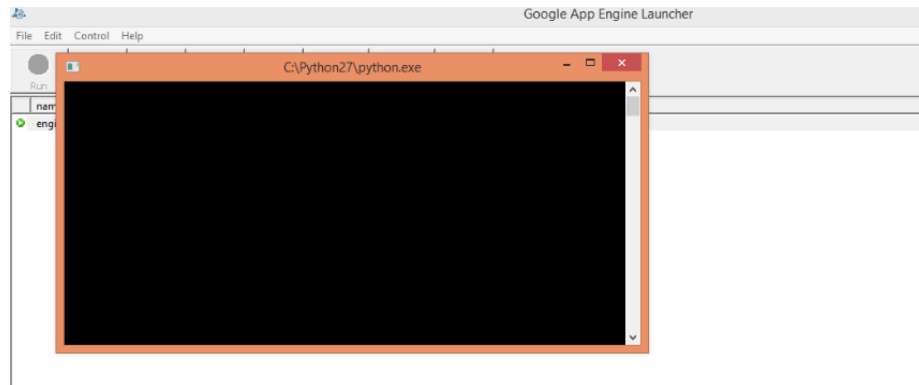
11. Create another application named engineapp, select the parent directory and set the port number as 8080.



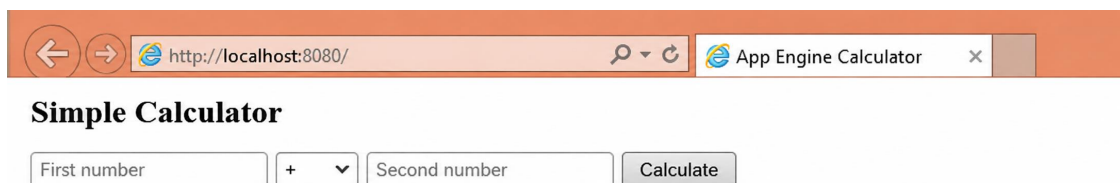
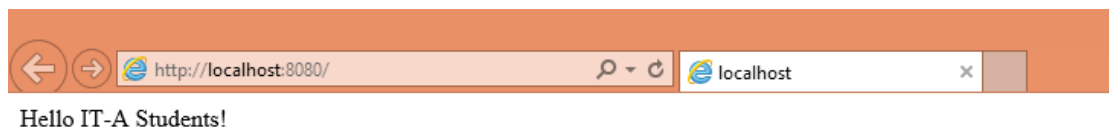
12. The application engineapp is created successfully and is displayed in the Google App Engine Launcher with port 8080.



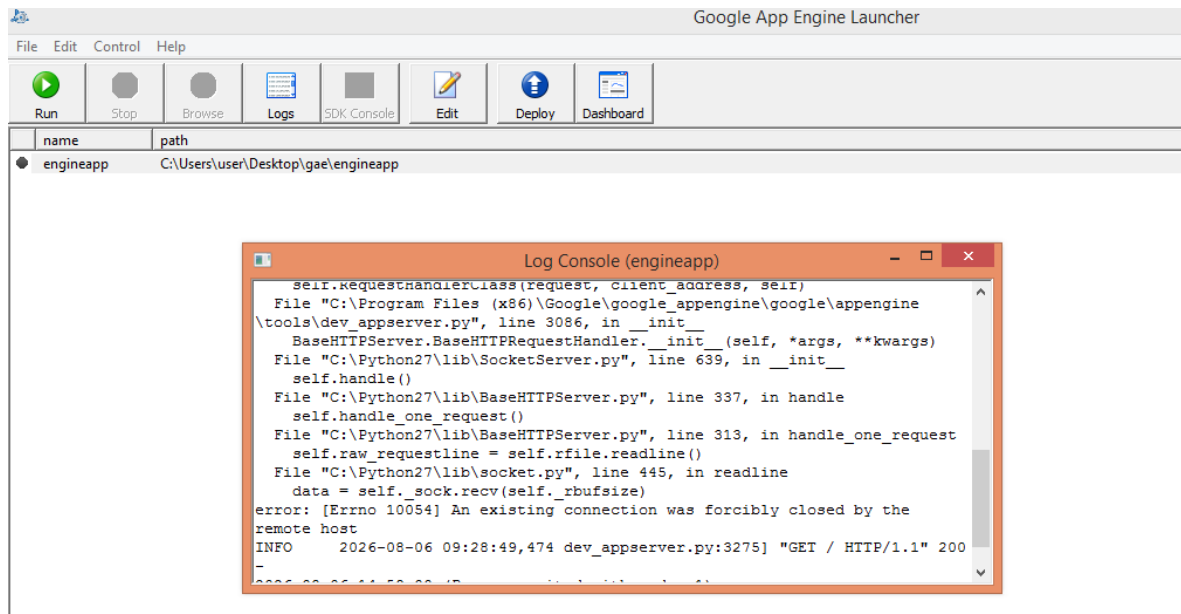
13. Click Run in Google App Engine Launcher. The Python development server starts and the application runs on the local machine.



14. Click Browse and open <http://localhost:8080/> in the web browser. The web application is executed successfully and the message “Hello IT-A Students!” and Simple Calculator is displayed in the browser.



15. Open the Log Console to view the application execution details and server request information.



RESULT: Thus, Google App Engine was successfully installed and configured, and simple web applications were created and executed successfully using Java/Python.

EXP NO: 9 A
DATE:

DOCKER COMMANDS

AIM: To install and configure Docker Desktop and study various Docker commands for creating, running, renaming, updating, listing, and managing Docker containers and images, and to execute Java commands interactively inside a Docker container.

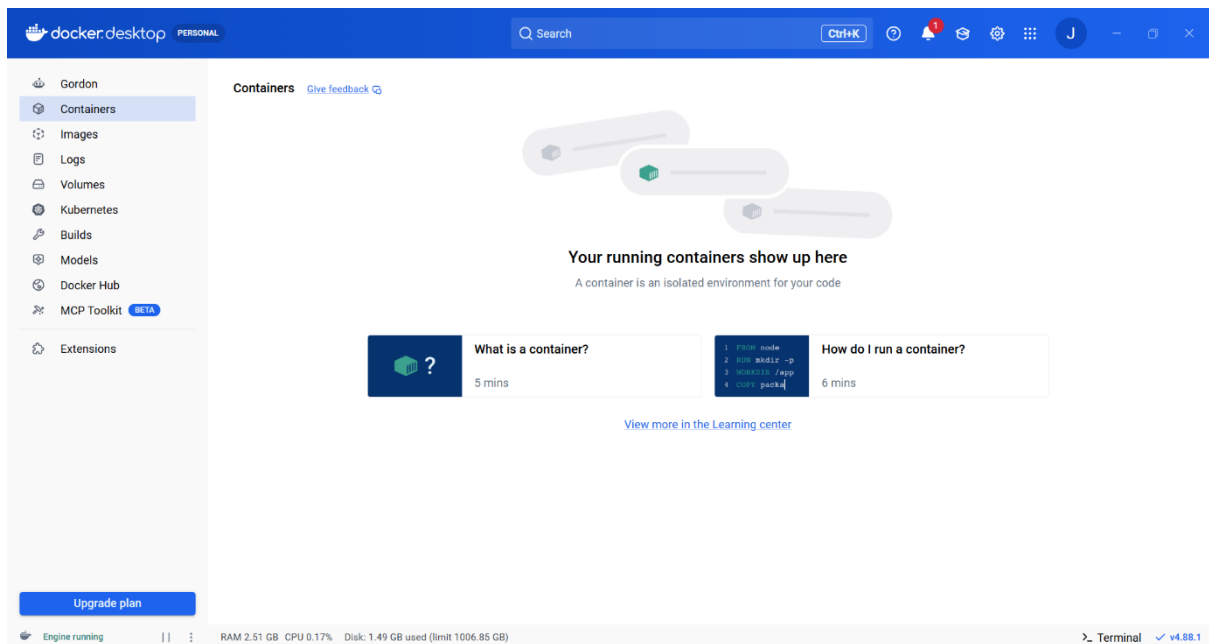
PROCEDURE:

1. Install and start Docker Desktop.
2. Verify that Docker is installed properly.
3. Create and run a hello-world Docker container.
4. List the available Docker images.
5. List running and stopped containers.
6. Rename one of the created containers.
7. Update the memory and swap limits of the container.
8. Start the renamed container normally and interactively.
9. Run a Java Docker image interactively.
10. Start JShell and execute a simple Java print statement.
11. Exit JShell.
12. Display the details of all containers.

COMMANDS:

- 1. Verify Docker installation:** `docker --version`
- 2. Create a hello-world container:** `docker create hello-world`
- 3. Run hello-world:** `docker run hello-world`
- 4. Display Docker images**
`docker image ls`
`docker image ls -a`
- 5. Display containers**
`docker container ls`
`docker container ls -a`
- 6. Display all containers:** `docker ps -a`
- 7. Rename a container:** `docker rename <old_container_name> <new_container_name>`
- 8. Update memory & swap:** `docker update --memory 512m --memory-swap 1g new_name`
- 9. Start the container:** `docker start intaa`
- 10. Start the container interactively:** `docker start -i intaa`
- 11. Run Java interactively:** `docker run -it eclipse-temurin:17-jdk jshell`
- 12. Execute Java code in JShell:** `System.out.println("HelloWorld");`
 Ctrl + D to terminate JShell
- 13. Final container status:** `docker ps -a`

OUTPUT:



```
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\senpa>docker --version
Docker version 29.7.2, build a7dcaa6

C:\Users\senpa>
```

```
C:\Users\senpa>docker create hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:5dd0d3e6e255913fc30f90b9f2b1d359cc2cbdb48090cc4b65f1676e203243cc
Status: Downloaded newer image for hello-world:latest
42e5bdde1d57056fbab48425363a9177a8a0f8800f7e9c613c55c3315032d8a3
```

```
C:\Users\senpa>docker run hello-world
```

```
Hello from Docker!
This message shows that your installation appears to be working correctly.
```

To generate this message, Docker took the following steps:

1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub. (amd64)
3. The Docker daemon created a new container from that image which runs the executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it to your terminal.

To try something more ambitious, you can run an Ubuntu container with:

```
$ docker run -it ubuntu bash
```

Share images, automate workflows, and more with a free Docker ID:

<https://hub.docker.com/>

For more examples and ideas, visit:

<https://docs.docker.com/get-started/>

```
C:\Users\senpa>docker image ls
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	5dd0d3e6e255	25.9kB	9.49kB	U

```
C:\Users\senpa>docker image ls -a
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	5dd0d3e6e255	25.9kB	9.49kB	U

```
C:\Users\senpa>docker container ls
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
804cca68430d	hello-world	"/hello"	About a minute ago	Exited (0)	About a minute ago	dreamy_black
42e5bdde1d57	hello-world	"/hello"	3 minutes ago	Created		sad_grothendieck

```
C:\Users\senpa>docker container ls -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
804cca68430d	hello-world	"/hello"	3 minutes ago	Exited (0)	3 minutes ago	dreamy_black
42e5bdde1d57	hello-world	"/hello"	5 minutes ago	Created		sad_grothendieck

```
C:\Users\senpa>docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
804cca68430d	hello-world	"/hello"	3 minutes ago	Exited (0)	3 minutes ago	dreamy_black
42e5bdde1d57	hello-world	"/hello"	5 minutes ago	Created		sad_grothendieck

```
C:\Users\senpa>docker rename dreamy_black johnee
```

```
C:\Users\senpa>docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
804cca68430d	hello-world	"/hello"	3 minutes ago	Exited (0)	3 minutes ago	johnee
42e5bdde1d57	hello-world	"/hello"	5 minutes ago	Created		sad_grothendieck

```
C:\Users\senpa>docker update --memory 512m --memory-swap 1g johnee
johnee

C:\Users\senpa>docker start johnee
johnee

C:\Users\senpa>docker start -i johnee

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

```
C:\Users\senpa>docker run -it eclipse-temurin:17-jdk jshell
Unable to find image 'eclipse-temurin:17-jdk' locally
17-jdk: Pulling from library/eclipse-temurin
06e9d71331fb: Pull complete
f3db1cd94078: Pull complete
ba5e48b2deeb: Pull complete
05ec4630b0c1: Pull complete
8b7049efea6c: Pull complete
9f9380f23d18: Pull complete
d96ac3614977: Download complete
7e08f63ee4cf: Download complete
Digest: sha256:a27c79d44326d5f689668df5fedfee487652066d2a91e172747056cc7fbee6fc
Status: Downloaded newer image for eclipse-temurin:17-jdk
Aug 28, 2026 7:42:01 PM java.util.prefs.FileSystemPreferences$1 run
INFO: Created user preferences directory.
| Welcome to JShell -- Version 17.0.20
| For an introduction type: /help intro

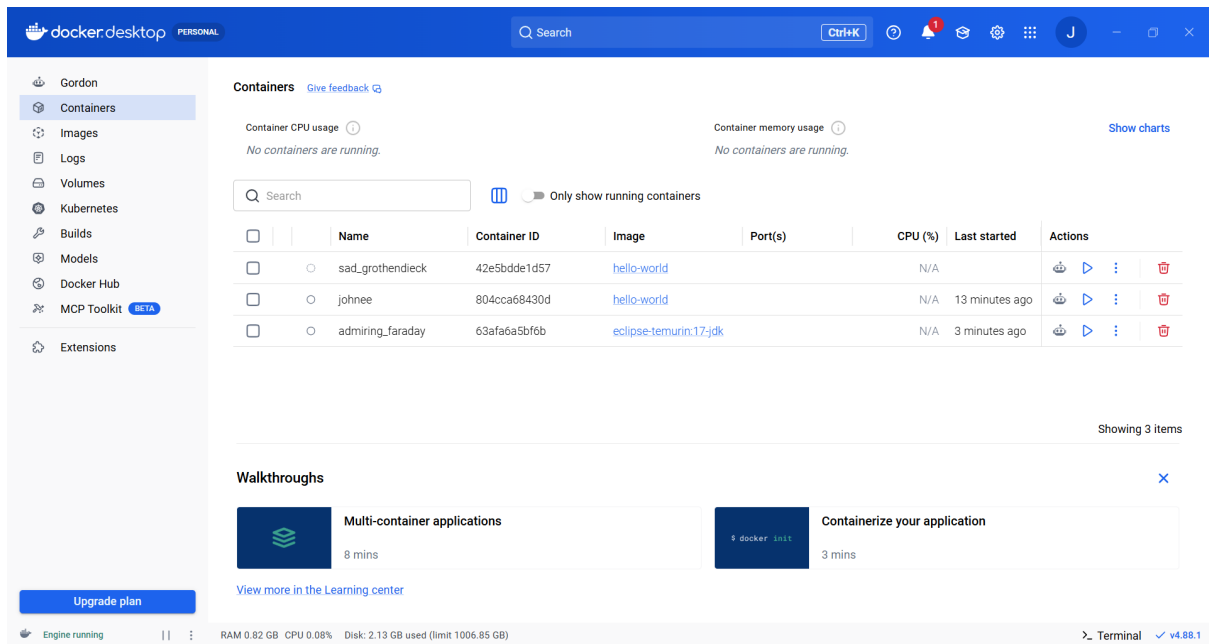
jshell> System.out.println("HelloWorld");
HelloWorld

jshell>

C:\Users\senpa>docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
63afa6a5bfb6	eclipse-temurin:17-jdk	"/__cacert_entrypoin..."	58 seconds ago	Exited (0) 11 seconds ago		admiring_faraday
804cca68430d	hello-world	"/hello"	16 minutes ago	Exited (0) 10 minutes ago		johnee
42e5bdde1d57	hello-world	"/hello"	18 minutes ago	Created		sad_grothendieck

```
C:\Users\senpa>
```



RESULT: Thus, Docker Desktop was successfully configured, and various Docker image and container management commands were executed. A Java container was also run interactively and a simple Java program was successfully executed using JShell.

EXP NO: 9 B
DATE:

**CREATE A NEW PROJECT AND BUILD AN IMAGE
 IN DOCKER**

AIM: To create a simple Python application, build a custom Docker image using a Dockerfile, create and run containers from the image, and execute a Python application continuously inside a Docker container.

PROCEDURE:

1. Create a directory named myapp.
2. Create a Python program named app.py.
3. Create a Dockerfile containing the required Docker instructions.
4. Rename Dockerfile.txt to Dockerfile.
5. Build a Docker image named mypythonapp.
6. Verify that the image has been created.
7. Create and run a container named mycontainer.
8. Check the status of the container.
9. Modify the Python program so that it runs continuously.
10. Rebuild the Docker image.
11. Create another container using the modified image.
12. Observe the continuously generated output.
13. Stop the running container and verify the container status.

COMMANDS & CODE:

1. Create project directory

```
cd C:\
mkdir myapp
cd myapp
```

2. Create app.py

```
notepad app.py
```

Enter:

```
print("Hello from my Docker container!")
```

3. Create Dockerfile

```
notepad Dockerfile.txt
```

Enter:

```
FROM python:3.9-slim
WORKDIR /app
COPY . /app
RUN pip install --no-cache-dir -r requirements.txt || true
CMD ["python", "app.py"]
```

4. Rename the Dockerfile

```
ren Dockerfile.txt Dockerfile
```

Verify the files:

```
dir
```

5. Build the Docker image

```
docker build -t mypythonapp .
```

6. Display images

docker images or: docker image ls

7. Run the container

```
docker run --name mycontainer mypythonapp
```

8. Check container status

```
docker ps -a
```

9. Modify app.py

```
notepad app.py
```

Replace the program with:

```
import time
```

```
while True:
```

```
    print("Hello, still running in Docker...", flush=True)
```

```
    time.sleep(5)
```

10. Rebuild the image

```
docker build -t mypythonapp .
```

11. Run another container

```
docker run --name mycontainer1 mypythonapp
```

12. Stop foreground execution

Press:Ctrl + C

If the container is still active, it can also be stopped using:

```
docker stop mycontainer1
```

13. Display container details

```
docker ps -a
```

OUTPUT:

A screenshot of a code editor window with multiple tabs: excel_test.py, app.py, App.js, Untitled, and app.py (active). The active tab shows the following code:

```
print("Hello from my Docker container!")
```

A screenshot of a code editor window with multiple tabs: excel_test.py, App.js, Untitled, app.py, and Dockerfile.txt (active). The active tab shows the following Dockerfile content:

```
FROM python:3.9-slim
WORKDIR /app
COPY . /app
RUN pip install --no-cache-dir -r requirements.txt || true
CMD ["python", "app.py"]
```

```
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\senpa>cd C:\
```

```
C:\>mkdir myapp
```

```
C:\>cd myapp
```

```
C:\myapp>notepad app.py
```

```
C:\myapp>notepad Dockerfile.txt
```

```
C:\myapp>ren Dockerfile.txt Dockerfile
```

```
C:\myapp>dir
Volume in drive C has no label.
Volume Serial Number is DC19-95AA
```

```
Directory of C:\myapp
```

```
29-08-2026  01:22    <DIR>          .
29-08-2026  01:16                  40 app.py
29-08-2026  01:21                 133 Dockerfile
                2 File(s)              173 bytes
                1 Dir(s)  348,106,440,704 bytes free
```

```
C:\myapp>docker build -t mypythonapp .
[+] Building 16.0s (10/10) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 172B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim 3.0s
=> [auth] library/python:pull token for registry-1.docker.io      0.0s
=> [internal] load .dockerignore                                    0.0s
=> => transferring context: 2B                                       0.0s
=> [1/4] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aablacdale13cf 9.2s
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aablacdale13cf 0.0s
=> => sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508 251B / 251B 0.3s
=> => sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 13.88MB / 13.88MB 6.1s
=> => sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 1.29MB / 1.29MB 2.3s
=> => sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 29.78MB / 29.78MB 7.7s
=> => extracting sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 0.7s
=> => extracting sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 0.1s
=> => extracting sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 0.5s
=> => extracting sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508 0.0s
=> [internal] load build context                                    0.1s
=> => transferring context: 247B                                       0.0s
=> [2/4] WORKDIR /app                                              0.2s
=> [3/4] COPY . /app                                              0.1s
=> [4/4] RUN pip install --no-cache-dir -r requirements.txt || true 2.2s
=> => exporting to image                                              0.9s
=> => exporting layers                                              0.5s
=> => exporting manifest sha256:acd4baa77c4aa94efacc95664f006faf4e9fae326e3865436c15b8892693994 0.0s
=> => exporting config sha256:3802506b98068546d1461ae1ba0daad00albeb32eca87a48360df54331e779e6 0.0s
=> => exporting attestation manifest sha256:5728640299f71e440c0a0c68ce2be62b86360529937d256b32961c6452ad2847 0.0s
=> => exporting manifest list sha256:2b420cb43fd0696732edf2f01c681f5908f9b3bc473bb3c7202d8cc008fed7c6 0.0s
=> => naming to docker.io/library/mypythonapp:latest              0.0s
=> => unpacking to docker.io/library/mypythonapp:latest            0.2s
```

```
C:\myapp>docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
eclipse-temurin:17-jdk	a27c79d44326	687MB	217MB	U
hello-world:latest	5dd0d3e6e255	25.9kB	9.49kB	U
mypythonapp:latest	2b420cb43fd0	194MB	47.5MB	

```
C:\myapp>docker run --name mycontainer mypythonapp
Hello from my Docker container!
```

The screenshot displays the Docker Desktop interface. At the top, a code editor shows a Python script named `app.py` with the following content:

```
import time

while True:
    print("Hello, still running in Docker...", flush=True)
    time.sleep(5)
```

Below the code editor, the terminal shows the execution of several Docker commands:

```
C:\myapp>docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
1e221130d25e   mypythonapp    "python app.py"         14 seconds ago    Exited (0) 13 seconds ago
63afa6a5bf6b   eclipse-temurin:17-jdk  "/_cacert_entrypoint..."  12 minutes ago    Exited (0) 11 minutes ago
804cca68430d   hello-world     "/hello"                28 minutes ago    Exited (0) 22 minutes ago
42e5b5dde1d57   hello-world     "/hello"                30 minutes ago    Created

C:\myapp>notepad app.py

C:\myapp>docker build -t mypythonapp .
[+] Building 4.0s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 172B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/4] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aablacdale13cf1731b1b
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aablacdale13cf1731b1b
=> [internal] load build context
=> => transferring context: 172B
=> CACHED [2/4] WORKDIR /app
=> [3/4] COPY . /app
=> [4/4] RUN pip install --no-cache-dir -r requirements.txt || true
=> => exporting image
=> => exporting manifest sha256:9d1b7731edf34714c6f7bae0f3b2270691ac97752d18c6f2718333db772b2b8b
=> => exporting config sha256:a1580af2572a23a0eee22781756c2ee5237b120907ea62f41b9df194859a6d8d
=> => exporting attestation manifest sha256:485e5d575e5f5edcbbe71a2290654eb98a938182af79f38de19f7366641e25dd2
=> => exporting manifest list sha256:4acec686c4e998324d85bca89b776b67c49862deaa28e9c59d40731a8d50eb5
=> => naming to docker.io/library/mypythonapp:latest
=> => unpacking to docker.io/library/mypythonapp:latest

C:\myapp>docker run --name mycontainer1 mypythonapp
Hello, still running in Docker...
Hello, still running in Docker...
Hello, still running in Docker...
Hello, still running in Docker...
Traceback (most recent call last):
  File "/app/app.py", line 5, in <module>
    time.sleep(5)
KeyboardInterrupt

C:\myapp>docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
022bc86e882a   mypythonapp    "python app.py"         26 seconds ago    Exited (130) 8 seconds ago
1e221130d25e   mypythonapp    "python app.py"         4 minutes ago    Exited (0) 4 minutes ago
63afa6a5bf6b   eclipse-temurin:17-jdk  "/_cacert_entrypoint..."  17 minutes ago    Exited (0) 16 minutes ago
804cca68430d   hello-world     "/hello"                32 minutes ago    Exited (0) 26 minutes ago
42e5b5dde1d57   hello-world     "/hello"                34 minutes ago    Created

C:\myapp>
```

The bottom part of the screenshot shows the Docker Desktop GUI. The left sidebar contains navigation options: Gordon, Containers, Images, Logs, Volumes, Kubernetes, Builds, Models, Docker Hub, MCP Toolkit, and Extensions. The main area displays the 'Containers' tab, showing a list of containers. The 'Containers' tab is selected, and the 'Only show running containers' filter is applied. The list shows five containers, with 'mycontainer1' (ID: 022bc86e882a) highlighted as the selected container. The status bar at the bottom indicates 'Engine running' and shows system resources: RAM 1.64 GB, CPU 0.00%, and Disk 2.33 GB used (limit 1006.85 GB).

RESULT: Thus, a simple Python project was successfully created, a custom Docker image was built using a Dockerfile, and containers were successfully created and executed from the image. The Python application was also modified and executed continuously inside a Docker container.

EXP NO: 9 C
DATE:

SIMPLE WORKING FLASK APPLICATION IN DOCKER

AIM: To create a simple Flask web application, build a Docker image for the application, run it inside a Docker container with port mapping, and access the application through a web browser.

PROCEDURE:

1. Create a directory named myflaskapp.
2. Create the Flask application in app.py.
3. Create requirements.txt containing the Flask dependency.
4. Create a Dockerfile.
5. Rename Dockerfile.txt to Dockerfile.
6. Build a Docker image for the Flask application.
7. Run the container in detached mode with port mapping.
8. Verify that the container is running.
9. Access the application through localhost:5000.
10. Check the container logs.
11. Stop the Flask container.
12. Remove the container.

COMMANDS & CODE:

1. Create project directory

```
cd C:\
mkdir myflaskapp
cd myflaskapp
```

2. Create app.py

```
notepad app.py
```

Enter:

```
from flask import Flask

app = Flask(__name__)

@app.route("/")
def home():
    return "Hello from Flask running inside Docker!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

3. Create requirements.txt

```
notepad requirements.txt
```

Enter:

```
flask
```

4. Create Dockerfile

```
notepad Dockerfile.txt
```

Enter:

```
FROM python:3.9-slim
```

```

WORKDIR /app
COPY requirements.txt .
RUN pip install -r requirements.txt
COPY . .
EXPOSE 5000
CMD ["python", "app.py"]

```

5. Rename Dockerfile

```
ren Dockerfile.txt Dockerfile
```

Verify:

```
dir
```

6. Build Docker image

```
docker build -t myflaskapp .
```

7. Verify image

```
docker images
```

8. Run Flask container

```
docker run -d --name myflaskcontainer -p 5000:5000 myflaskapp
```

9. Check running containers

```
docker ps
```

10. Access Flask applicationOpen a browser and enter: <http://localhost:5000>**11. View Flask logs**

```
docker logs myflaskcontainer
```

12. Stop the container

```
docker stop myflaskcontainer
```

13. Check stopped container

```
docker ps -a
```

14. Remove container

```
docker rm myflaskcontainer
```

OUTPUT:

```

from flask import Flask

app = Flask(__name__)

@app.route("/")
def home():
    return "Hello from Flask running inside Docker!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)

```

```

flask

```

```

excel_test.py | App.js | Untitled | app.py | app.py | requirements | Docker | X | +
File Edit View H1 |≡| B I S ↻ |
FROM python:3.9-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install -r requirements.txt
COPY . .
EXPOSE 5000
CMD ["python", "app.py"]

```

docker desktop PERSONAL Q Search Ctrl+K 🔔 📧 ⚙️ 🌐 J – 🗖 ✕

Containers [Give feedback](#)

Container CPU usage 📉 **0.04% / 1200% (12 CPUs available)** Container memory usage 📉 **19.69MB / 7.42GB** [Show charts](#)

Q Search 📋 🔍 Only show running containers

☐	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	sad_grothendieck	42e5bde1d57	hello-world		0%		🔍 ▶ ⋮ 🗑
☐	johnee	804cca68430d	hello-world		0%	40 minutes ago	🔍 ▶ ⋮ 🗑
☐	admiring_faraday	63afa6a5bf6b	eclipse-temurin:17-jdk		0%	30 minutes ago	🔍 ▶ ⋮ 🗑
☐	mycontainer	1e221130d25e	mypythontapp		0%	17 minutes ago	🔍 ▶ ⋮ 🗑
☐	mycontainer1	022bc86e882a	mypythontapp		0%	13 minutes ago	🔍 ▶ ⋮ 🗑
☐	myflaskcontainer	06e3c0005969	myflaskapp	5000:5000	0.03%	2 minutes ago	🔍 ▶ ⋮ 🗑

[Upgrade plan](#) Showing 6 items

🔧 Engine running | RAM 1.51 GB CPU 5.44% Disk: 2.37 GB used (limit 1006.85 GB) 🔗 Terminal v4.28.1



```

Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\senpa>cd C:\
C:\>mkdir myflaskapp
C:\>cd myflaskapp
C:\myflaskapp>notepad app.py
C:\myflaskapp>notepad requirements.txt
C:\myflaskapp>notepad Dockerfile.txt
C:\myflaskapp>ren Dockerfile.txt Dockerfile
C:\myflaskapp>dir
Volume in drive C has no label.
Volume Serial Number is DC19-95AA

Directory of C:\myflaskapp

29-08-2026 01:38 <DIR> .
29-08-2026 01:35      204 app.py
29-08-2026 01:37      145 Dockerfile
29-08-2026 01:35         5 requirements.txt
                3 File(s)      354 bytes
                1 Dir(s)  347,777,155,072 bytes free

C:\myflaskapp>docker build -t myflaskapp .
[+] Building 9.7s (11/11) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                0.1s
=> == transferring dockerfile: 180B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim  1.8s
=> [auth] library/python:pull token for registry-1.docker.io       0.0s
=> [internal] load .dockerignore                                    0.1s
=> == transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aablacdale13cf1731b1b  0.1s
=> == resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aablacdale13cf1731b1b  0.1s
=> [internal] load build context                                    0.1s

```

```
=> => unpacking to docker.io/library/myflaskapp:latest 0.5s

C:\myflaskapp>docker images

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
eclipse-temurin:17-jdk               a27c79d44326                     687MB       217MB         U
hello-world:latest                   5dd0d3e0e255                     25.9kB      9.49kB        U
myflaskapp:latest                    803f3fdb05ae                     203MB       49.7MB        U
mypythonapp:latest                   4acec686c4e9                     194MB       47.5MB        U

C:\myflaskapp>docker run -d --name myflaskcontainer -p 5000:5000 myflaskapp
06e3c000596942a591ac54cfe8c54f589a2ec06cb7f94437402257d12e58a26

C:\myflaskapp>docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
06e3c0005969   myflaskapp "python app.py"         19 seconds ago Up 19 seconds  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp  myflaskcontainer

C:\myflaskapp>docker logs myflaskcontainer
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [28/Aug/2026 20:10:23] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [28/Aug/2026 20:10:23] "GET /favicon.ico HTTP/1.1" 404 -

C:\myflaskapp>docker stop myflaskcontainer
myflaskcontainer

C:\myflaskapp>docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
06e3c0005969   myflaskapp "python app.py"         3 minutes ago Exited (137) 8 seconds ago          myflaskcontainer
022bc86e882a   mypythonapp "python app.py"         14 minutes ago Exited (130) 14 minutes ago          mycontainer1
1e221130d25e   2b420cb43fd0 "python app.py"         18 minutes ago Exited (0) 18 minutes ago          mycontainer
63afa6a5bf6b   eclipse-temurin:17-jdk "/_cacert_entrypoint..." 31 minutes ago Exited (0) 30 minutes ago          admiring_faraday
804cca68430d   hello-world "/hello"                46 minutes ago Exited (0) 40 minutes ago          johnee
42e5bdd1d57    hello-world "/hello"                48 minutes ago Created                                     sad_grothendieck
```

```
C:\myflaskapp>docker rm myflaskcontainer
myflaskcontainer

C:\myflaskapp>docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
022bc86e882a   mypythonapp "python app.py"         14 minutes ago Exited (130) 14 minutes ago          mycontainer1
1e221130d25e   2b420cb43fd0 "python app.py"         19 minutes ago Exited (0) 19 minutes ago          mycontainer
63afa6a5bf6b   eclipse-temurin:17-jdk "/_cacert_entrypoint..." 31 minutes ago Exited (0) 30 minutes ago          admiring_faraday
804cca68430d   hello-world "/hello"                46 minutes ago Exited (0) 41 minutes ago          johnee
42e5bdd1d57    hello-world "/hello"                48 minutes ago Created                                     sad_grothendieck
```

The screenshot shows the Docker Desktop application window. On the left is a sidebar with navigation options: Containers (selected), Images, Logs, Volumes, Kubernetes, Builds, Models, Docker Hub, MCP Toolkit (beta), and Extensions. The main area is titled 'Containers' and shows 'No containers are running.' for both CPU and memory usage. Below this is a table of containers. At the bottom, there's a status bar showing 'Engine running', RAM usage (1.50 GB), CPU usage (0.08%), and disk usage (2.37 GB used of 1006.85 GB limit). A terminal icon is visible in the bottom right corner.

	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	sad_grothendieck	42e5bdd1d57	hello-world		N/A		
☐	johnee	804cca68430d	hello-world		N/A	44 minutes ago	
☐	admiring_faraday	63afa6a5bf6b	eclipse-temurin:17-jdk		N/A	34 minutes ago	
☐	mycontainer	1e221130d25e	mypythonapp		N/A	21 minutes ago	
☐	mycontainer1	022bc86e882a	mypythonapp		N/A	17 minutes ago	

Showing 5 items

RESULT: Thus, a simple Flask web application was successfully containerized using Docker. The Docker image and container were created successfully, port 5000 was mapped between the host and container, and the Flask application was successfully accessed through the web browser using localhost:5000.

EXP NO: 9 D
DATE:

MANAGING DOCKER CONTAINERS AND IMAGES

AIM: To study Docker container and image management and demonstrate persistent data storage using Docker volumes by creating a Flask application that stores data inside a named Docker volume.

PROCEDURE:

1. Create a project directory named myvolapp.
2. Create a Flask application that writes timestamps into /data/log.txt.
3. Create the required dependency file.
4. Create a Dockerfile.
5. Build an image named myvolapp.
6. Run a container with port 5006 and mount the named volume mydata to /data.
7. Verify that the volume has been created.
8. Access the Flask application through the browser.
9. Refresh the application multiple times to write entries to log.txt.
10. Use an Alpine container to access and display the contents of the Docker volume.
11. Stop and remove the original application container.
12. Verify that the data remains available in the Docker volume.
13. Create a new container using the same volume.
14. Verify that old and new data are available.
15. Verify the created images and containers using Docker Desktop.

COMMANDS & CODE:

1. Create project directory

```
cd C:\
mkdir myvolapp
cd myvolapp
```

2. Create app.py

```
notepad app.py
```

Enter:

```
from flask import Flask
from datetime import datetime
import os

app = Flask(__name__)

@app.route("/")
def home():
    os.makedirs("/data", exist_ok=True)

    with open("/data/log.txt", "a") as f:
        f.write("Visited at: " + str(datetime.now()) + "\n")

    return "Data written to Docker volume successfully!"
```

```
if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5006)
```

3. Create requirements file

```
notepad requirements.txt
```

Enter:

```
flask
```

4. Create Dockerfile

```
notepad Dockerfile.txt
```

Enter:

```
FROM python:3.9-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install -r requirements.txt
COPY . .
EXPOSE 5006
CMD ["python", "app.py"]
```

Rename:

```
ren Dockerfile.txt Dockerfile
```

Verify:

```
dir
```

5. Build Docker image

```
docker build -t myvolapp .
```

6. Verify image

```
docker images
```

7. Run container with named volume

```
docker run -d --name myvolcontainer -p 5006:5006 -v mydata:/data myvolapp
```

8. List Docker volumes

```
docker volume ls
```

9. Check running containers

```
docker ps
```

10. Access Flask applicationOpen: <http://localhost:5006>

Refresh the page multiple times.

11. Read the volume using Alpine

```
docker run --rm -it -v mydata:/data alpine cat /data/log.txt
```

12. Stop the container

```
docker stop myvolcontainer
```

13. Remove the container

```
docker rm myvolcontainer
```

Verify:

```
docker ps -a
```

14. Verify volume persistence

```
docker run --rm -it -v mydata:/data alpine cat /data/log.txt
```

15. Create a new container using the same volume

```
docker run -d --name myvolcontainer2 -p 5006:5006 -v mydata:/data myvolapp
```

Check:

```
docker ps
```

16. Access the application again<http://localhost:5006>

Refresh the page and again check the log file:

```
docker run --rm -it -v mydata:/data alpine cat /data/log.txt
```

17. Verify using Docker Desktop

Open: Docker Desktop → Images

Verify:

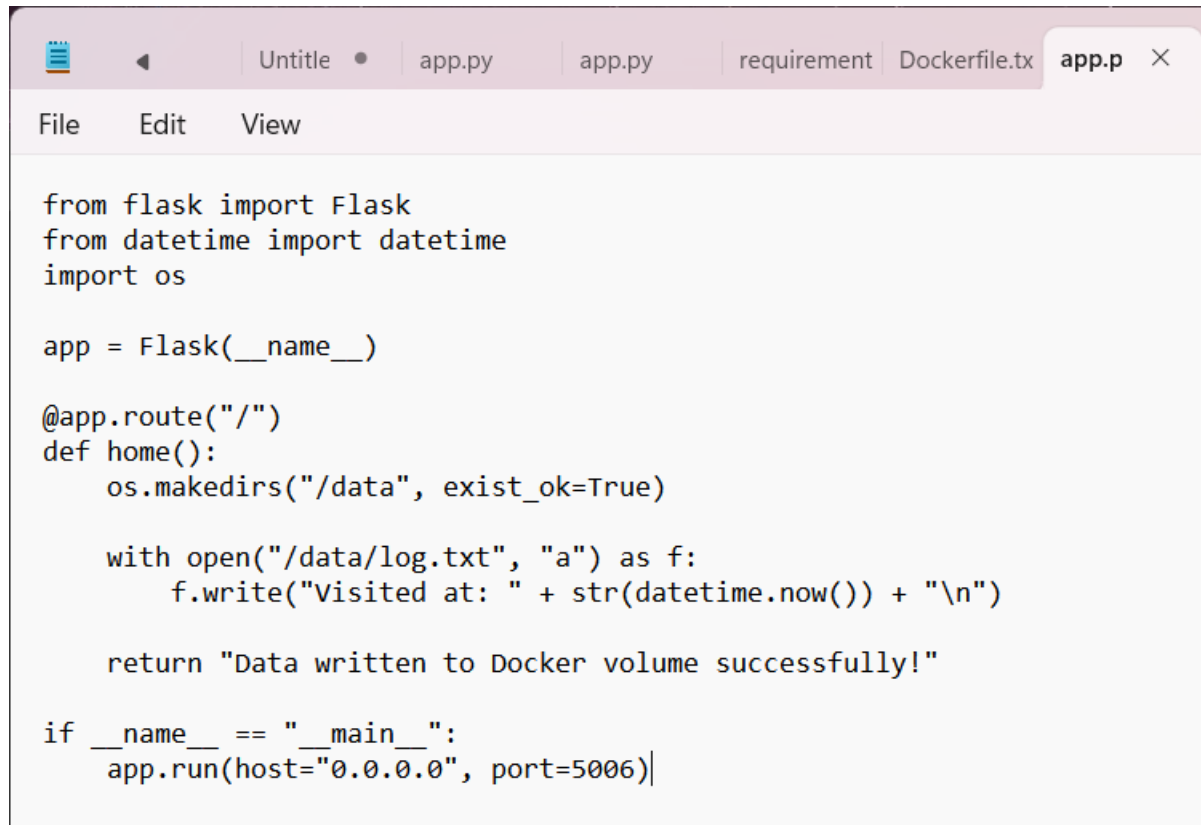
myvolapp

alpine

Then open: Docker Desktop → Containers

Verify that myvolcontainer2 is running with port mapping: 5006:5006

OUTPUT:



```

from flask import Flask
from datetime import datetime
import os

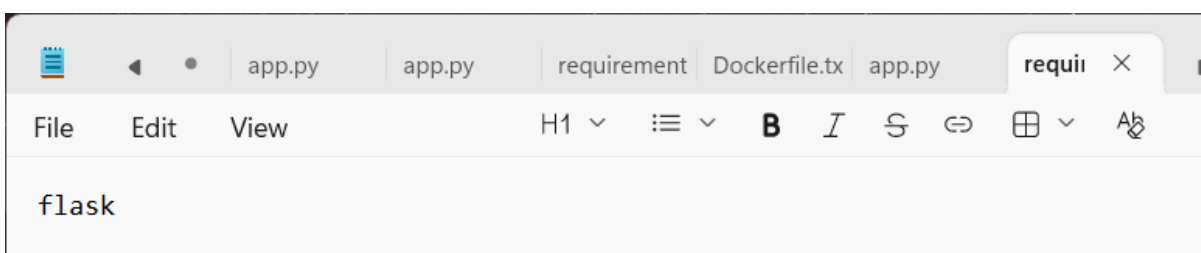
app = Flask(__name__)

@app.route("/")
def home():
    os.makedirs("/data", exist_ok=True)

    with open("/data/log.txt", "a") as f:
        f.write("Visited at: " + str(datetime.now()) + "\n")

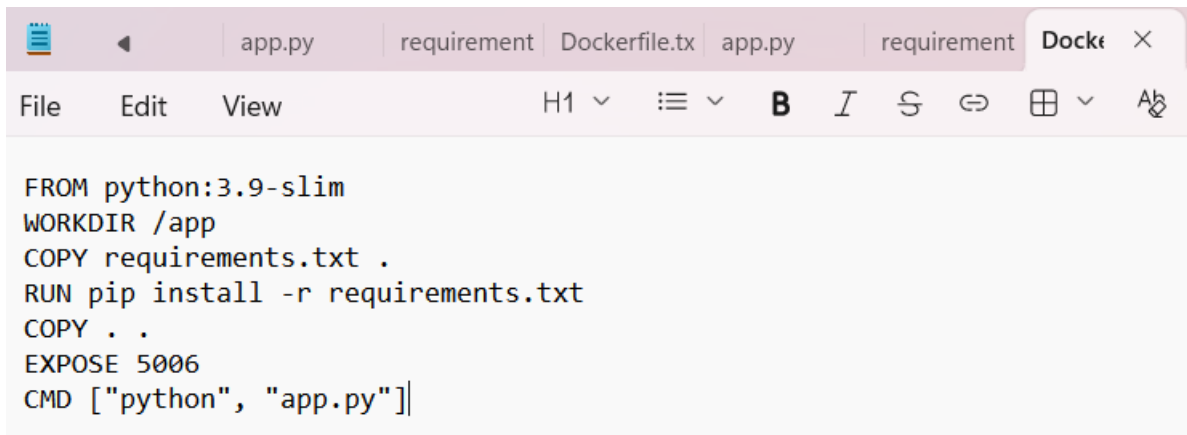
    return "Data written to Docker volume successfully!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5006)
  
```



```

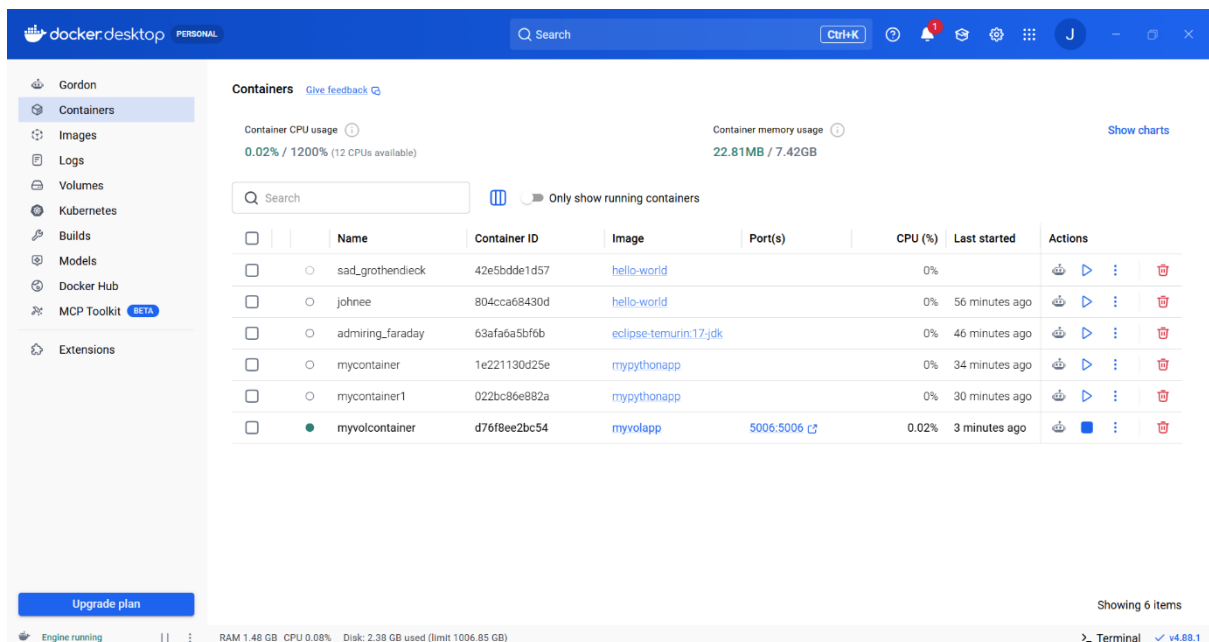
flask
  
```



```

FROM python:3.9-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install -r requirements.txt
COPY . .
EXPOSE 5006
CMD ["python", "app.py"]

```

Containers Give feedback

Container CPU usage: 0.02% / 1200% (12 CPUs available) Container memory usage: 22.81MB / 7.42GB

Search Only show running containers

	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	sad_grothendieck	42e5bdde1d57	hello-world		0%		
☐	johniee	804cca68430d	hello-world		0%	56 minutes ago	
☐	admiring_faraday	63afa6a5bf6b	eclipse-temurin:17-jdk		0%	46 minutes ago	
☐	mycontainer	1e221130d25e	mypythonapp		0%	34 minutes ago	
☐	mycontainer1	022bc86e882a	mypythonapp		0%	30 minutes ago	
☒	myvolcontainer	d76f8ee2bc54	myvolapp	5006:5006	0.02%	3 minutes ago	

Showing 6 items

Engine running RAM 1.48 GB CPU 0.08% Disk: 2.38 GB used (limit 1006.85 GB)

Terminal v4.88.1

The screenshot shows the Docker Desktop interface with the 'Containers' tab selected. The left sidebar contains navigation options: Gordon, Containers, Images, Logs, Volumes, Kubernetes, Builds, Models, Docker Hub, MCP Toolkit (BETA), and Extensions. The main area displays container statistics: Container CPU usage (0.02% / 1200% (12 CPUs available)) and Container memory usage (20.44MB / 7.42GB). A search bar and a toggle for 'Only show running containers' are present. Below is a table of containers:

☐	☐	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	☐	sad_grothendieck	42e5bde1d57	hello-world		0%		
☐	☐	johnnee	804cca68430d	hello-world		0%	1 hour ago	
☐	☐	admiring_faraday	63afa6a5bf6b	eclipse-temurin:17-jdk		0%	51 minutes ago	
☐	☐	mycontainer	1e221130d25e	mypythonapp		0%	39 minutes ago	
☐	☐	mycontainer1	022bc86e882a	mypythonapp		0%	34 minutes ago	
☐	☒	myvolcontainer2	403d83cfa14d	myvolapp	5006:5006	0.02%	1 minute ago	

Showing 6 items

Engine running | RAM 1.49 GB CPU 0.42% Disk: 2.38 GB used (limit 1006.85 GB) | Terminal v4.88.1

The screenshot shows the Docker Desktop interface with the 'Images' tab selected. The left sidebar is the same as the previous screenshot. The main area displays image statistics: 0 Bytes / 0 Bytes in use, 6 images. A search bar and sort icons are present. Below is a table of images:

☐	Name	Tag	Image ID	Created	Size	Actions
☐	hello-world	latest	5dd0d3e6e255	5 months ago	25.87 KB	
☐	eclipse-temurin	17-jdk	a27c79d44326	7 days ago	686.69 MB	
☐	mypythonapp	latest	4acec686c4e9	36 minutes ago	193.7 MB	
☐	myflaskapp	latest	803f3fdb05ae	25 minutes ago	202.6 MB	
☐	myvolapp	latest	4fa729fc214a	10 minutes ago	202.6 MB	
☐	alpine	latest	28bd5fe8b56d	2 months ago	13 MB	

Showing 6 items

Engine running | RAM 0.88 GB CPU 0.17% Disk: 2.38 GB used (limit 1006.85 GB) | Terminal v4.88.1

```

Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\senpa>cd C:\

C:\>mkdir myvolapp

C:\>cd myvolapp

C:\myvolapp>notepad app.py

C:\myvolapp>notepad requirements.txt

C:\myvolapp>notepad Dockerfile.txt

C:\myvolapp>ren Dockerfile.txt Dockerfile

C:\myvolapp>dir
Volume in drive C has no label.
Volume Serial Number is DC19-95AA

Directory of C:\myvolapp

29-08-2026  01:53    <DIR>          .
29-08-2026  01:51                400 app.py
29-08-2026  01:52                145 Dockerfile
29-08-2026  01:51                5 requirements.txt
               3 File(s)                550 bytes
               1 Dir(s)  347,378,794,496 bytes free

C:\myvolapp>docker build -t myvolapp .
[+] Building 2.1s (11/11) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile               0.1s
=> => transferring dockerfile: 184B                               0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim 1.6s
=> [auth] library/python:pull token for registry-1.docker.io     0.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                     0.0s
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acd1e13cf 0.0s
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acd1e13cf 0.0s
=> [internal] load build context                                  0.0s

```

```

C:\myvolapp>docker images
 Info →  In Use


| IMAGE                  | ID           | DISK USAGE | CONTENT SIZE | EXTRA                                                                               |
|------------------------|--------------|------------|--------------|-------------------------------------------------------------------------------------|
| eclipse-temurin:17-jdk | a27c79d44326 | 687MB      | 217MB        |  |
| hello-world:latest     | 5dd0d3e6e255 | 25.9KB     | 9.49KB       |  |
| myFlaskapp:latest      | 803f3fdb05ae | 203MB      | 49.7MB       |                                                                                     |
| mypythonapp:latest     | 4acec686c4e9 | 194MB      | 47.5MB       |  |
| myvolapp:latest        | 4fa729fc214a | 203MB      | 49.7MB       |                                                                                     |



C:\myvolapp>docker run -d --name myvolcontainer -p 5006:5006 -v mydata:/data myvolapp
d76f8ee2bc54197863b74dd88d64a6fe661dd21bf6d0cb46b1b2424a34d76a7d

C:\myvolapp>docker volume ls


| DRIVER | VOLUME NAME |
|--------|-------------|
| local  | mydata      |



C:\myvolapp>docker ps


| CONTAINER ID | IMAGE    | COMMAND         | CREATED        | STATUS        | PORTS                                       | NAMES          |
|--------------|----------|-----------------|----------------|---------------|---------------------------------------------|----------------|
| d76f8ee2bc54 | myvolapp | "python app.py" | 18 seconds ago | Up 16 seconds | 0.0.0.0:5006->5006/tcp, [::]:5006->5006/tcp | myvolcontainer |



C:\myvolapp>docker run --rm -it -v mydata:/data alpine cat /data/log.txt
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
55afa1ecc21d: Pull complete
56dceff11b33: Download complete
f5124fb579e2: Download complete
Digest: sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f8b
Status: Downloaded newer image for alpine:latest
Visited at: 2026-08-28 20:25:28.232612
Visited at: 2026-08-28 20:25:59.442494
Visited at: 2026-08-28 20:26:00.341195
Visited at: 2026-08-28 20:26:01.003277
Visited at: 2026-08-28 20:26:01.639162
Visited at: 2026-08-28 20:26:02.253512

C:\myvolapp>docker ps


| CONTAINER ID | IMAGE    | COMMAND         | CREATED       | STATUS       | PORTS                                       | NAMES          |
|--------------|----------|-----------------|---------------|--------------|---------------------------------------------|----------------|
| d76f8ee2bc54 | myvolapp | "python app.py" | 2 minutes ago | Up 2 minutes | 0.0.0.0:5006->5006/tcp, [::]:5006->5006/tcp | myvolcontainer |



C:\myvolapp>docker stop myvolcontainer
myvolcontainer

```

```

C:\myvolapp>docker rm myvolcontainer
myvolcontainer

C:\myvolapp>docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
022bc86e882a   mypythonapp   "python app.py"         31 minutes ago Exited (130) 30 minutes ago
1e221130d25e   2b420cb43fd0   "python app.py"         35 minutes ago Exited (0) 35 minutes ago
63afa6a5bf6b   eclipse-temurin:17-jdk  "/__cacert_entrypoint..." 47 minutes ago Exited (0) 47 minutes ago
804cca68430d   hello-world    "/hello"                 About an hour ago Exited (0) 57 minutes ago
42e5bde1d57    hello-world    "/hello"                 About an hour ago Created
sad_grothendieck

C:\myvolapp>docker run --rm -it -v mydata:/data alpine cat /data/log.txt
Visited at: 2026-08-28 20:25:28.232612
Visited at: 2026-08-28 20:25:59.442494
Visited at: 2026-08-28 20:26:00.341195
Visited at: 2026-08-28 20:26:01.003277
Visited at: 2026-08-28 20:26:01.639162
Visited at: 2026-08-28 20:26:02.253512

C:\myvolapp>docker run -d --name myvolcontainer2 -p 5006:5006 -v mydata:/data myvolapp
403d83cfa14dd91c7c179c8e2cb3d89858ead7c88f7c9a3533c003faf7bdee72

C:\myvolapp>docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
403d83cfa14d   myvolapp      "python app.py"         6 seconds ago Up 5 seconds   0.0.0.0:5006->5006/tcp, [::]:5006->5006/tcp   myvolcontainer2

C:\myvolapp>docker run --rm -it -v mydata:/data alpine cat /data/log.txt
Visited at: 2026-08-28 20:25:28.232612
Visited at: 2026-08-28 20:25:59.442494
Visited at: 2026-08-28 20:26:00.341195
Visited at: 2026-08-28 20:26:01.003277
Visited at: 2026-08-28 20:26:01.639162
Visited at: 2026-08-28 20:26:02.253512
Visited at: 2026-08-28 20:31:51.775204
Visited at: 2026-08-28 20:31:52.753162
Visited at: 2026-08-28 20:31:55.357458
Visited at: 2026-08-28 20:32:02.802126

```

RESULT: Thus, Docker containers, images, and volumes were successfully created and managed. A named Docker volume mydata was successfully mounted to the container, and the stored data remained persistent even after the original container was stopped and removed. The persistent data was successfully accessed using an Alpine container and reused by a newly created container.

EXP NO: 10 MINI PROJECT – DIGITAL CLASSROOM QUIZ APPLICATION
DATE:

AIM: To design, develop, and deploy a cloud-based Digital Classroom Quiz Application that enables teachers to create multiple-choice quizzes and allows students to attempt them online with instant automated evaluation.

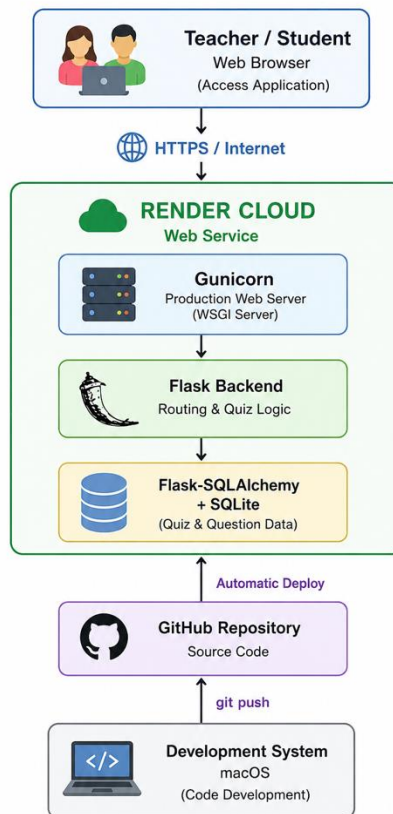
PROBLEM STATEMENT: Traditional classroom assessments often require teachers to manually prepare questions, distribute quizzes, evaluate responses, and calculate scores. This process can be time-consuming and inefficient, especially when quick assessment of student understanding is required. The proposed Digital Classroom Quiz Application provides a web-based solution where teachers can create quizzes and add multiple-choice questions, while students can access the available quizzes, submit their answers, and receive their scores instantly. The application is hosted as a cloud web service so that it can be accessed through the Internet without requiring the application to run on the user's local computer.

OBJECTIVES:

- To develop a web-based classroom quiz application using Flask.
- To provide a separate interface for teachers and students.
- To allow teachers to create quizzes dynamically.
- To allow teachers to add multiple-choice questions and specify correct answers.
- To enable students to select and attempt available quizzes.
- To automatically evaluate submitted answers.
- To display quiz scores instantly.
- To maintain quiz and question data using SQLite and SQLAlchemy.
- To use Git and GitHub for source-code version control.
- To deploy the application as a cloud-based Web Service using Render.
- To provide a publicly accessible URL for the deployed application.

SOFTWARE AND CLOUD REQUIREMENTS:

Component	Technology Used
Operating System	macOS
Programming Language	Python
Web Framework	Flask
ORM	Flask-SQLAlchemy
Database	SQLite
Frontend	HTML5, CSS3
UI Framework	Bootstrap 5
Icons	Bootstrap Icons
Production Server	Gunicorn
Version Control	Git
Source Repository	GitHub
Cloud Platform	Render
Cloud Service	Web Service
Deployment Model	PaaS-style managed cloud deployment
Browser	Google Chrome

SYSTEM ARCHITECTURE:**MODULES:****Module 1 – Teacher and Quiz Management**

The Teacher module provides an interface through which teachers can view existing quizzes and create new quizzes. Each quiz is assigned a unique identifier and stored in the database.

Module 2 – Question Management

This module allows teachers to add multiple-choice questions to a selected quiz. Each question contains four options and one correct answer. Questions are associated with their corresponding quiz using the quiz ID.

Module 3 – Student Quiz Interface

The Student module displays all currently available quizzes. A student can select a quiz and answer the associated multiple-choice questions through an interactive web interface.

Module 4 – Automated Evaluation

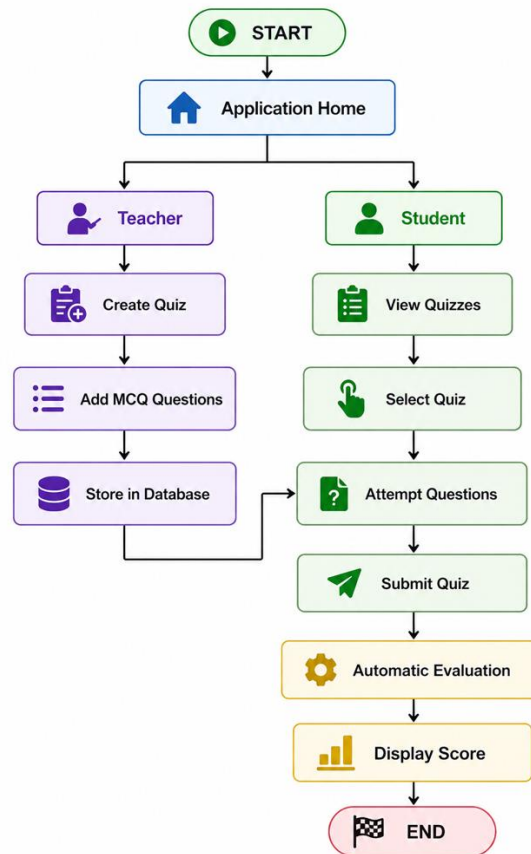
When a student submits a quiz, the Flask backend retrieves each selected answer and compares it with the correct answer stored in the database. The total number of correct responses is calculated automatically and the result is displayed immediately.

Module 5 – Cloud Deployment

The application source code is maintained using Git and GitHub. The GitHub repository is connected to a Render Web Service. Render builds the application, installs the required Python

packages, starts the application using Gunicorn, and provides a public URL through which the application can be accessed over the Internet.

APPLICATION WORKFLOW:



PROJECT DIRECTORY STRUCTURE:

```

digital-classroom-quiz/
├── app.py
├── requirements.txt
├── templates/
│   ├── base.html
│   ├── index.html
│   ├── teacher.html
│   ├── create_quiz.html
│   ├── add_question.html
│   ├── student.html
│   ├── quiz.html
│   └── result.html
├── static/
│   └── style.css

```

PROCEDURE:**Step 1 – Create the Project Directory**

Open Terminal and execute:

```
mkdir digital-classroom-quiz
cd digital-classroom-quiz
```

Step 2 – Create and Activate Python Virtual Environment

```
python3 -m venv venv
source venv/bin/activate
```

The virtual environment isolates the project dependencies from other Python installations.

Step 3 – Install Required Packages

```
pip install flask flask-sqlalchemy gunicorn
```

The packages perform the following functions:

- **Flask** – Web application framework.
- **Flask-SQLAlchemy** – Database ORM integration.
- **Gunicorn** – Production WSGI server used for cloud deployment.

Step 4 – Create the Project Directories

```
mkdir templates
mkdir static
```

IMPLEMENTATION:**1. Flask Backend – app.py**

```
from flask import Flask, render_template, request, redirect, url_for
from flask_sqlalchemy import SQLAlchemy
```

```
app = Flask(__name__)
```

```
app.config["SQLALCHEMY_DATABASE_URI"] = "sqlite:///quiz.db"
app.config["SQLALCHEMY_TRACK_MODIFICATIONS"] = False
```

```
db = SQLAlchemy(app)
```

```
class Quiz(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    title = db.Column(db.String(100), nullable=False)
```

```
class Question(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    quiz_id = db.Column(
        db.Integer,
        db.ForeignKey("quiz.id"),
        nullable=False
    )
```

```

question = db.Column(db.String(300), nullable=False)

option1 = db.Column(db.String(100), nullable=False)
option2 = db.Column(db.String(100), nullable=False)
option3 = db.Column(db.String(100), nullable=False)
option4 = db.Column(db.String(100), nullable=False)

correct_answer = db.Column(
    db.String(100),
    nullable=False
)

@app.route("/")
def home():
    return render_template("index.html")

@app.route("/teacher")
def teacher():

    quizzes = Quiz.query.all()

    return render_template(
        "teacher.html",
        quizzes=quizzes
    )

@app.route("/create_quiz", methods=["GET", "POST"])
def create_quiz():

    if request.method == "POST":

        title = request.form["title"]

        new_quiz = Quiz(title=title)

        db.session.add(new_quiz)
        db.session.commit()

        return redirect(
            url_for(
                "add_question",
                quiz_id=new_quiz.id
            )
        )

    return render_template("create_quiz.html")

```

```

@app.route(
    "/add_question/<int:quiz_id>",
    methods=["GET", "POST"]
)
def add_question(quiz_id):

    quiz = Quiz.query.get_or_404(quiz_id)

    if request.method == "POST":

        question = request.form["question"]
        option1 = request.form["option1"]
        option2 = request.form["option2"]
        option3 = request.form["option3"]
        option4 = request.form["option4"]

        correct_answer = request.form[
            "correct_answer"
        ]

        new_question = Question(
            quiz_id=quiz_id,
            question=question,
            option1=option1,
            option2=option2,
            option3=option3,
            option4=option4,
            correct_answer=correct_answer
        )

        db.session.add(new_question)
        db.session.commit()

        return redirect(
            url_for(
                "add_question",
                quiz_id=quiz_id
            )
        )

    questions = Question.query.filter_by(
        quiz_id=quiz_id
    ).all()

    return render_template(
        "add_question.html",
        quiz=quiz,
        questions=questions
    )

```

```
@app.route("/student")
def student():

    quizzes = Quiz.query.all()

    return render_template(
        "student.html",
        quizzes=quizzes
    )

@app.route(
    "/quiz/<int:quiz_id>",
    methods=["GET", "POST"]
)
def attempt_quiz(quiz_id):

    quiz = Quiz.query.get_or_404(quiz_id)

    questions = Question.query.filter_by(
        quiz_id=quiz_id
    ).all()

    if request.method == "POST":

        score = 0

        for q in questions:

            answer = request.form.get(
                f"question_{q.id}"
            )

            if answer == q.correct_answer:
                score += 1

        return render_template(
            "result.html",
            quiz=quiz,
            score=score,
            total=len(questions)
        )

    return render_template(
        "quiz.html",
        quiz=quiz,
        questions=questions
    )
```

```
with app.app_context():
    db.create_all()
```

```
if __name__ == "__main__":
    app.run(debug=True)
```

FRONTEND IMPLEMENTATION:

2. Common Layout – templates/base.html

```
<!DOCTYPE html>
<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport"
    content="width=device-width, initial-scale=1.0">

<title>
{% block title %}
Digital Classroom Quiz
{% endblock %}
</title>

<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">

<link
rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.3/font/bootstrap-icons.min.css">

<link
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swap"
rel="stylesheet">

<link
rel="stylesheet"
href="{{ url_for('static', filename='style.css') }}">

</head>

<body>

<nav class="navbar navbar-expand-lg navbar-dark custom-navbar">
```

```
<div class="container">

<a class="navbar-brand d-flex align-items-center gap-2"
  href="/">

<div class="brand-icon">
<i class="bi bi-mortarboard-fill"></i>
</div>

<div>

<div class="brand-title">
Digital Classroom
</div>

<div class="brand-subtitle">
Quiz App
</div>

</div>

</a>

<button
class="navbar-toggler"
type="button"
data-bs-toggle="collapse"
data-bs-target="#navbarNav">

<span class="navbar-toggler-icon"></span>

</button>

<div class="collapse navbar-collapse"
  id="navbarNav">

<ul class="navbar-nav ms-auto
  align-items-lg-center gap-lg-3">

<li class="nav-item">

<a class="nav-link" href="/">

<i class="bi bi-house-door me-1"></i>
Home

</a>
```

```
</li>
```

```
<li class="nav-item">
```

```
<a class="nav-link" href="/teacher">
```

```
<i class="bi bi-person-workspace me-1"></i>  
Teacher
```

```
</a>
```

```
</li>
```

```
<li class="nav-item">
```

```
<a class="nav-link" href="/student">
```

```
<i class="bi bi-person me-1"></i>  
Student
```

```
</a>
```

```
</li>
```

```
</ul>
```

```
</div>
```

```
</div>
```

```
</nav>
```

```
{% block content %}  
{% endblock %}
```

```
<footer class="app-footer">
```

```
<div class="container d-flex flex-column  
flex-md-row justify-content-between  
align-items-center gap-2">
```

```
<span>
```

```
© 2026 Digital Classroom Quiz App
```

```
</span>
```

```

<span>
Empowering learning through technology
</span>

</div>

</footer>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js">
</script>

</body>

</html>

```

3. Home Page – templates/index.html

```

{% extends "base.html" %}

{% block title %}
Digital Classroom Quiz
{% endblock %}

{% block content %}

<section class="hero-section">

<div class="container">

<div class="row align-items-center g-5">

<div class="col-lg-7">

<div class="hero-badge">

<i class="bi bi-stars"></i>

Welcome to Digital Learning

</div>

<h1 class="hero-title">

Learn. Quiz.<br>

Grow.

<span class="gradient-text">

```

Succeed.

</h1>

<p class="hero-description">

Create, share and attempt interactive quizzes.
Experience instant evaluation and make classroom
learning smarter and more engaging.

</p>

<div class="row g-3 mt-3">

<div class="col-md-6">

<div class="role-card">

<div class="role-icon teacher-icon">

<i class="bi bi-mortarboard-fill"></i>

</div>

<h4>I'm a Teacher</h4>

<p>

Create quizzes, add multiple-choice questions
and manage classroom assessments effortlessly.

</p>

<a href="/teacher"
class="btn btn-purple w-100">

Go to Teacher Panel

<i class="bi bi-arrow-right ms-2"></i>

</div>

</div>

```
<div class="col-md-6">

<div class="role-card">

<div class="role-icon student-icon">

<i class="bi bi-person-fill"></i>

</div>

<h4>I'm a Student</h4>

<p>

Choose available quizzes, answer questions
and instantly view your performance.

</p>

<a href="/student"
  class="btn btn-green w-100">

Go to Student Panel

<i class="bi bi-arrow-right ms-2"></i>

</a>

</div>

</div>

</div>

</div>

<div class="col-lg-5">

<div class="hero-visual">

<div class="hero-graduation-icon">

<i class="bi bi-mortarboard-fill"></i>

</div>

<h3 class="fw-bold mt-4">
Smart Learning
</h3>
```

```
<p class="text-secondary">
```

Interactive quizzes with instant results

```
</p>
```

```
<div class="mock-result">
```

```
<small class="text-secondary">
```

Sample Quiz Result

```
</small>
```

```
<div class="result-circle">
```

```
<span>85%</span>
```

```
</div>
```

```
<h5 class="fw-bold">
```

Great Job! 🎉

```
</h5>
```

```
<p class="mb-0 text-secondary">
```

17 / 20 Correct

```
</p>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
<section class="features-section">
```

```
<div class="container">
```

```
<h2 class="section-title">
```

```
<i class="bi bi-bar-chart-fill me-2"></i>
```

```
App Features
```

```
</h2>
```

```
<div class="row g-4">
```

```
<div class="col-md-6 col-lg-3">
```

```
<div class="feature-card">
```

```
<div class="feature-icon purple-bg">
```

```
<i class="bi bi-file-earmark-plus"></i>
```

```
</div>
```

```
<h5>Easy Quiz Creation</h5>
```

```
<p>
```

```
Teachers can create quizzes and add MCQ  
questions within seconds.
```

```
</p>
```

```
</div>
```

```
</div>
```

```
<div class="col-md-6 col-lg-3">
```

```
<div class="feature-card">
```

```
<div class="feature-icon green-bg">
```

```
<i class="bi bi-lightning-charge-fill"></i>
```

```
</div>
```

```
<h5>Instant Evaluation</h5>
```

```
<p>
```

Students receive their score immediately after submitting a quiz.

</p>

</div>

</div>

<div class="col-md-6 col-lg-3">

<div class="feature-card">

<div class="feature-icon blue-bg">

<i class="bi bi-graph-up-arrow"></i>

</div>

<h5>Performance Insights</h5>

<p>

Quiz scores help students understand their performance instantly.

</p>

</div>

</div>

<div class="col-md-6 col-lg-3">

<div class="feature-card">

<div class="feature-icon orange-bg">

<i class="bi bi-cloud-check-fill"></i>

</div>

<h5>Cloud Accessible</h5>

<p>

The application is hosted online and can be accessed through any modern web browser.

```
</p>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
{% endblock %}
```

4. Teacher Dashboard – templates/teacher.html

```
{% extends "base.html" %}
```

```
{% block title %}
```

```
Teacher Dashboard
```

```
{% endblock %}
```

```
{% block content %}
```

```
<section class="page-section">
```

```
<div class="container">
```

```
<div class="d-flex flex-column flex-md-row
    justify-content-between
    align-items-md-center mb-4">
```

```
<div>
```

```
<h1 class="page-heading">
```



```
Teacher Dashboard
```

```
</h1>
```

```
<p class="page-subtitle">
```

```
Create quizzes and manage questions
for your students.
```

```
</p>
```

```
</div>
```

```
<a href="/create_quiz"
    class="btn btn-purple">
```

```
<i class="bi bi-plus-lg me-2"></i>
```

```
Create New Quiz
```

```
</a>
```

```
</div>
```

```
<div class="app-card">
```

```
<h4 class="fw-bold mb-4">
```

```
Available Quizzes
```

```
</h4>
```

```
{% if quizzes %}
```

```
{% for quiz in quizzes %}
```

```
<div class="quiz-list-card">
```

```
<div>
```

```
<div class="fw-bold">
```

```
{{ quiz.title }}
```

```
</div>
```

```
<small class="text-secondary">
```

```
Quiz ID: {{ quiz.id }}
```

```
</small>
```

```
</div>
```

```
<a href="/add_question/{{ quiz.id }}"
  class="btn btn-outline-primary">
```

```
Add Questions
```

```
</a>
```

```
</div>
```

```
{% endfor %}
```

```

{% else %}

<div class="text-center py-5">

<h5 class="fw-bold">

No quizzes created yet

</h5>

<p class="text-secondary">

Create your first quiz to get started.

</p>

</div>

{% endif %}

</div>

</div>

</section>

{% endblock %}

```

5. Create Quiz – templates/create_quiz.html

```

{% extends "base.html" %}

{% block title %}
Create Quiz
{% endblock %}

{% block content %}

<section class="page-section">

<div class="container">

<div class="row justify-content-center">

<div class="col-lg-7">

<h1 class="page-heading">
Create a New Quiz
</h1>

```

```
<p class="page-subtitle">
```

Give your quiz a meaningful title to get started.

```
</p>
```

```
<div class="app-card">
```

```
<form method="POST">
```

```
<div class="mb-4">
```

```
<label class="form-label">
```

Quiz Title

```
</label>
```

```
<input  
type="text"  
name="title"  
class="form-control"  
placeholder="Example: Cloud Computing Quiz"  
required>
```

```
</div>
```

```
<button type="submit"  
class="btn btn-purple">
```

Create Quiz

```
</button>
```

```
<a href="/teacher"  
class="btn btn-outline-secondary">
```

Cancel

```
</a>
```

```
</form>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
{% endblock %}
```

6. Add Question – templates/add_question.html

```
{% extends "base.html" %}
```

```
{% block title %}
```

```
Add Questions
```

```
{% endblock %}
```

```
{% block content %}
```

```
<section class="page-section">
```

```
<div class="container">
```

```
<h1 class="page-heading">
```

```
Add Questions
```

```
</h1>
```

```
<p class="page-subtitle">
```

```
Quiz: <strong>{{ quiz.title }}</strong>
```

```
</p>
```

```
<div class="row g-4">
```

```
<div class="col-lg-7">
```

```
<div class="app-card">
```

```
<h4 class="fw-bold mb-4">
```

```
New Question
```

```
</h4>
```

```
<form method="POST">
```

```
<div class="mb-3">
```

```
<label class="form-label">
```

```
Question
```

```
</label>
```

```
<input  
type="text"  
name="question"  
class="form-control"  
required>
```

```
</div>
```

```
<div class="row g-3">
```

```
<div class="col-md-6">
```

```
<label class="form-label">  
Option 1  
</label>
```

```
<input type="text"  
      name="option1"  
      class="form-control"  
      required>
```

```
</div>
```

```
<div class="col-md-6">
```

```
<label class="form-label">  
Option 2  
</label>
```

```
<input type="text"  
      name="option2"  
      class="form-control"  
      required>
```

```
</div>
```

```
<div class="col-md-6">
```

```
<label class="form-label">  
Option 3  
</label>
```

```
<input type="text"  
      name="option3"  
      class="form-control"  
      required>
```

```
</div>
```

```
<div class="col-md-6">
```

```
<label class="form-label">
```

Option 4

```
</label>
```

```
<input type="text"
```

```
  name="option4"
```

```
  class="form-control"
```

```
  required>
```

```
</div>
```

```
</div>
```

```
<div class="mt-3 mb-4">
```

```
<label class="form-label">
```

Correct Answer

```
</label>
```

```
<select name="correct_answer"
```

```
  class="form-select"
```

```
  required>
```

```
<option value="option1">Option 1</option>
```

```
<option value="option2">Option 2</option>
```

```
<option value="option3">Option 3</option>
```

```
<option value="option4">Option 4</option>
```

```
</select>
```

```
</div>
```

```
<button type="submit"
```

```
  class="btn btn-green">
```

Add Question

```
</button>
```

```
</form>
```

```
</div>
```

```
</div>
```

```
<div class="col-lg-5">
```

```
<div class="app-card">
```

```
<h4 class="fw-bold">
```

Questions Added

```
<span class="badge text-bg-primary">
```

```
{{ questions|length }}
```

```
</span>
```

```
</h4>
```

```
<hr>
```

```
{% if questions %}
```

```
{% for q in questions %}
```

```
<div class="quiz-list-card">
```

```
<div>
```

```
<strong>
```

```
{{ loop.index }}.
```

```
</strong>
```

```
{{ q.question }}
```

```
</div>
```

```
</div>
```

```
{% endfor %}
```

```
{% else %}
```

```
<p class="text-secondary">
```

No questions have been added yet.

</p>

{% endif %}

<a href="/teacher"
class="btn btn-purple w-100 mt-3">

Finish Quiz

</div>

</div>

</div>

</div>

</section>

{% endblock %}

7. Student Dashboard – templates/student.html

{% extends "base.html" %}

{% block title %}

Student Dashboard

{% endblock %}

{% block content %}

<section class="page-section">

<div class="container">

<h1 class="page-heading">



Student Dashboard

</h1>

<p class="page-subtitle">

Choose a quiz below and start testing your knowledge.

</p>

```
<div class="app-card">
```

```
<h4 class="fw-bold mb-4">
```

```
Available Quizzes
```

```
</h4>
```

```
{% if quizzes %}
```

```
{% for quiz in quizzes %}
```

```
<div class="quiz-list-card">
```

```
<div>
```

```
<div class="fw-bold">
```

```
{{ quiz.title }}
```

```
</div>
```

```
<small class="text-secondary">
```

```
Ready to attempt
```

```
</small>
```

```
</div>
```

```
<a href="/quiz/{{ quiz.id }}"
  class="btn btn-green">
```

```
Attempt Quiz
```

```
</a>
```

```
</div>
```

```
{% endfor %}
```

```
{% else %}
```

```
<div class="text-center py-5">
```

```
<h5 class="fw-bold">
```

No quizzes available

</h5>

<p class="text-secondary">

Please check again later.

</p>

</div>

{% endif %}

</div>

</div>

</section>

{% endblock %}

8. Quiz Attempt Page – templates/quiz.html

{% extends "base.html" %}

{% block title %}

{{ quiz.title }}

{% endblock %}

{% block content %}

<section class="page-section">

<div class="container">

<div class="row justify-content-center">

<div class="col-lg-9">

<h1 class="page-heading">

 {{ quiz.title }}

</h1>

<p class="page-subtitle">

Choose one answer for each question
and submit your quiz.

```
</p>
```

```
<form method="POST">
```

```
{% for q in questions %}
```

```
<div class="question-card">
```

```
<h5 class="fw-bold">
```

```
<span class="question-number">
```

```
{{ loop.index }}
```

```
</span>
```

```
{{ q.question }}
```

```
</h5>
```

```
<label class="option-box">
```

```
<input
class="form-check-input me-2"
type="radio"
name="question_{{ q.id }}"
value="option1"
required>
```

```
{{ q.option1 }}
```

```
</label>
```

```
<label class="option-box">
```

```
<input
class="form-check-input me-2"
type="radio"
name="question_{{ q.id }}"
value="option2">
```

```
{{ q.option2 }}
```

```
</label>
```

```
<label class="option-box">
```

```
<input
class="form-check-input me-2"
type="radio"
name="question_{{ q.id }}"
value="option3">

{{ q.option3 }}

</label>

<label class="option-box">

<input
class="form-check-input me-2"
type="radio"
name="question_{{ q.id }}"
value="option4">

{{ q.option4 }}

</label>

</div>

{% endfor %}

<button
type="submit"
class="btn btn-green btn-lg w-100">

Submit Quiz

</button>

</form>

</div>

</div>

</div>

</section>

{% endblock %}
```

9. Result Page – templates/result.html

```
{% extends "base.html" %}
```

```
{% block title %}
```

```
Quiz Result
```

```
{% endblock %}
```

```
{% block content %}
```

```
<section class="page-section">
```

```
<div class="container">
```

```
<div class="score-box">
```

```
<div style="font-size:55px;">
```



```
</div>
```

```
<h2 class="fw-bold">
```

```
Quiz Completed!
```

```
</h2>
```

```
<p class="text-secondary">
```

```
{{ quiz.title }}
```

```
</p>
```

```
<div class="score-circle">
```

```
<h1>
```

```
{{ score }}/{{ total }}
```

```
</h1>
```

```
<span>
```

```
Correct Answers
```

```
</span>
```

```
</div>
```

```
{% if total > 0 %}
```

```
<h3 class="fw-bold">
```

```
{{ ((score / total) * 100)|round(1) }}%
```

```
</h3>
```

```
{% if (score / total) >= 0.8 %}
```

```
<p class="text-success fw-semibold">
```

Excellent work! Keep it up 🚀

```
</p>
```

```
{% elif (score / total) >= 0.5 %}
```

```
<p class="text-primary fw-semibold">
```

Good job! You're making progress 👍

```
</p>
```

```
{% else %}
```

```
<p class="text-warning fw-semibold">
```

Keep practicing — you'll improve! 📖

```
</p>
```

```
{% endif %}
```

```
{% endif %}
```

```
<a href="/student"
  class="btn btn-green">
```

Attempt Another Quiz

```
</a>
```

```
<a href="/"
  class="btn btn-purple">
```

Home

```
</a>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
{% endblock %}
```

CSS IMPLEMENTATION:

10. static/style.css

Since the complete CSS is comparatively long, the important styling components implemented are:

```
:root {
  --primary: #5b3df5;
  --green: #11a861;
  --navy: #071936;
  --text-dark: #0d1735;
  --text-muted: #65708b;
  --background: #f8f9ff;
}

* {
  box-sizing: border-box;
}

body {
  margin: 0;
  font-family: "Inter", sans-serif;
  background: var(--background);
  color: var(--text-dark);
}

.custom-navbar {
  background: linear-gradient(
    100deg,
    #06142f,
    #071936,
    #081c3e
  );
  padding: 12px 0;
}

.hero-section {
  min-height: 520px;
  display: flex;
  align-items: center;
```

```
background:
  radial-gradient(
    circle at 90% 75%,
    rgba(173,255,222,.62),
    transparent 30%
  ),
  linear-gradient(
    120deg,
    #fbf8ff,
    #eef2ff 55%,
    #ecfcff
  );
}

.hero-title {
  font-size: clamp(42px, 5vw, 68px);
  font-weight: 800;
}

.gradient-text {
  background: linear-gradient(
    90deg,
    #6647f7,
    #7559ff
  );

  -webkit-background-clip: text;
  -webkit-text-fill-color: transparent;
}

.role-card,
.feature-card,
.app-card,
.question-card {
  background: white;
  border-radius: 18px;
  padding: 24px;

  box-shadow:
    0 12px 35px
    rgba(34,40,80,.09);
}

.btn-purple {
  border: none;
  color: white;

  background:
    linear-gradient(
      90deg,
```

```
        #5130eb,  
        #7648ff  
    );  
  
    border-radius: 10px;  
    padding: 11px 20px;  
    font-weight: 600;  
}  
  
.btn-green {  
    border: none;  
    color: white;  
  
    background:  
        linear-gradient(  
            90deg,  
            #08a85e,  
            #12be71  
        );  
  
    border-radius: 10px;  
    padding: 11px 20px;  
    font-weight: 600;  
}  
  
.page-section {  
    min-height: calc(100vh - 155px);  
    padding: 60px 0;  
  
    background:  
        linear-gradient(  
            135deg,  
            #f9f8ff,  
            #f1f6ff  
        );  
}  
  
.quiz-list-card {  
    background: white;  
    border: 1px solid #e8e9f2;  
    border-radius: 14px;  
    padding: 18px 20px;  
    margin-bottom: 14px;  
  
    display: flex;  
    justify-content: space-between;  
    align-items: center;  
}  
  
.option-box {
```

```

display: block;
border: 1px solid #e4e6ef;
border-radius: 10px;
padding: 12px 15px;
margin-top: 10px;
cursor: pointer;
}

.option-box:hover {
  background: #f6f4ff;
  border-color: #cfc5ff;
}

.score-box {
  text-align: center;
  background: white;
  border-radius: 22px;
  padding: 42px;
  max-width: 550px;
  margin: auto;
}

.app-footer {
  padding: 22px 0;
  color: rgba(255,255,255,.78);

  background:
    linear-gradient(
      90deg,
      #06132e,
      #071a39
    );
}

```

DEPENDENCY FILE:

11. requirements.txt

```

Flask
Flask-SQLAlchemy
gunicorn

```

The dependency file can also be generated using:
 pip freeze > requirements.txt

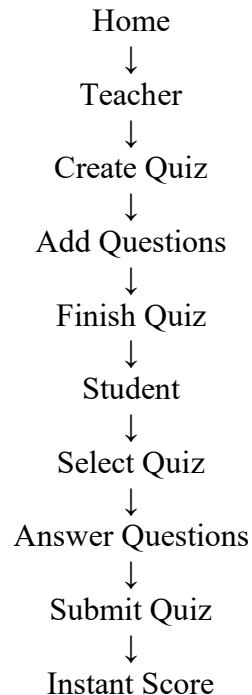
LOCAL EXECUTION:

The Flask application was first tested locally before cloud deployment.

Run: python3 app.py

The application becomes available at: <http://127.0.0.1:5000>

The following workflow was tested:



VERSION CONTROL USING GIT AND GITHUB:

Step 1 – Initialize Git Repository

```
git init
```

Step 2 – Add Project Files

```
git add .
```

Step 3 – Commit the Project

```
git commit -m "Working Digital Classroom Quiz App"
```

Step 4 – Configure Main Branch

```
git branch -M main
```

Step 5 – Connect GitHub Repository

```
git remote add origin https://github.com/JohnBellar/digital-classroom-quiz.git
```

Step 6 – Push Source Code

```
git push -u origin main
```

The complete application source code is therefore stored remotely in the GitHub repository.

CLOUD DEPLOYMENT USING RENDER

Step 1 – Create Render Web Service

The Render dashboard was opened and a new: **Web Service** was created.

The GitHub repository containing the Digital Classroom Quiz Application was connected to Render.

Step 2 – Configure the Cloud Service

The following deployment configuration was used:

Service Type : Web Service

Language : Python

Branch : main

Build Command:

```
pip install -r requirements.txt
```

Start Command:

```
gunicorn app:app
```

The free cloud instance was selected for the academic prototype.

Step 3 – Build the Application

During deployment, Render cloned the GitHub repository and executed:

```
pip install -r requirements.txt
```

The required packages were successfully installed.

The deployment log displayed:

Successfully installed Flask

Flask-SQLAlchemy

gunicorn

sqlalchemy

werkzeug

followed by:

Build successful

Step 4 – Start Flask Using Gunicorn

Render executed:

```
gunicorn app:app
```

The production server started successfully and listened on the port supplied by the Render environment.

The deployment log displayed:

Starting gunicorn

Listening at: http://0.0.0.0:10000

Using worker: sync

Booting worker

Step 5 – Verify Cloud Deployment

The application responded successfully to Render's health check: HEAD / HTTP/1.1 200 and the deployment status displayed: Your service is live

Thus, the Flask application was successfully deployed as a cloud-based Web Service.

PUBLIC CLOUD APPLICATION:

The deployed Digital Classroom Quiz Application is available at:

<https://digital-classroom-quiz.onrender.com>

The application can be accessed from any Internet-connected device using a web browser, independent of the local development machine.

UPDATING THE CLOUD APPLICATION

After improving the user interface, the updated source code was committed using:

```
git add .
```

```
git commit -m "Upgrade Digital Classroom Quiz UI"
```

and pushed using:

git push

Render detected the updated GitHub commit and redeployed the application.

When required, a manual deployment was also performed using:

Render Dashboard

↓
Manual Deploy

↓
Deploy Latest Commit

or:

Manual Deploy

↓
Clear Build Cache & Deploy

The updated application was subsequently available through the same public URL.

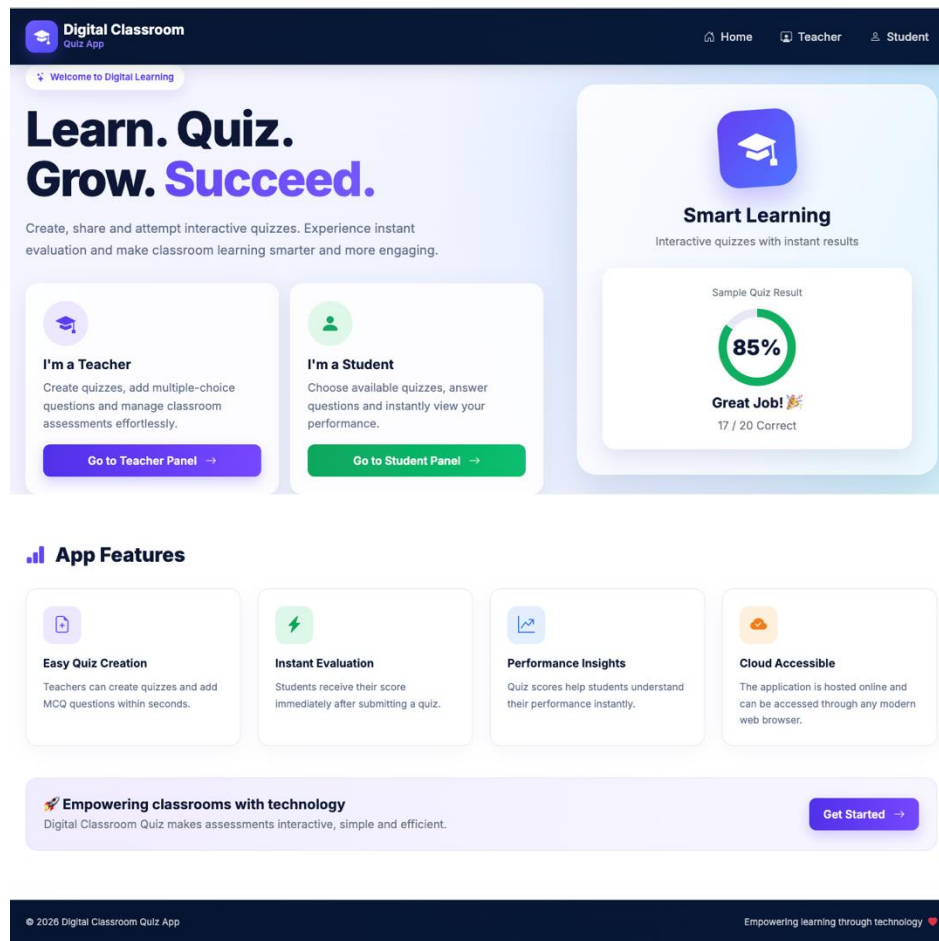
OUTPUT:

The screenshot displays the Render Dashboard interface for a workspace named 'John's workspace'. The application 'digital-classroom-quiz' is selected. The left sidebar shows navigation options: Dashboard, digital-classroom-quiz, Events, Settings, MONITOR (Logs, Metrics), and MANAGE (Environment, Shell). The main panel shows the 'Application logs' for August 29, 2026. The logs include the following entries:

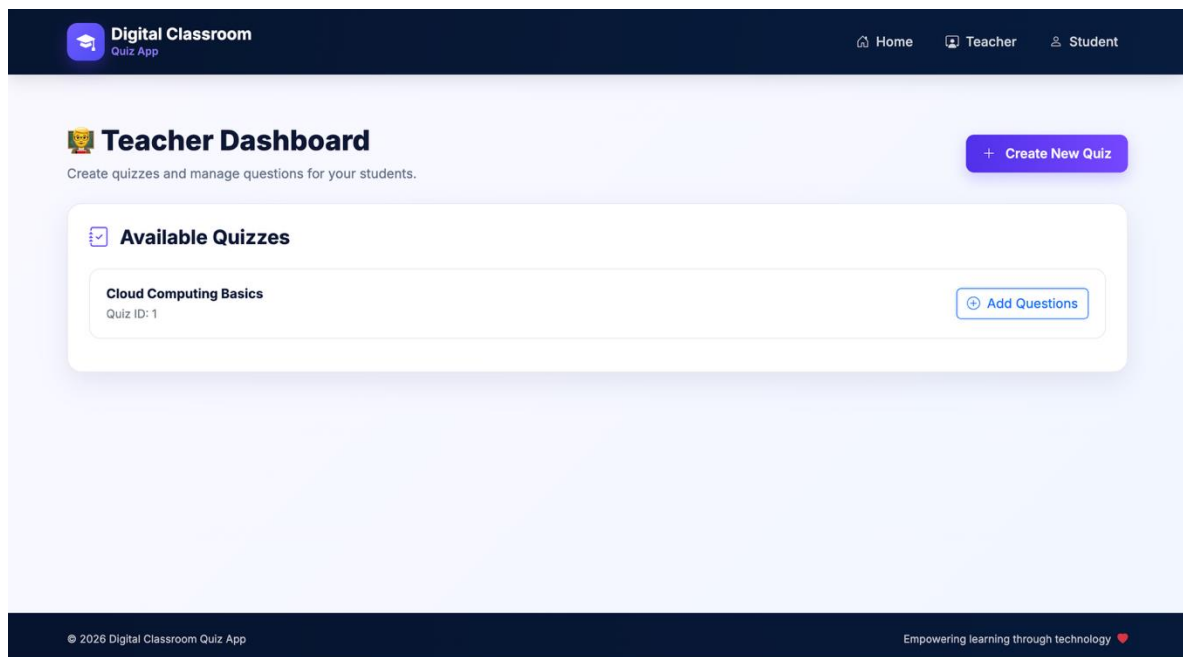
```

08:18:01 PM    [4dr7w]    [29/Aug/2026:14:55:54 +0000] [58] [INFO] Detected service running on port 10000
08:18:01 PM    [4dr7w]    [29/Aug/2026:14:55:54 +0000] [58] [INFO] Docs on specifying a port: https://render.com/docs/web-services#port-binding
08:25:30 PM    [4dr7w]    [29/Aug/2026:14:55:55 +0000] [58] [INFO] Deploying...
08:25:30 PM    [4dr7w]    [29/Aug/2026:14:55:55 +0000] [58] [INFO] Setting WEB_CONCURRENCY=1 by default, based on available CPUs in the instance
08:25:54 PM    [4dr7w]    [29/Aug/2026:14:55:55 +0000] [58] [INFO] 127.0.0.1 - - [29/Aug/2026:14:55:55 +0000] "GET / HTTP/1.1" 200 781 "https://digital-classroom-quiz.onrender.com/student"
08:25:54 PM    [4dr7w]    [29/Aug/2026:14:55:55 +0000] [58] [INFO] Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36
08:25:55 PM    [4dr7w]    [29/Aug/2026:14:55:55 +0000] [58] [INFO] 127.0.0.1 - - [29/Aug/2026:14:55:55 +0000] "GET /static/style.css HTTP/1.1" 304 0 "https://digital-classroom-quiz.onrender.com/"
08:25:55 PM    [4dr7w]    [29/Aug/2026:14:55:55 +0000] [58] [INFO] Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36
08:25:57 PM    [4dr7w]    [29/Aug/2026:14:55:57 +0000] [58] [INFO] 127.0.0.1 - - [29/Aug/2026:14:55:57 +0000] "GET / HTTP/1.1" 200 781 "https://digital-classroom-quiz.onrender.com/student"
08:25:57 PM    [4dr7w]    [29/Aug/2026:14:55:57 +0000] [58] [INFO] Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36
08:25:58 PM    [4dr7w]    [29/Aug/2026:14:55:58 +0000] [58] [INFO] 127.0.0.1 - - [29/Aug/2026:14:55:58 +0000] "GET /static/style.css HTTP/1.1" 304 0 "https://digital-classroom-quiz.onrender.com/"
08:25:58 PM    [4dr7w]    [29/Aug/2026:14:55:58 +0000] [58] [INFO] Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36
08:26:00 PM    [4dr7w]    [2026-08-29 14:56:00 +0000] [58] [INFO] Handling signal: term
08:26:00 PM    [4dr7w]    [2026-08-29 14:56:00 +0000] [60] [INFO] Worker exiting (pid: 60)
08:26:01 PM    [4dr7w]    [2026-08-29 14:56:00 +0000] [58] [INFO] Your service is live
08:26:01 PM    [4dr7w]    [2026-08-29 14:56:00 +0000] [58] [INFO] Available at your primary URL https://digital-classroom-quiz.onrender.com
08:26:01 PM    [4dr7w]    [2026-08-29 14:56:00 +0000] [58] [INFO] Shutting down: Master
08:26:02 PM    [4dr7w]    [2026-08-29 14:56:02 +0000] [58] [INFO] Detected service running on port 10000
08:31:02 PM    [4dr7w]    [29/Aug/2026:15:02:08 +0000] [58] [INFO] Docs on specifying a port: https://render.com/docs/web-services#port-binding
08:31:03 PM    [4dr7w]    [29/Aug/2026:15:02:08 +0000] [58] [INFO] 127.0.0.1 - - [29/Aug/2026:15:02:08 +0000] "GET /teacher HTTP/1.1" 200 3763 "https://digital-classroom-quiz.onrender.com/"
08:32:08 PM    [4dr7w]    [29/Aug/2026:15:02:08 +0000] [58] [INFO] Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36
  
```

Successful deployment of the Flask application on Render Cloud



Digital Classroom Quiz Application Home Page



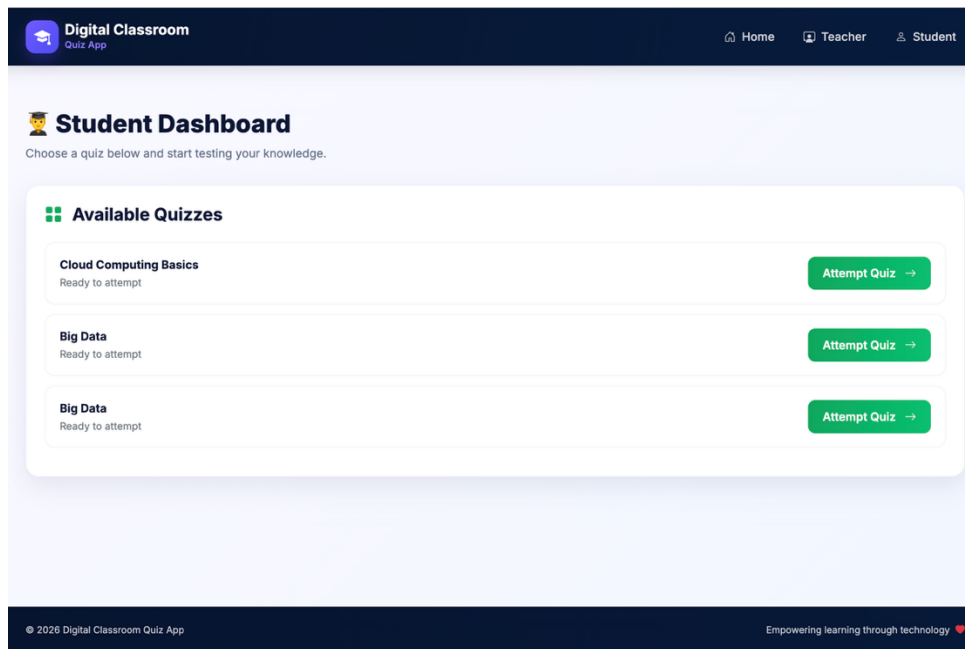
Teacher Dashboard displaying quiz management functionality

The screenshot shows the 'Create a New Quiz' page of the Digital Classroom Quiz App. The browser address bar displays 'digital-classroom-quiz.onrender.com/create_quiz'. The app's header includes the logo and navigation links for Home, Teacher, and Student. The main heading is 'Create a New Quiz' with a subtext 'Give your quiz a meaningful title to get started.' Below this is a form with a 'Quiz Title' label and a text input field containing 'Big Data'. At the bottom of the form are two buttons: 'Create Quiz' (with a plus icon) and 'Cancel'. The footer contains the copyright notice '© 2026 Digital Classroom Quiz App' and the tagline 'Empowering learning through technology' with a heart icon.

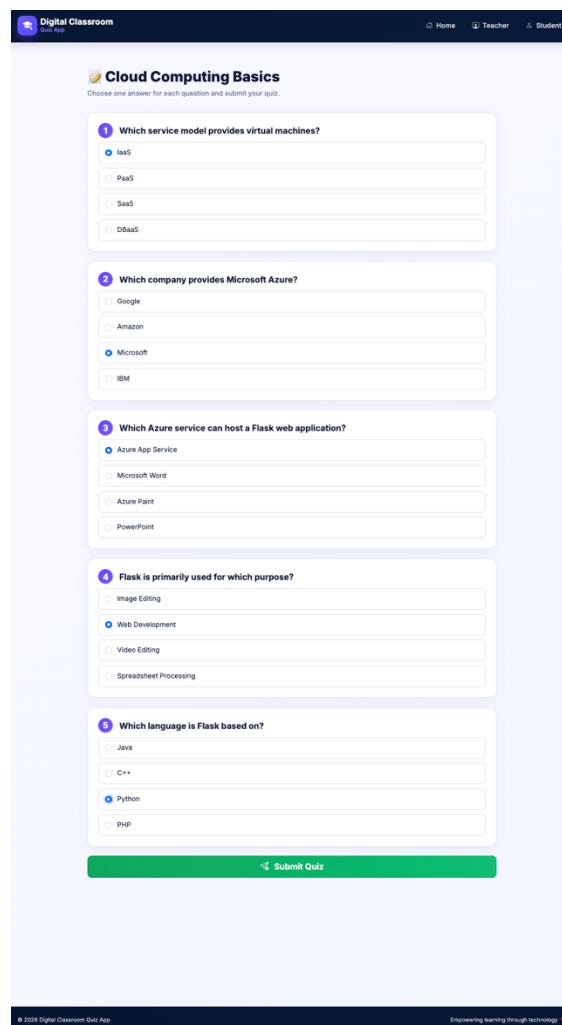
Interface for creating a new classroom quiz

The screenshot shows the 'Add Questions' page for a quiz titled 'Big Data'. The app's header is consistent with the previous screen. The main heading is 'Add Questions' with the subtitle 'Quiz: Big Data'. The interface is divided into two main sections. The 'New Question' section on the left contains a 'Question' input field with the text 'Which service model provides virtual machines?'. Below this are four option fields: 'Option 1' (Google), 'Option 2' (Amazon), 'Option 3' (Azure Paint), and 'Option 4' (IBM). A 'Correct Answer' dropdown menu is set to 'Option 2'. A green '+ Add Question' button is at the bottom. The 'Questions Added' section on the right shows a count of '1' and a list containing the question '1. Which service model provides virtual machines?'. A purple 'Finish Quiz' button with a checkmark icon is at the bottom of this section. The footer is identical to the first screenshot.

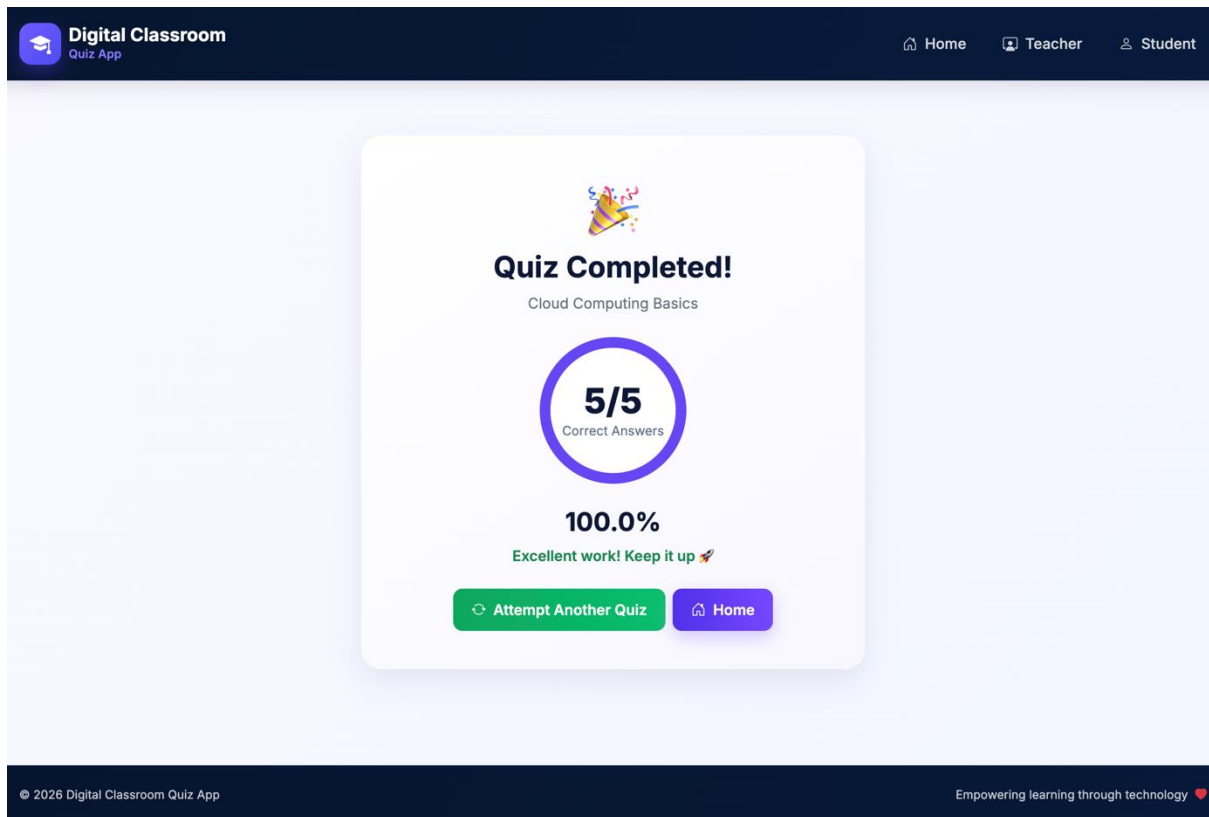
MCQ question creation and correct-answer configuration



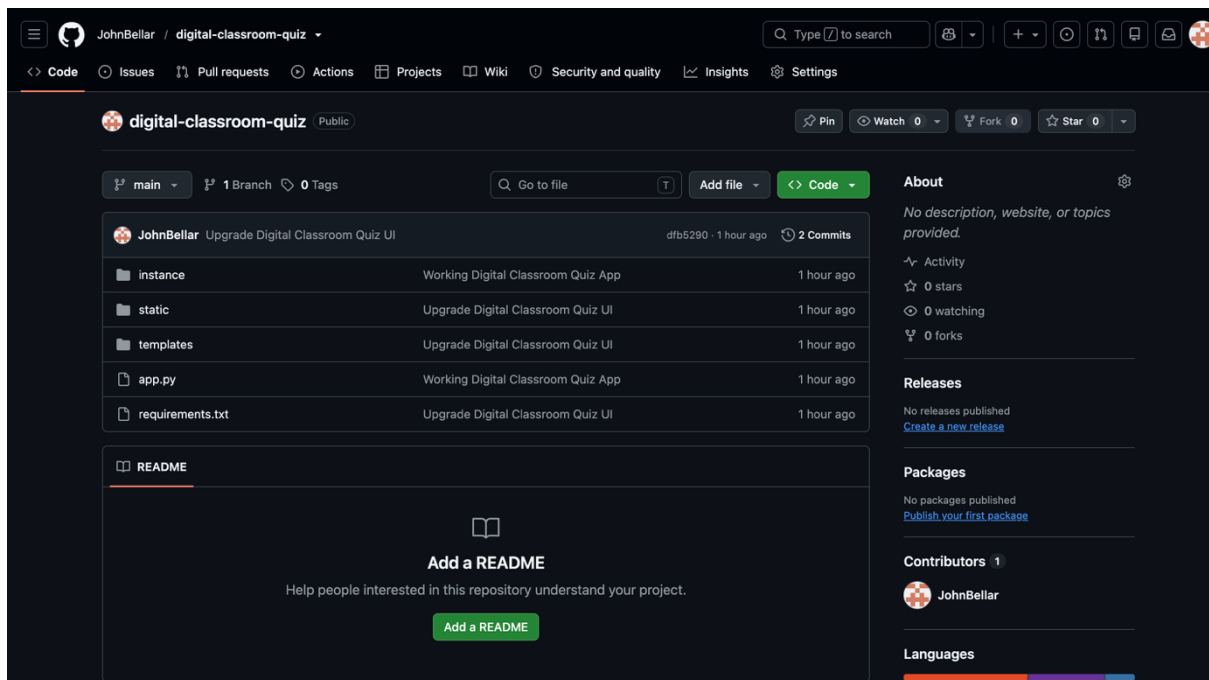
Student Dashboard displaying available quizzes



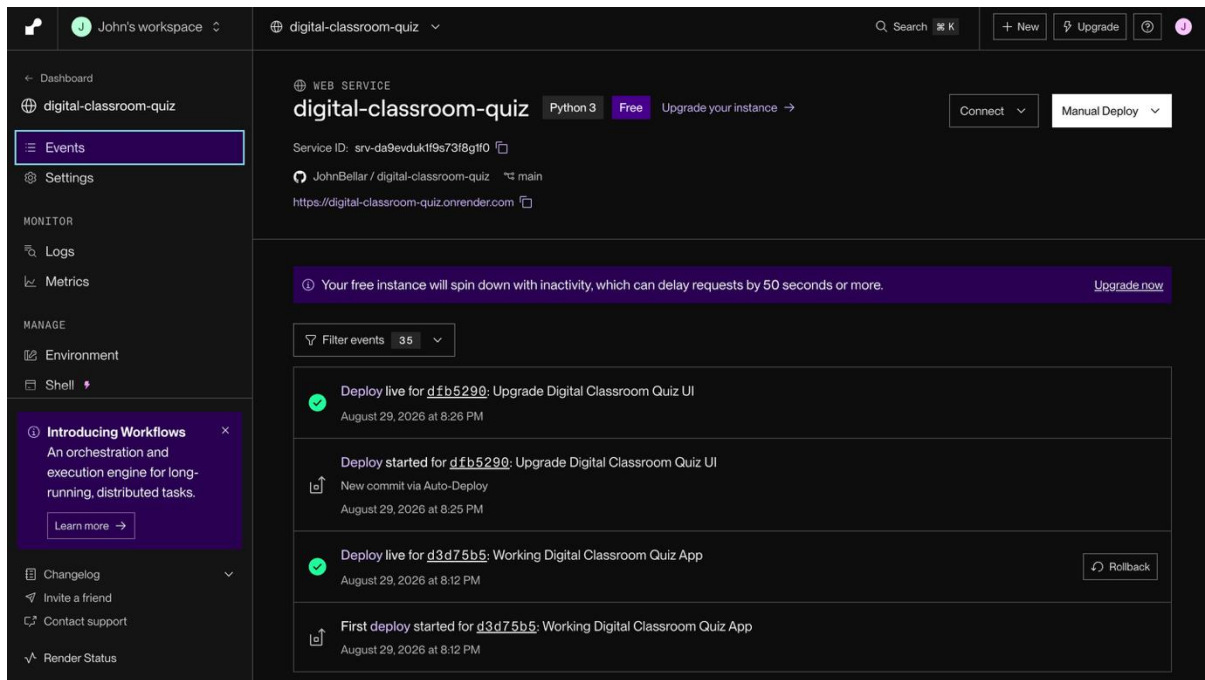
Student interface for attempting a multiple-choice quiz



Automated evaluation and instant quiz result



Application source code maintained using GitHub



Digital Classroom Quiz Application running as a Render cloud Web Service

TEST CASES:

Test Case	Expected Result	Status
Open cloud URL	Home page should load	Passed
Open Teacher Dashboard	Teacher interface should display	Passed
Create a quiz	Quiz should be created	Passed
Add MCQ	Question should be stored	Passed
Open Student Dashboard	Available quizzes should appear	Passed
Select quiz	Questions should be displayed	Passed
Submit correct answer	Score should increase	Passed
Submit incorrect answer	Score should not increase	Passed
Submit complete quiz	Score and percentage should display	Passed
Access from Internet	Application should load without local Flask server	Passed

CLOUD COMPUTING CONCEPTS DEMONSTRATED:

The mini-project demonstrates several fundamental cloud-computing concepts:

- **Cloud Application Hosting:** The Flask application executes on cloud infrastructure instead of relying on the developer's local computer.
- **Platform as a Service:** Render manages infrastructure-related activities such as application deployment, runtime environment, networking, HTTPS access, and service execution.
- **Public Web Access:** The application is made available through a public Internet endpoint.
- **Production Application Server:** Gunicorn is used instead of Flask's development server for the deployed application.
- **Git-based Deployment:** Application source code is stored in GitHub and used as the source for cloud deployment.

- **Continuous Redeployment:** Updated code pushed to the connected Git repository can be redeployed to the cloud application.

LIMITATION: The prototype uses SQLite for lightweight database management. On a free cloud service with an ephemeral filesystem, locally written SQLite data is not suitable for guaranteed long-term persistent storage. For an academic prototype, SQLite provides a simple implementation for demonstrating quiz creation, retrieval, and evaluation. For a production system, the database can be migrated to a managed persistent cloud database such as PostgreSQL.

FUTURE ENHANCEMENTS: The application can be enhanced by implementing teacher and student authentication, user accounts, persistent cloud database storage, student result history, leaderboards, quiz timers, randomized questions, subject-wise quiz categories, teacher analytics dashboards, downloadable reports, role-based access control, and managed PostgreSQL database integration.

RESULT: The Digital Classroom Quiz Application was successfully designed, developed, tested, version-controlled, and deployed as a cloud-based web application. The system enables teachers to create quizzes and add multiple-choice questions with correct answers. Students can access available quizzes through a web browser, submit their responses, and receive automatically calculated scores instantly. The application was implemented using Python Flask, Flask-SQLAlchemy, SQLite, HTML, CSS and Bootstrap, maintained using Git and GitHub, served in the cloud using Gunicorn, and successfully deployed as a Render Web Service. The application is publicly accessible over the Internet, thereby demonstrating the successful implementation and deployment of a functional cloud-based digital classroom assessment system.